

Chapter 11 – Training Deep Neural Networks

This chapter introduces techniques for training deep neural networks effectively, addressing challenges like vanishing and exploding gradients. You'll explore initialization methods, activation functions (such as ReLU variants), batch normalization, gradient clipping, transfer learning, and advanced optimizers like Adam. These tools help build stable, high-performing models on complex datasets.

This notebook contains all the sample code and solutions to the exercises in chapter 11.

[Open in Colab](#)[Open in Kaggle](#)

Setup

This project requires Python 3.7 or above:

```
In [2]: import sys

assert sys.version_info >= (3, 7)
```

And TensorFlow ≥ 2.8 :

```
In [3]: from packaging import version
import tensorflow as tf

assert version.parse(tf.__version__) >= version.parse("2.8.0")
```

As we did in previous chapters, let's define the default font sizes to make the figures prettier:

```
In [4]: import matplotlib.pyplot as plt

plt.rc('font', size=14)
plt.rc('axes', labelsz=14, titlesz=14)
plt.rc('legend', fontsize=14)
plt.rc('xtick', labelsz=10)
plt.rc('ytick', labelsz=10)
```

And let's create the `images/deep` folder (if it doesn't already exist), and define the `save_fig()` function which is used through this notebook to save the figures in high-res for the book:

```
In [5]: from pathlib import Path

IMAGES_PATH = Path() / "images" / "deep"
IMAGES_PATH.mkdir(parents=True, exist_ok=True)

def save_fig(fig_id, tight_layout=True, fig_extension="png", resolution=300):
    path = IMAGES_PATH / f"{fig_id}.{fig_extension}"
```

```
if tight_layout:
    plt.tight_layout()
plt.savefig(path, format=fig_extension, dpi=resolution)
```

Vanishing/Exploding Gradients Problem

When training deep neural networks, gradients can become **very small (vanish)** or **very large (explode)** as they are backpropagated through many layers. This can lead to slow learning or unstable updates, making it difficult for the network to converge to a good solution.

In [5]: *# extra code - this cell generates and saves Figure 11-1*

```
import numpy as np

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

z = np.linspace(-5, 5, 200)

plt.plot([-5, 5], [0, 0], 'k-')
plt.plot([-5, 5], [1, 1], 'k--')
plt.plot([0, 0], [-0.2, 1.2], 'k-')
plt.plot([-5, 5], [-3/4, 7/4], 'g--')
plt.plot(z, sigmoid(z), "b-", linewidth=2,
        label=r"$\sigma(z) = \frac{1}{1+e^{-z}}$")
props = dict(facecolor='black', shrink=0.1)
plt.annotate('Saturating', xytext=(3.5, 0.7), xy=(5, 1), arrowprops=props,
        fontsize=14, ha="center")
plt.annotate('Saturating', xytext=(-3.5, 0.3), xy=(-5, 0), arrowprops=props,
        fontsize=14, ha="center")
plt.annotate('Linear', xytext=(2, 0.2), xy=(0, 0.5), arrowprops=props,
        fontsize=14, ha="center")
plt.grid(True)
plt.axis([-5, 5, -0.2, 1.2])
plt.xlabel("$z$")
plt.legend(loc="upper left", fontsize=16)

save_fig("sigmoid_saturation_plot")
plt.show()
```


This code creates a dense (fully connected) layer with 50 neurons, using the ReLU activation function and He normal initialization for the weights. He initialization helps prevent vanishing gradients in deep networks by scaling the weights appropriately for ReLU activations, maintaining stable variance across layers. This is a common technique for training deep neural networks effectively.

`kernel_initializer="he_normal"` specifies that the weights should be initialized using the He normal method.

```
In [7]: def show_initial_weights(layer, input_shape, fan_in=None):
        """Build `layer` (if not yet built), print some stats and draw a histogram.

        Args:
            layer: a Keras layer instance (Dense, Conv2D, ...).
            input_shape: shape tuple (batch dim may be None) to build the layer.
            fan_in: optional integer; only used when the initializer expects it
                    (e.g. `mode='fan_in'`) or to override the automatic value.
        """
        layer.build(input_shape)
        print("built layer:", layer)

        # infer fans from the kernel shape
        kernel_shape = layer.kernel.shape
        inferred_fan_in = int(kernel_shape[0])
        inferred_fan_out = int(kernel_shape[1])
        if fan_in is None:
            fan_in = inferred_fan_in

        # try to compute a theoretical stddev according to the initializer
        init = layer.kernel_initializer
        stddev = None
        if isinstance(init, tf.keras.initializers.VarianceScaling):
            scale, mode, dist = init.scale, init.mode, init.distribution

            # choose the appropriate "fan" depending on the mode
            if mode == "fan_in":
                divisor = fan_in
            elif mode == "fan_out":
                divisor = inferred_fan_out
            elif mode == "fan_avg":
                divisor = (fan_in + inferred_fan_out) / 2
            else:
                divisor = fan_in

            # variance = scale / divisor for both normal and uniform,
            # but if the distribution is uniform we can also report the
            # limit that Keras uses (a = sqrt(3*scale/divisor)).
            stddev = np.sqrt(scale / divisor)
            print("kernel initializer distribution:", dist)
            if dist == "uniform":
                limit = np.sqrt(3.0 * scale / divisor)
                print("uniform limit:", limit)
        elif init.__class__.__name__.lower().startswith("he"):
            # fair approximation for He initializers
            stddev = np.sqrt(2.0 / fan_in)

        if stddev is not None:
            print("theoretical stddev:", stddev)
```

```

print("kernel shape:", layer.kernel.shape,
      "bias shape:", layer.bias.shape)

kernel, bias = layer.get_weights()
print("kernel mean:", kernel.mean(), "std:", kernel.std())
print("first 10 values:", np.round(kernel.flatten()[:10], 4))

plt.hist(kernel.flatten(), bins=50)
plt.title(f"{layer.kernel_initializer.__class__.__name__} initialized weight")
plt.show()

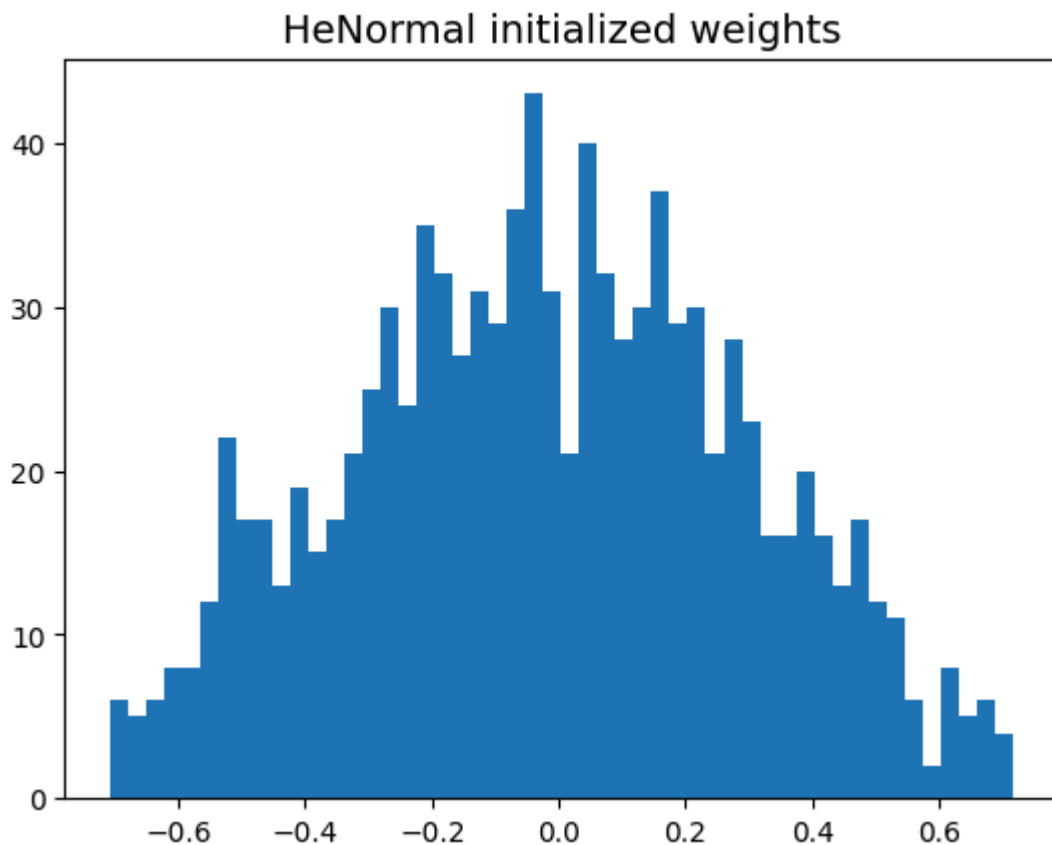
fan_in = 20
show_initial_weights(dense, input_shape=(None, 20), fan_in=fan_in)

```

```

built layer: <Dense name=dense, built=True>
kernel initializer distribution: truncated_normal
theoretical stddev: 0.31622776601683794
kernel shape: (20, 50) bias shape: (50,)
kernel mean: -0.016442059 std: 0.3108163
first 10 values: [ 0.0023  0.184  -0.1803  0.2686 -0.5326  0.1735  0.2378  0.2095
 0.599
 -0.0645]

```



The `.kernel_initializer` object itself also exposes its configuration; you can query `dense.kernel_initializer` for the seed, scale, etc., and even call it manually to sample a tensor:

```

In [8]: init = tf.keras.initializers.he_normal()
w = init(shape=(fan_in, 50))
print(w.numpy().std(), np.sqrt(2.0 / fan_in))

```

```

0.31687748 0.31622776601683794

```

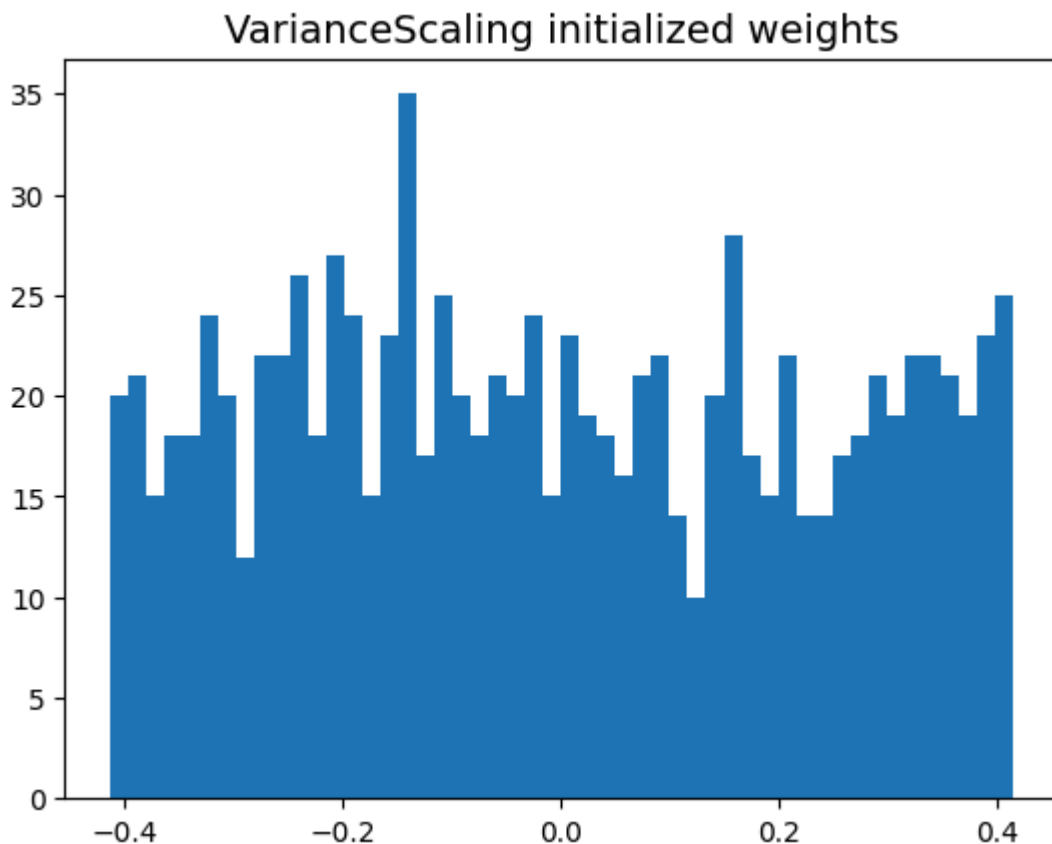
Custom Initialization

Furthermore, a custom He initialization variant for a dense layer using the sigmoid activation function is instantiated. The `VarianceScaling` initializer is configured with `scale=2.0` (suitable for sigmoid, which is similar to Xavier/Glorot initialization), `mode="fan_avg"` (averages fan-in and fan-out), and `distribution="uniform"` for uniform weight distribution. This helps stabilize gradients in deep networks, though He initialization is more commonly used with ReLU. The layer has 50 neurons and sigmoid activation.

```
In [9]: he_avg_init = tf.keras.initializers.VarianceScaling(scale=2., mode="fan_avg",  
                                                           distribution="uniform")  
dense = tf.keras.layers.Dense(50, activation="sigmoid",  
                               kernel_initializer=he_avg_init)
```

```
In [10]: show_initial_weights(dense, input_shape=(None, 20), fan_in=fan_in)
```

```
built layer: <Dense name=dense_1, built=True>  
kernel initializer distribution: uniform  
uniform limit: 0.4140393356054125  
theoretical stddev: 0.23904572186687872  
kernel shape: (20, 50) bias shape: (50,)  
kernel mean: -0.0031661207 std: 0.23937334  
first 10 values: [-0.104 -0.216 -0.1485 -0.4098 -0.1564 0.2005 -0.3475 0.251  
0.4112  
-0.1511]
```



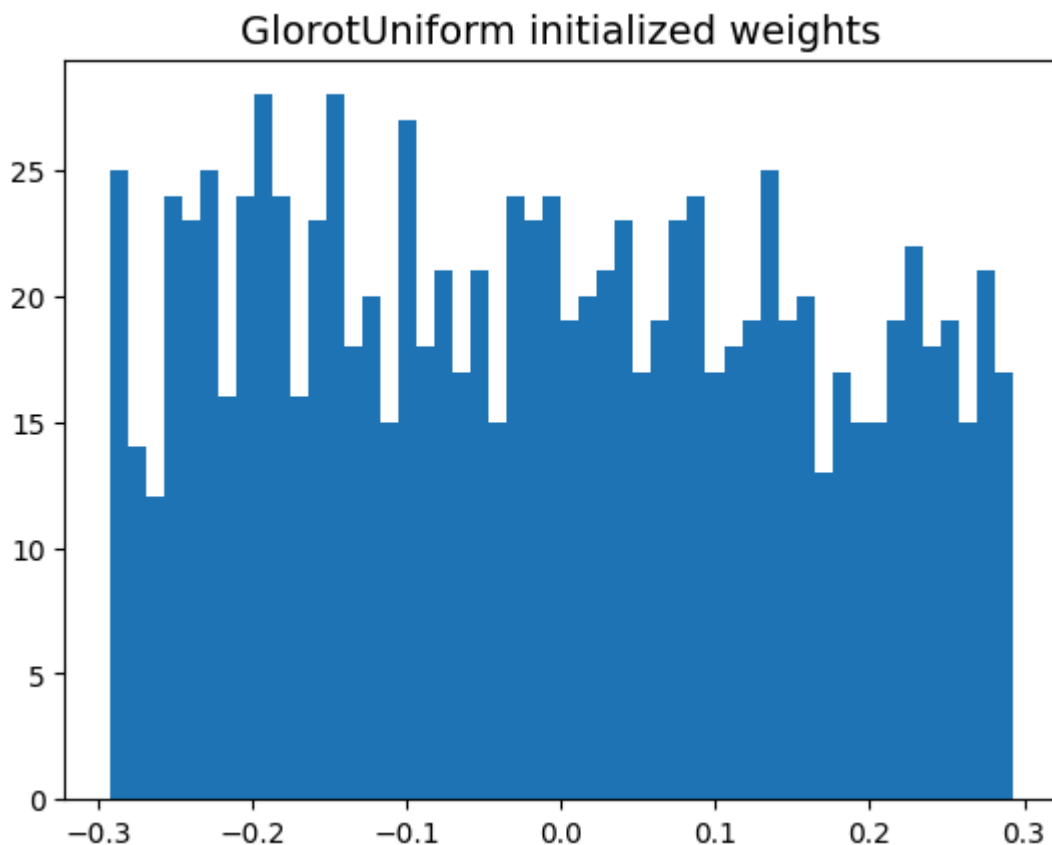
Secondly, we can use **Xavier initialization** for a dense layer with the tanh activation function as follows:

```
In [11]: from tensorflow import keras
layer = keras.layers.Dense(
    50,
    activation="tanh",
    kernel_initializer=keras.initializers.GlorotUniform(),
)
```

where `kernel_initializer=keras.initializers.GlorotUniform()` specifies Xavier uniform initialization. Equivalently, you could use `GlorotNormal()` for normal distribution.

```
In [12]: show_initial_weights(layer, input_shape=(None, 20), fan_in=fan_in)
```

```
built layer: <Dense name=dense_2, built=True>
kernel initializer distribution: uniform
uniform limit: 0.29277002188455997
theoretical stddev: 0.1690308509457033
kernel shape: (20, 50) bias shape: (50,)
kernel mean: -0.009273 std: 0.16631311
first 10 values: [-0.12    0.2114  0.1765 -0.2867 -0.0129  0.2536 -0.222  -0.2325
-0.0585
-0.0242]
```



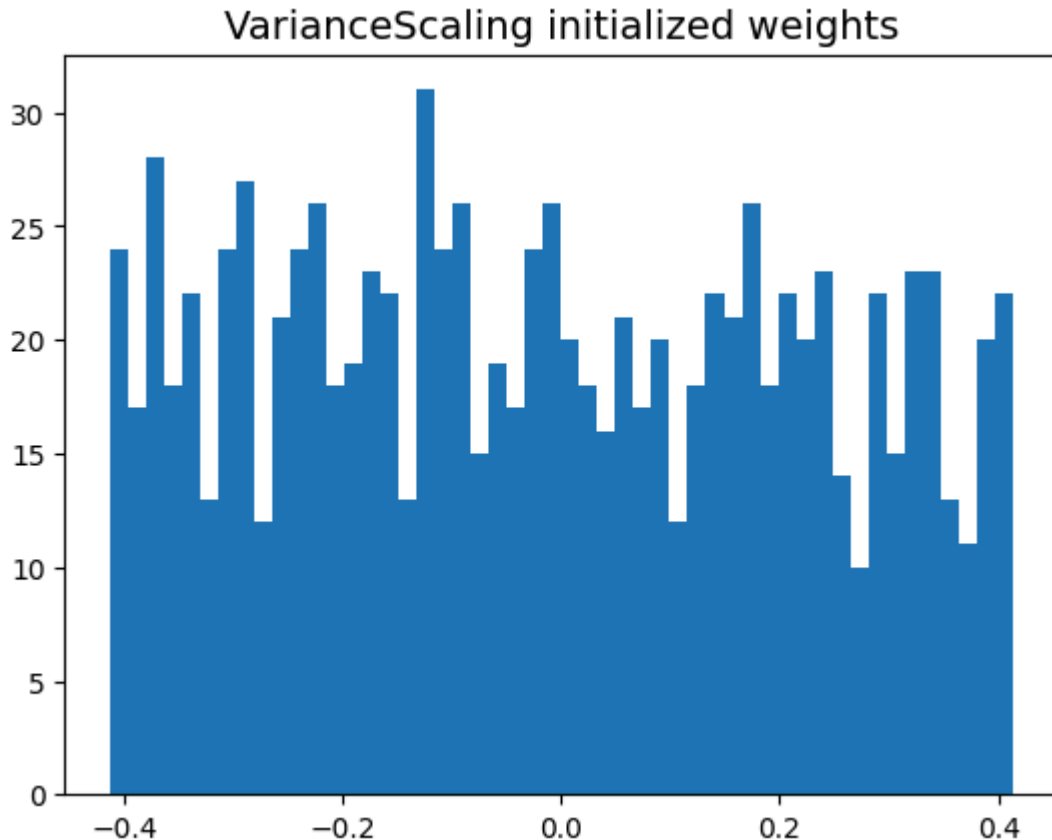
```
In [13]: layer = keras.layers.Dense(
    50,
    activation="sigmoid",
    kernel_initializer=keras.initializers.VarianceScaling(
        scale=2.0, mode="fan_avg", distribution="uniform"
    ),
)
```

```
In [14]: show_initial_weights(layer, input_shape=(None, 20), fan_in=fan_in)
```

```

built layer: <Dense name=dense_3, built=True>
kernel initializer distribution: uniform
uniform limit: 0.4140393356054125
theoretical stddev: 0.23904572186687872
kernel shape: (20, 50) bias shape: (50,)
kernel mean: -0.013412503 std: 0.23670338
first 10 values: [-0.1282  0.3177 -0.164   0.1715 -0.0418  0.0047 -0.2984 -0.3687
0.1566
-0.152 ]

```



This code creates a dense layer with 50 neurons using the sigmoid activation function and a custom Xavier initialization variant. The `VarianceScaling` initializer is set with `scale=2.0` (appropriate for sigmoid, matching Xavier/Glorot scaling), `mode="fan_avg"` (averaging fan-in and fan-out), and `distribution="uniform"` for uniform weight distribution. This helps stabilize gradients in deep networks, though Xavier is typically used with tanh or sigmoid. The layer is defined using TensorFlow's Keras API.

Conclusion

- He initialization is especially appropriate for neural networks using ReLU activations, as it helps keep the signals stable during forward and backward propagation, promoting faster convergence and better performance. `he_normal` is one implementation of He initialization; it samples weights from a normal distribution whose **variance** is computed from the number of incoming connections (`fan_in`):

$$\text{Var}(W) = \frac{2}{\text{fan_in}}$$

- Xavier initialization is suitable for networks using Sigmoid or Tanh activations, because it helps maintain stable signal variance during forward and backward passes.

The variance in this case is

$$\text{Var}(W) = \frac{1}{f_{\text{an}} \backslash \text{in}} \quad (\text{or, equivalently, } \text{Var}(W) = \frac{2}{f_{\text{an}} \backslash \text{in} + f_{\text{an}} \backslash \text{out}})$$

According to the chosen probability distribution, the initialization parameters are set as follows:

- **Normal distribution** – weights are drawn from $\mathcal{N}(0, \sigma^2)$ with

$$\sigma = \sqrt{\frac{2}{n_{\text{in}} + n_{\text{out}}}}.$$

- **Uniform distribution** – weights are sampled uniformly from $[-r, r]$ where

$$r = \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}.$$

Nonsaturating Activation Functions

Using nonsaturating activation functions like ReLU and its variants (Leaky ReLU, ELU, SELU) helps mitigate the vanishing gradients problem. These functions do not saturate for positive inputs, allowing gradients to flow more easily through deep networks.

Leaky ReLU

This code creates a dense layer with 50 neurons using the Leaky ReLU activation function, which allows a small, non-zero gradient when the input is negative. This helps prevent neurons from "dying" during training, addressing issues associated with the standard ReLU function.

```
In [15]: # extra code - this cell generates and saves Figure 11-2

def leaky_relu(z, alpha):
    return np.maximum(alpha * z, z)

z = np.linspace(-5, 5, 200)
plt.plot(z, leaky_relu(z, 0.1), "b-", linewidth=2, label=r"$LeakyReLU(z) = \max(\backslash$
plt.plot([-5, 5], [0, 0], 'k-')
plt.plot([0, 0], [-1, 3.7], 'k-')
plt.grid(True)
props = dict(facecolor='black', shrink=0.1)
```



```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/activations/leaky_relu.p
y:41: UserWarning: Argument `alpha` is deprecated. Use `negative_slope` instead.
warnings.warn(
```

```
In [17]: model = tf.keras.models.Sequential([
# [...] # more layers
tf.keras.layers.Dense(50, kernel_initializer="he_normal"), # no activation
tf.keras.layers.LeakyReLU(alpha=0.2), # activation as a separate layer
# [...] # more layers
])
```

ELU (Exponential Linear Unit)

This code creates a dense layer with 50 neurons using the **Exponential Linear Unit (ELU)** activation function. ELU helps mitigate the vanishing gradients problem by allowing negative outputs for negative inputs, which helps maintain a non-zero gradient. This can lead to faster convergence and improved performance in deep neural networks compared to standard ReLU. The ELU is defined as:

$$\text{ELU}_{\alpha}(z) = \begin{cases} z & \text{if } z \geq 0, \\ \alpha(e^z - 1) & \text{if } z < 0, \end{cases}$$

and its derivative is

$$\text{ELU}'_{\alpha}(z) = \begin{cases} 1 & \text{if } z \geq 0, \\ \alpha e^z & \text{if } z < 0. \end{cases}$$

Notes: the negative part is bounded below by $-\alpha$, the derivative is continuous at 0 when $\alpha = 1$ (the common default), and ELU yields nonzero gradients for $z < 0$, helping training stability.

Implementing ELU in TensorFlow is trivial, just specify the activation function when building each layer, and use He initialization:

```
In [18]: dense = tf.keras.layers.Dense(50, activation="elu",
kernel_initializer="he_normal")
```

You don't have to stick with `LeakyReLU` – you can switch to ELU and tune its `alpha` parameter directly. `tf.keras.layers.ELU` is a layer class, so you can either instantiate it and pass the object as the `activation`, or create it inline.

For example, to use an ELU with `alpha=0.5` (the default is `1.0`):

```
In [19]: model = tf.keras.models.Sequential([
# ...existing code...
tf.keras.layers.Dense(50, kernel_initializer="he_normal"), # no activation
tf.keras.layers.ELU(alpha=0.5), # ELU activation as a separate layer with c
# ...existing code...
])
```

SELU (Scaled ELU)

By default, the SELU hyperparameters (`scale` and `alpha`) are tuned in such a way that the mean output of each neuron remains close to 0, and the standard deviation remains close to 1 (assuming the inputs are standardized with mean 0 and standard deviation 1 too, and other constraints are respected, as explained in the book).

Using this activation function, even a 1,000 layer deep neural network preserves roughly mean 0 and standard deviation 1 across all layers, avoiding the exploding/vanishing gradients problem:

```
In [20]: # extra code - this cell generates and saves Figure 11-3

from scipy.special import erfc

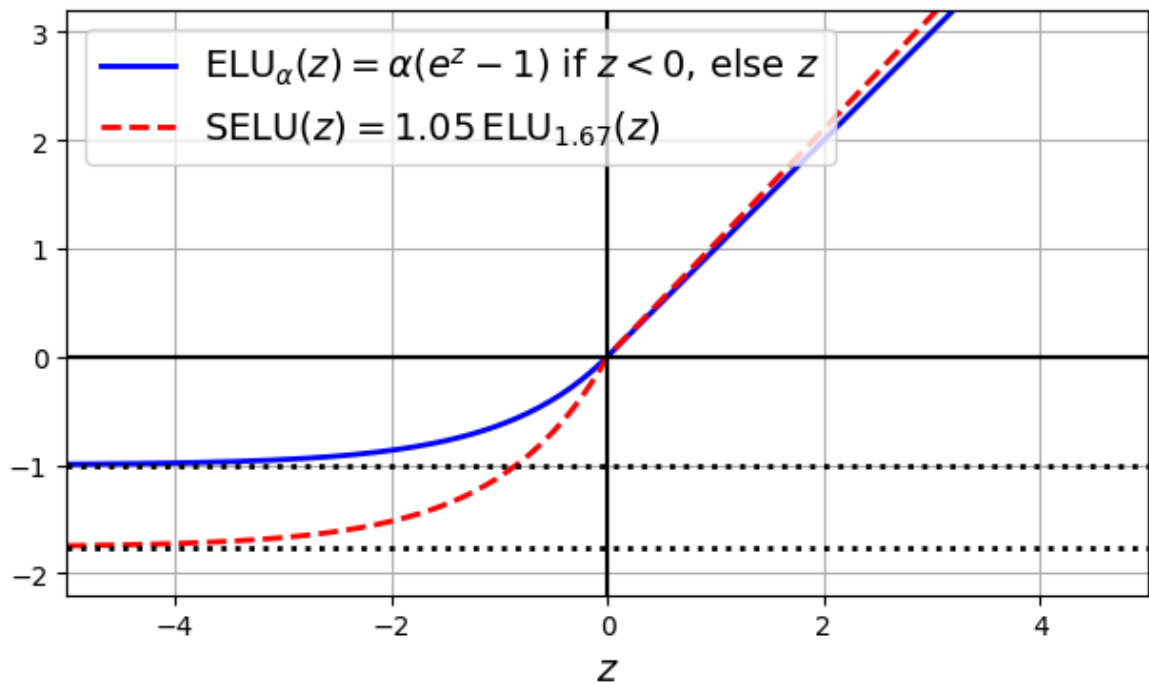
# alpha and scale to self normalize with mean 0 and standard deviation 1
# (see equation 14 in the paper):
alpha_0_1 = -np.sqrt(2 / np.pi) / (erfc(1 / np.sqrt(2)) * np.exp(1 / 2) - 1)
scale_0_1 = (
    (1 - erfc(1 / np.sqrt(2)) * np.sqrt(np.e))
    * np.sqrt(2 * np.pi)
    * (
        2 * erfc(np.sqrt(2)) * np.e ** 2
        + np.pi * erfc(1 / np.sqrt(2)) ** 2 * np.e
        - 2 * (2 + np.pi) * erfc(1 / np.sqrt(2)) * np.sqrt(np.e)
        + np.pi
        + 2
    ) ** (-1 / 2)
)

def elu(z, alpha=1):
    return np.where(z < 0, alpha * (np.exp(z) - 1), z)

def selu(z, scale=scale_0_1, alpha=alpha_0_1):
    return scale * elu(z, alpha)

z = np.linspace(-5, 5, 200)
plt.plot(z, elu(z), "b-", linewidth=2, label=r"ELU$_\alpha(z) = \alpha (e^z - 1)$")
plt.plot(z, selu(z), "r--", linewidth=2, label=r"SELU$(z) = 1.05 \setminus, \text{\$ELU}_{\{1.67\}}$")
plt.plot([-5, 5], [0, 0], 'k-')
plt.plot([-5, 5], [-1, -1], 'k:', linewidth=2)
plt.plot([-5, 5], [-1.758, -1.758], 'k:', linewidth=2)
plt.plot([0, 0], [-2.2, 3.2], 'k-')
plt.grid(True)
plt.axis((-5, 5, -2.2, 3.2))
plt.xlabel("$z$")
plt.gca().set_aspect("equal")
plt.legend()

save_fig("elu_selu_plot")
plt.show()
```



Using SELU is straightforward:

```
In [21]: dense = tf.keras.layers.Dense(50, activation="selu",
                                         kernel_initializer="lecun_normal")
```

You would typically integrate it directly into the `Dense` layer definition, along with `lecun_normal` initialization, as this combination is crucial for achieving the self-normalizing properties of SELU.

Extra material – an example of a self-regularized network using SELU

Let's create a neural net for Fashion MNIST with 100 hidden layers, using the SELU activation function:

```
In [22]: tf.random.set_seed(42)
model = tf.keras.Sequential()
model.add(tf.keras.layers.Input(shape=(28, 28)))
model.add(tf.keras.layers.Flatten())
for layer in range(100):
    model.add(tf.keras.layers.Dense(100, activation="selu",
                                     kernel_initializer="lecun_normal"))
model.add(tf.keras.layers.Dense(10, activation="softmax"))
```

```
In [23]: model.compile(loss="sparse_categorical_crossentropy",
                      optimizer=tf.keras.optimizers.SGD(learning_rate=0.001),
                      metrics=["accuracy"])
```

Now let's train it. Do not forget to scale the inputs to mean 0 and standard deviation 1:

```
In [7]: fashion_mnist = tf.keras.datasets.fashion_mnist.load_data()
(X_train_full, y_train_full), (X_test, y_test) = fashion_mnist
X_train, y_train = X_train_full[:-5000], y_train_full[:-5000]
```

```
X_valid, y_valid = X_train_full[-5000:], y_train_full[-5000:]
X_train, X_valid, X_test = X_train / 255, X_valid / 255, X_test / 255
```

```
In [25]: class_names = ["T-shirt/top", "Trouser", "Pullover", "Dress", "Coat",
                        "Sandal", "Shirt", "Sneaker", "Bag", "Ankle boot"]
```

```
In [8]: pixel_means = X_train.mean(axis=0, keepdims=True)
pixel_stds = X_train.std(axis=0, keepdims=True)
X_train_scaled = (X_train - pixel_means) / pixel_stds
X_valid_scaled = (X_valid - pixel_means) / pixel_stds
X_test_scaled = (X_test - pixel_means) / pixel_stds
```

```
In [27]: history = model.fit(X_train_scaled, y_train, epochs=5,
                             validation_data=(X_valid_scaled, y_valid))
```

Epoch 1/5

1719/1719 ————— 39s 19ms/step - accuracy: 0.3897 - loss: 1.6379 - val_accuracy: 0.6994 - val_loss: 0.8218

Epoch 2/5

1719/1719 ————— 31s 18ms/step - accuracy: 0.6724 - loss: 0.8839 - val_accuracy: 0.7122 - val_loss: 0.8037

Epoch 3/5

1719/1719 ————— 30s 18ms/step - accuracy: 0.7300 - loss: 0.7452 - val_accuracy: 0.6962 - val_loss: 0.8994

Epoch 4/5

1719/1719 ————— 32s 19ms/step - accuracy: 0.6445 - loss: 0.9682 - val_accuracy: 0.7092 - val_loss: 0.8186

Epoch 5/5

1719/1719 ————— 31s 18ms/step - accuracy: 0.6681 - loss: 0.8906 - val_accuracy: 0.7430 - val_loss: 0.7250

```
In [28]: # extra code - this cell defines a new function to compute and print gradients f
def compute_layer_gradients(model, target_layer, X_sample, y_sample):
    """
    Computes and prints the gradients of the loss with respect to the trainable
    variables of a specific target layer for a given sample of data.

    Args:
        model (tf.keras.Model): The Keras model.
        target_layer (tf.keras.layers.Layer): The layer for which to compute gra
        X_sample (tf.Tensor): A small batch of input features for demonstration.
        y_sample (tf.Tensor): A small batch of corresponding labels for demonstr

    Returns:
        list: A list of gradients (tf.Tensor) for each trainable variable in the
            target_layer. Returns an empty list if the layer has no trainable
            variables or if gradients cannot be computed.
    """
    if not target_layer.trainable_variables:
        print(f"Layer '{target_layer.name}' does not have trainable variables to
        return [])

    # Instantiate the loss function for direct calculation
    # Since the model uses 'softmax' activation in the output layer, from_logits
    loss_fn = tf.keras.losses.SparseCategoricalCrossentropy()

    with tf.GradientTape() as tape:
        # Forward pass: get the model's predictions
        y_pred = model(X_sample, training=True)
```

```

        # Calculate the loss using the instantiated loss function
        loss = loss_fn(y_sample, y_pred)

        # Get gradients of the loss with respect to the trainable variables of the target layer
        trainable_vars_of_layer = target_layer.trainable_variables
        gradients = tape.gradient(loss, trainable_vars_of_layer)

        print(f"\nGradients for layer '{target_layer.name}':")
        for var, grad in zip(trainable_vars_of_layer, gradients):
            print(f"  Variable: {var.name}")
            print(f"  Shape: {var.shape}")
            if grad is not None:
                print(f"    Gradient (first 5 elements): {grad.numpy().flatten()[:5]}")
                print(f"    Gradient Norm (L2): {tf.norm(grad).numpy()}")
            else:
                print(f"    Gradient: None (variable not contributing to this loss path)")
        print("-" * 30)

        return gradients

# Example usage of the new function:
# Assuming 'model', 'X_train_scaled', 'y_train' are defined from previous cells.
# Get the first Dense layer (model.layers[1] in your current setup)
target_layer_selu = model.layers[1]
X_sample = X_train_scaled[:10]
y_sample = y_train[:10]

# Compute and print gradients using the refactored function
computed_gradients_selu = compute_layer_gradients(model, target_layer_selu, X_sample, y_sample)

```

```

Gradients for layer 'dense_9':
  Variable: kernel
  Shape: (784, 100)
  Gradient (first 5 elements): [ 1.0870647e-04 -1.8343849e-06  1.1079906e-04  9.5
245894e-05
-1.5951735e-05]
  Gradient Norm (L2): 3.7559328079223633
-----
  Variable: bias
  Shape: (100,)
  Gradient (first 5 elements): [-0.01278107  0.00021568 -0.01302711 -0.01119846
0.00187551]
  Gradient Norm (L2): 0.1613408327102661
-----

```

The network managed to learn, despite how deep it is. Now look at what happens if we try to use the ReLU activation function instead:

```
In [29]: tf.random.set_seed(42)
```

```
In [30]: model = tf.keras.Sequential()
model.add(tf.keras.layers.Input(shape=[28, 28]))
model.add(tf.keras.layers.Flatten())
for layer in range(100):
    model.add(tf.keras.layers.Dense(100, activation="relu",
                                     kernel_initializer="he_normal"))
model.add(tf.keras.layers.Dense(10, activation="softmax"))

```

```
In [31]: model.compile(loss="sparse_categorical_crossentropy",
                      optimizer=tf.keras.optimizers.SGD(learning_rate=0.001),
                      metrics=["accuracy"])
```

```
In [32]: history = model.fit(X_train_scaled, y_train, epochs=5,
                             validation_data=(X_valid_scaled, y_valid))
```

```
Epoch 1/5
1719/1719 ————— 38s 18ms/step - accuracy: 0.2095 - loss: 2.0111 -
val_accuracy: 0.3680 - val_loss: 1.4998
Epoch 2/5
1719/1719 ————— 39s 17ms/step - accuracy: 0.4403 - loss: 1.2937 -
val_accuracy: 0.5098 - val_loss: 1.0590
Epoch 3/5
1719/1719 ————— 30s 17ms/step - accuracy: 0.5474 - loss: 1.0719 -
val_accuracy: 0.5732 - val_loss: 1.0589
Epoch 4/5
1719/1719 ————— 29s 17ms/step - accuracy: 0.6404 - loss: 0.9059 -
val_accuracy: 0.6640 - val_loss: 0.8915
Epoch 5/5
1719/1719 ————— 29s 17ms/step - accuracy: 0.7031 - loss: 0.7839 -
val_accuracy: 0.7580 - val_loss: 0.7074
```

Not great at all, we suffered from the vanishing/exploding gradients problem.

```
In [33]: target_layer_relu = model.layers[1]

# Compute and print gradients using the refactored function
computed_gradients_relu = compute_layer_gradients(model, target_layer_relu, X_sa
```

```
Gradients for layer 'dense_110':
  Variable: kernel
  Shape: (784, 100)
  Gradient (first 5 elements): [-5.2931409e-05  6.3393003e-05 -2.5682637e-04  8.9
688896e-05  1.5162128e-05]
  Gradient Norm (L2): 4.366199016571045
-----
  Variable: bias
  Shape: (100,)
  Gradient (first 5 elements): [ 0.00622337 -0.00745338  0.03019615 -0.0105451 -
0.00178267]
  Gradient Norm (L2): 0.1733173131942749
-----
```

```
In [34]: model.layers[1].get_weights()[0]
```



```
Out[34]: array([[ 0.04127103,  0.09410827, -0.04553709, ...,  0.01995559,
                  0.02428236, -0.01700601],
                [-0.02497083,  0.01605151, -0.03279416, ...,  0.04592589,
                  0.00551896, -0.05351187],
                [ 0.0587437 ,  0.09145813, -0.00695916, ...,  0.09943683,
                  0.01988941, -0.05342944],
                ...,
                [-0.11076238,  0.00306231,  0.03301753, ...,  0.02957512,
                  -0.07546408,  0.05741999],
                [ 0.04381795,  0.03568869, -0.00842719, ..., -0.03900368,
                  0.04032969, -0.00660047],
                [-0.0230844 ,  0.00930108,  0.00191613, ...,  0.07464728,
                  0.01472195, -0.01715416]], dtype=float32)
```

GELU, Swish and Mish

The **GELU (Gaussian Error Linear Unit)** activation function is a smooth, non-linear activation function that combines properties of *ReLU* and *sigmoid functions*. It helps mitigate the vanishing gradients problem by allowing small negative outputs, which can improve training stability and performance in deep neural networks.

where the GELU activation is defined as:

$$\text{GELU}(z) = 0.5z \left(1 + \tanh \left[\sqrt{\frac{2}{\pi}} (z + 0.044715z^3) \right] \right)$$

```
In [35]: # extra code - this cell generates and saves Figure 11-4

def swish(z, beta=1):
    return z * sigmoid(beta * z)

def approx_gelu(z):
    return swish(z, beta=1.702)

def softplus(z):
    return np.log(1 + np.exp(z))

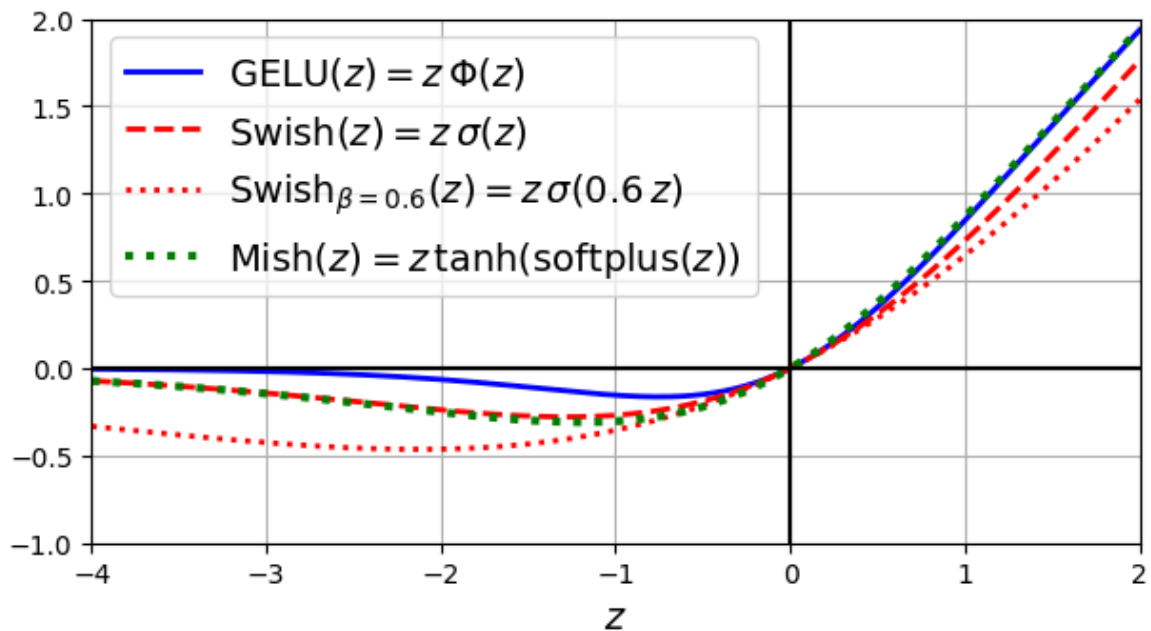
def mish(z):
    return z * np.tanh(softplus(z))

z = np.linspace(-4, 2, 200)

beta = 0.6
plt.plot(z, approx_gelu(z), "b-", linewidth=2,
         label=r"GELU$(z) = z\,\Phi(z)$")
plt.plot(z, swish(z), "r--", linewidth=2,
         label=r"Swish$(z) = z\,\sigma(z)$")
plt.plot(z, swish(z, beta), "r:", linewidth=2,
         label=r"Swish$_{\{\beta=\{beta\}\}}(z)=z\,\sigma(\{beta\},z)$")
plt.plot(z, mish(z), "g:", linewidth=3,
         label=r"Mish$(z) = z\,\tanh(\text{softplus}(z))$")
plt.plot([-4, 2], [0, 0], 'k-')
plt.plot([0, 0], [-2.2, 3.2], 'k-')
plt.grid(True)
plt.axis([-4, 2, -1, 2])
plt.gca().set_aspect("equal")
```

```
plt.xlabel("$z$")
plt.legend(loc="upper left")

save_fig("gelu_swish_mish_plot")
plt.show()
```



This plot compares three smooth, non-saturating activations (and a Swish variant) over $z \in [-4, 2]$:

- Blue (approx GELU): $\text{GELU} \approx z \cdot \Phi(z)$. smooth, nearly linear for positive z , yields small negative outputs for negative z — used in Transformers (BERT).
- Red (Swish, $\beta=1$ — dashed; $\beta=0.6$ — dotted): $\text{swish}(z)=z \cdot \sigma(\beta z)$. smooth, can be non-monotonic; β controls the “leak” for negatives (smaller $\beta \rightarrow$ smaller negative response). Swish with $\beta \approx 1.702$ closely approximates GELU.
- Green (Mish): $z \cdot \tanh(\text{softplus}(z))$. smooth, non-monotonic, produces mild negative tail and very smooth transitions.

Key observations and practical implications

- All three preserve nonzero gradients for negative inputs (unlike ReLU), improving gradient flow and representation richness in deep nets.
- Differences are subtle for moderate z but can affect optimization dynamics and generalization; non-monotonicity can act like a mild implicit regularizer.
- Computational cost: $\text{GELU/Swish/Mish} > \text{ReLU}$; GELU and Swish are commonly implemented and well-optimized (GELU standard in NLP; Swish used in state-of-the-art vision nets like EfficientNet). Mish is promising but less ubiquitous.
- Recommendation: prefer GELU for Transformer/NLP work, try Swish (tune β) for vision, consider Mish as an experimental alternative — always validate on your task and monitor training speed and generalization.

Batch Normalization

Batch normalization **standardizes the inputs to each layer**, stabilizing learning and allowing for higher learning rates. It helps mitigate internal covariate shift, improving convergence speed and model performance.

```
In [36]: # extra code - clear the name counters and set the random seed
tf.keras.backend.clear_session()
tf.random.set_seed(42)
```

```
In [37]: model = tf.keras.Sequential([
    tf.keras.layers.Input(shape=[28, 28]),
    tf.keras.layers.Flatten(),
    tf.keras.layers.BatchNormalization(), # standardize inputs
    tf.keras.layers.Dense(300, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.BatchNormalization(), # standardize inputs
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.BatchNormalization(), # standardize inputs
    tf.keras.layers.Dense(10, activation="softmax")
])
```

```
In [38]: model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	
flatten (Flatten)	(None, 784)	
batch_normalization (BatchNormalization)	(None, 784)	
dense (Dense)	(None, 300)	
batch_normalization_1 (BatchNormalization)	(None, 300)	
dense_1 (Dense)	(None, 100)	
batch_normalization_2 (BatchNormalization)	(None, 100)	
dense_2 (Dense)	(None, 10)	

Total params: 271,346 (1.04 MB)

Trainable params: 268,978 (1.03 MB)

Non-trainable params: 2,368 (9.25 KB)

```
In [39]: [(var.name, var.trainable) for var in model.layers[1].variables]
```

```
Out[39]: [('gamma', True),
          ('beta', True),
          ('moving_mean', False),
          ('moving_variance', False)]
```

```
In [40]: # extra code - just show that the model works! 😊
model.compile(loss="sparse_categorical_crossentropy", optimizer="sgd",
              metrics=["accuracy"])
model.fit(X_train, y_train, epochs=2, validation_data=(X_valid, y_valid))
```

```
Epoch 1/2
1719/1719 ————— 15s 8ms/step - accuracy: 0.7544 - loss: 0.7397 - v
al_accuracy: 0.8580 - val_loss: 0.3930
Epoch 2/2
1719/1719 ————— 14s 8ms/step - accuracy: 0.8563 - loss: 0.4076 - v
al_accuracy: 0.8674 - val_loss: 0.3615
```

```
Out[40]: <keras.src.callbacks.history.History at 0x7e421ad36660>
```

```
In [41]: [(var.name, var.trainable) for var in model.layers[1].variables]
```

```
Out[41]: [('gamma', True),
          ('beta', True),
          ('moving_mean', False),
          ('moving_variance', False)]
```

Sometimes applying BN *before the activation function works better* (there's a debate on this topic). Moreover, the layer before a `BatchNormalization` layer does not need to have bias terms, since the `BatchNormalization` layer does as well, it would be a waste of parameters, so you can set `use_bias=False` when creating those layers:

```
In [42]: # extra code - clear the name counters and set the random seed
tf.keras.backend.clear_session()
tf.random.set_seed(42)
```

```
In [43]: model = tf.keras.Sequential([
    tf.keras.layers.Input(shape=[28, 28]),
    tf.keras.layers.Flatten(),

    tf.keras.layers.Dense(300, kernel_initializer="he_normal", use_bias=False),
    tf.keras.layers.BatchNormalization(), # BN before activation
    tf.keras.layers.Activation("relu"),

    tf.keras.layers.Dense(100, kernel_initializer="he_normal", use_bias=False),
    tf.keras.layers.BatchNormalization(), # BN before activation
    tf.keras.layers.Activation("relu"),

    tf.keras.layers.Dense(10, activation="softmax")
])
```

```
In [44]: # extra code - just show that the model works! 😊
model.compile(loss="sparse_categorical_crossentropy", optimizer="sgd",
              metrics=["accuracy"])
model.fit(X_train, y_train, epochs=2, validation_data=(X_valid, y_valid))
```

```
Epoch 1/2
1719/1719 ————— 12s 7ms/step - accuracy: 0.7265 - loss: 0.8343 - val_accuracy: 0.8462 - val_loss: 0.4342
Epoch 2/2
1719/1719 ————— 11s 6ms/step - accuracy: 0.8478 - loss: 0.4365 - val_accuracy: 0.8602 - val_loss: 0.3890
```

```
Out[44]: <keras.src.callbacks.history.History at 0x7e4213aae0f0>
```

Gradient Clipping

To prevent exploding gradients during training, we can use gradient clipping. This technique limits the maximum value of gradients, ensuring stable updates to the model weights.

All `tf.keras.optimizers` accept `clipnorm` or `clipvalue` arguments:

```
In [45]: optimizer = tf.keras.optimizers.SGD(clipvalue=1.0)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer)
```

`clipnorm` (concise explanation)

- What it does

Clips the global ℓ_2 norm of the gradient vector for an update. If the total gradient norm $\|g\|_2$ exceeds the threshold c , all gradients are rescaled by $c/\|g\|_2$:

$$g \leftarrow \begin{cases} g, & \text{if } \|g\|_2 \leq c, \\ g \cdot \frac{c}{\|g\|_2}, & \text{otherwise.} \end{cases}$$

- How Keras applies it

`tf.keras` uses a global-norm clip (`tf.clip_by_global_norm`) on the gradients computed for the current batch — i.e., it scales the whole gradient tuple together (based on $\|g\|_2$).

- Difference vs. `clipvalue`

- `clipnorm`: rescales the entire gradient vector when its ℓ_2 norm is too large (preserves direction).
- `clipvalue`: clips each gradient component element-wise to the range $[-c, c]$ (may change direction).

- When to use

- Prevent exploding gradients (RNNs, very deep nets, unstable training).
- Use when you observe very large gradient norms or unstable loss/weight updates.

- Typical settings and tips

- Common starting value: `clipnorm ≈ 1.0` (tune for your model).

- Works with any optimizer (SGD, Adam, etc.).
- Does not fix vanishing gradients — only bounds explosion.
- Monitor gradient norms with e.g. `tf.linalg.global_norm` to choose an appropriate threshold.
- Keras example
 - `optimizer = tf.keras.optimizers.SGD(clipnorm=1.0)`

```
In [46]: optimizer = tf.keras.optimizers.SGD(clipnorm=1.0)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer)
```

Reusing Pretrained Layers

Reusing a Keras model

Let's split the fashion MNIST training set in two:

- `X_train_A` : all images of all items except for T-shirts/tops and pullovers (classes 0 and 2).
- `X_train_B` : a much smaller training set of just the first 200 images of T-shirts/tops and pullovers.

The validation set and the test set are also split this way, but without restricting the number of images.

We will train a model on set A (classification task with 8 classes), and try to reuse it to tackle set B (binary classification). We hope to transfer a little bit of knowledge from task A to task B, since classes in set A (trousers, dresses, coats, sandals, shirts, sneakers, bags, and ankle boots) are somewhat similar to classes in set B (T-shirts/tops and pullovers). However, since we are using `Dense` layers, only patterns that occur at the same location can be reused (in contrast, convolutional layers will transfer much better, since learned patterns can be detected anywhere on the image, as we will see in the chapter 14).

```
In [47]: # extra code - split Fashion MNIST into tasks A and B, then train and save
#           model A to "my_model_A".

pos_class_id = class_names.index("Pullover")
neg_class_id = class_names.index("T-shirt/top")

def split_dataset(X, y):
    """Split a Fashion-MNIST dataset into two tasks A and B.

    Task A contains all samples whose label is not Pullover or
    T-shirt/top; the labels are remapped to 0-7 in the order of the
    remaining classes.
    Task B contains only the Pullover vs. T-shirt/top examples; its
    labels are binary (1 for Pullover, 0 for T-shirt/top).

    Args:
        X (np.ndarray): array of input images, shape `(n_samples, ...)`.
```

```

y (np.ndarray): array of integer labels in `0..9`, shape
                `(n_samples,)`.

Returns:
    ((X_A, y_A), (X_B, y_B)):
        * `X_A`, `y_A` - data and 8-class labels for task A.
        * `X_B`, `y_B` - data and binary labels for task B.
    """
    y_for_B = (y == pos_class_id) | (y == neg_class_id)
    y_A = y[~y_for_B]
    y_B = (y[y_for_B] == pos_class_id).astype(np.float32)
    old_class_ids = list(set(range(10)) - set([neg_class_id, pos_class_id]))
    for old_class_id, new_class_id in zip(old_class_ids, range(8)):
        y_A[y_A == old_class_id] = new_class_id # reorder class ids for A
    X_A = X[~y_for_B]
    X_B = X[y_for_B]
    return ((X_A, y_A), (X_B, y_B))

(X_train_A, y_train_A), (X_train_B, y_train_B) = split_dataset(X_train, y_train)
(X_valid_A, y_valid_A), (X_valid_B, y_valid_B) = split_dataset(X_valid, y_valid)
(X_test_A, y_test_A), (X_test_B, y_test_B) = split_dataset(X_test, y_test)
X_train_B = X_train_B[:200]
y_train_B = y_train_B[:200]

```

```

In [48]: tf.keras.backend.clear_session()
         tf.random.set_seed(42)

model_A = tf.keras.Sequential([
    tf.keras.layers.Input(shape=[28, 28]),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dense(8, activation="softmax")
])

model_A.compile(loss="sparse_categorical_crossentropy",
                optimizer=tf.keras.optimizers.SGD(learning_rate=0.001),
                metrics=["accuracy"])
history = model_A.fit(X_train_A, y_train_A, epochs=20,
                     validation_data=(X_valid_A, y_valid_A))
model_A.save("my_model_A.keras")

```

Epoch 1/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.4816 - loss: 1.6029 - val_accuracy: 0.7829 - val_loss: 0.7436

Epoch 2/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.7978 - loss: 0.6782 - val_accuracy: 0.8433 - val_loss: 0.5259

Epoch 3/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.8441 - loss: 0.5082 - val_accuracy: 0.8609 - val_loss: 0.4423

Epoch 4/20
1376/1376 ————— 10s 4ms/step - accuracy: 0.8627 - loss: 0.4339 - val_accuracy: 0.8739 - val_loss: 0.3977

Epoch 5/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.8732 - loss: 0.3915 - val_accuracy: 0.8807 - val_loss: 0.3699

Epoch 6/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.8802 - loss: 0.3637 - val_accuracy: 0.8832 - val_loss: 0.3505

Epoch 7/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.8861 - loss: 0.3436 - val_accuracy: 0.8867 - val_loss: 0.3360

Epoch 8/20
1376/1376 ————— 6s 5ms/step - accuracy: 0.8898 - loss: 0.3284 - val_accuracy: 0.8907 - val_loss: 0.3249

Epoch 9/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.8933 - loss: 0.3162 - val_accuracy: 0.8930 - val_loss: 0.3157

Epoch 10/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.8959 - loss: 0.3060 - val_accuracy: 0.8952 - val_loss: 0.3081

Epoch 11/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.8989 - loss: 0.2972 - val_accuracy: 0.8957 - val_loss: 0.3015

Epoch 12/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.9005 - loss: 0.2896 - val_accuracy: 0.8982 - val_loss: 0.2959

Epoch 13/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.9022 - loss: 0.2830 - val_accuracy: 0.8997 - val_loss: 0.2910

Epoch 14/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.9042 - loss: 0.2771 - val_accuracy: 0.9012 - val_loss: 0.2867

Epoch 15/20
1376/1376 ————— 6s 5ms/step - accuracy: 0.9058 - loss: 0.2719 - val_accuracy: 0.9022 - val_loss: 0.2826

Epoch 16/20
1376/1376 ————— 9s 4ms/step - accuracy: 0.9079 - loss: 0.2671 - val_accuracy: 0.9035 - val_loss: 0.2790

Epoch 17/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.9092 - loss: 0.2627 - val_accuracy: 0.9047 - val_loss: 0.2757

Epoch 18/20
1376/1376 ————— 6s 4ms/step - accuracy: 0.9105 - loss: 0.2586 - val_accuracy: 0.9062 - val_loss: 0.2726

Epoch 19/20
1376/1376 ————— 10s 4ms/step - accuracy: 0.9119 - loss: 0.2549 - val_accuracy: 0.9070 - val_loss: 0.2698

Epoch 20/20
1376/1376 ————— 7s 5ms/step - accuracy: 0.9132 - loss: 0.2514 - val_accuracy: 0.9082 - val_loss: 0.2672

In [49]: *# extra code - train and evaluate model B, without reusing model A*

```
tf.random.set_seed(42)
model_B = tf.keras.Sequential([
    tf.keras.layers.Input(shape=[28, 28]),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dense(1, activation="sigmoid")
])

model_B.compile(loss="binary_crossentropy",
                optimizer=tf.keras.optimizers.SGD(learning_rate=0.001),
                metrics=["accuracy"])
history = model_B.fit(X_train_B, y_train_B, epochs=20,
                     validation_data=(X_valid_B, y_valid_B))
model_B.evaluate(X_test_B, y_test_B)
```

Epoch 1/20
7/7  1s 68ms/step - accuracy: 0.4649 - loss: 0.7090 - val_accuracy: 0.5193 - val_loss: 0.6892

Epoch 2/20
7/7  0s 29ms/step - accuracy: 0.5302 - loss: 0.6899 - val_accuracy: 0.5895 - val_loss: 0.6743

Epoch 3/20
7/7  0s 32ms/step - accuracy: 0.6078 - loss: 0.6729 - val_accuracy: 0.6726 - val_loss: 0.6604

Epoch 4/20
7/7  0s 30ms/step - accuracy: 0.7206 - loss: 0.6569 - val_accuracy: 0.7399 - val_loss: 0.6474

Epoch 5/20
7/7  0s 30ms/step - accuracy: 0.7864 - loss: 0.6417 - val_accuracy: 0.8091 - val_loss: 0.6351

Epoch 6/20
7/7  0s 30ms/step - accuracy: 0.8179 - loss: 0.6277 - val_accuracy: 0.8516 - val_loss: 0.6235

Epoch 7/20
7/7  0s 28ms/step - accuracy: 0.8552 - loss: 0.6147 - val_accuracy: 0.8754 - val_loss: 0.6123

Epoch 8/20
7/7  0s 34ms/step - accuracy: 0.8857 - loss: 0.6021 - val_accuracy: 0.8892 - val_loss: 0.6013

Epoch 9/20
7/7  0s 31ms/step - accuracy: 0.9159 - loss: 0.5898 - val_accuracy: 0.8991 - val_loss: 0.5904

Epoch 10/20
7/7  0s 32ms/step - accuracy: 0.9108 - loss: 0.5778 - val_accuracy: 0.9090 - val_loss: 0.5797

Epoch 11/20
7/7  0s 30ms/step - accuracy: 0.9409 - loss: 0.5660 - val_accuracy: 0.9149 - val_loss: 0.5691

Epoch 12/20
7/7  0s 33ms/step - accuracy: 0.9517 - loss: 0.5544 - val_accuracy: 0.9179 - val_loss: 0.5587

Epoch 13/20
7/7  0s 35ms/step - accuracy: 0.9517 - loss: 0.5431 - val_accuracy: 0.9219 - val_loss: 0.5486

Epoch 14/20
7/7  0s 32ms/step - accuracy: 0.9517 - loss: 0.5321 - val_accuracy: 0.9268 - val_loss: 0.5385

Epoch 15/20
7/7  0s 30ms/step - accuracy: 0.9625 - loss: 0.5213 - val_accuracy: 0.9288 - val_loss: 0.5286

Epoch 16/20
7/7  0s 32ms/step - accuracy: 0.9625 - loss: 0.5108 - val_accuracy: 0.9327 - val_loss: 0.5188

Epoch 17/20
7/7  0s 33ms/step - accuracy: 0.9652 - loss: 0.5004 - val_accuracy: 0.9347 - val_loss: 0.5091

Epoch 18/20
7/7  0s 30ms/step - accuracy: 0.9652 - loss: 0.4903 - val_accuracy: 0.9347 - val_loss: 0.4995

Epoch 19/20
7/7  0s 29ms/step - accuracy: 0.9652 - loss: 0.4802 - val_accuracy: 0.9347 - val_loss: 0.4900

Epoch 20/20
7/7  0s 30ms/step - accuracy: 0.9679 - loss: 0.4703 - val_accuracy: 0.9679 - val_loss: 0.4703

uracy: 0.9347 - val_loss: 0.4807

63/63 ————— 0s 4ms/step - accuracy: 0.9255 - loss: 0.4884

Out[49]: [0.48864567279815674, 0.9225000143051147]

Model B reaches 91.85% accuracy on the test set. Now let's try reusing the pretrained model A.

```
In [50]: # reuse model A to create model B_on_A, then train and evaluate it
model_A = tf.keras.models.load_model("my_model_A.keras")
model_B_on_A = tf.keras.Sequential(model_A.layers[:-1])

# the last layer of model A is a Dense layer with 8 outputs, but we need only 1
model_B_on_A.add(tf.keras.layers.Dense(1, activation="sigmoid"))
```

Note that `model_B_on_A` and `model_A` actually share layers now, so when we train one, it will update both models. If we want to avoid that, we need to build `model_B_on_A` on top of a *clone* of `model_A`:

```
In [51]: tf.random.set_seed(42) # extra code - ensure reproducibility
```

The cell that follows will re-seed the RNG and then make a **clone** of `model_A` before re-using its layers. Cloning and copying the weights guarantees that the layers used to build `model_B_on_A` are independent objects, so training the new model won't inadvertently update the original `model_A`.

reset the random seed for reproducibility, clone `model_A`, copy its weights, then slice off its last layer and add a new single-output Dense layer for task B.

```
In [52]: model_A_clone = tf.keras.models.clone_model(model_A)
model_A_clone.set_weights(model_A.get_weights())
```

```
In [53]: # extra code - creating model_B_on_A just like in the previous cell
model_B_on_A = tf.keras.Sequential(model_A_clone.layers[:-1])
model_B_on_A.add(tf.keras.layers.Dense(1, activation="sigmoid"))
```

```
In [54]: # freeze all layers of model_B_on_A except the last one (the new Dense layer we
for layer in model_B_on_A.layers[:-1]:
    layer.trainable = False

optimizer = tf.keras.optimizers.SGD(learning_rate=0.001)
model_B_on_A.compile(loss="binary_crossentropy", optimizer=optimizer,
                    metrics=["accuracy"])
history = model_B_on_A.fit(X_train_B, y_train_B, epochs=4,
                        validation_data=(X_valid_B, y_valid_B))
```

Epoch 1/4

7/7 ————— 1s 66ms/step - accuracy: 0.5640 - loss: 1.1100 - val_accuracy: 0.5618 - val_loss: 0.7083

Epoch 2/4

7/7 ————— 0s 28ms/step - accuracy: 0.6460 - loss: 0.5768 - val_accuracy: 0.7646 - val_loss: 0.5206

Epoch 3/4

7/7 ————— 0s 29ms/step - accuracy: 0.8239 - loss: 0.4680 - val_accuracy: 0.8051 - val_loss: 0.4992

Epoch 4/4

7/7 ————— 0s 44ms/step - accuracy: 0.8513 - loss: 0.4609 - val_accuracy: 0.8140 - val_loss: 0.4943

```
In [55]: # unfreeze all layers of model_B_on_A and continue training for a few more epoch
        for layer in model_B_on_A.layers[:-1]:
            layer.trainable = True

        optimizer = tf.keras.optimizers.SGD(learning_rate=0.001)
        model_B_on_A.compile(loss="binary_crossentropy", optimizer=optimizer,
                             metrics=["accuracy"])
        history = model_B_on_A.fit(X_train_B, y_train_B, epochs=16,
                                   validation_data=(X_valid_B, y_valid_B))
```

Epoch 1/16
 7/7 ————— 2s 119ms/step - accuracy: 0.8526 - loss: 0.4524 - val_accuracy: 0.8279 - val_loss: 0.4721
 Epoch 2/16
 7/7 ————— 0s 49ms/step - accuracy: 0.8747 - loss: 0.4268 - val_accuracy: 0.8487 - val_loss: 0.4397
 Epoch 3/16
 7/7 ————— 1s 33ms/step - accuracy: 0.9082 - loss: 0.3937 - val_accuracy: 0.8645 - val_loss: 0.4121
 Epoch 4/16
 7/7 ————— 0s 32ms/step - accuracy: 0.9139 - loss: 0.3658 - val_accuracy: 0.8783 - val_loss: 0.3889
 Epoch 5/16
 7/7 ————— 0s 31ms/step - accuracy: 0.9139 - loss: 0.3424 - val_accuracy: 0.8912 - val_loss: 0.3694
 Epoch 6/16
 7/7 ————— 0s 31ms/step - accuracy: 0.9170 - loss: 0.3226 - val_accuracy: 0.9021 - val_loss: 0.3525
 Epoch 7/16
 7/7 ————— 0s 29ms/step - accuracy: 0.9220 - loss: 0.3053 - val_accuracy: 0.9090 - val_loss: 0.3379
 Epoch 8/16
 7/7 ————— 0s 32ms/step - accuracy: 0.9328 - loss: 0.2904 - val_accuracy: 0.9169 - val_loss: 0.3251
 Epoch 9/16
 7/7 ————— 0s 33ms/step - accuracy: 0.9377 - loss: 0.2773 - val_accuracy: 0.9209 - val_loss: 0.3140
 Epoch 10/16
 7/7 ————— 0s 30ms/step - accuracy: 0.9377 - loss: 0.2658 - val_accuracy: 0.9219 - val_loss: 0.3042
 Epoch 11/16
 7/7 ————— 0s 29ms/step - accuracy: 0.9446 - loss: 0.2557 - val_accuracy: 0.9228 - val_loss: 0.2954
 Epoch 12/16
 7/7 ————— 0s 30ms/step - accuracy: 0.9446 - loss: 0.2467 - val_accuracy: 0.9248 - val_loss: 0.2877
 Epoch 13/16
 7/7 ————— 0s 30ms/step - accuracy: 0.9516 - loss: 0.2386 - val_accuracy: 0.9278 - val_loss: 0.2807
 Epoch 14/16
 7/7 ————— 0s 32ms/step - accuracy: 0.9624 - loss: 0.2312 - val_accuracy: 0.9298 - val_loss: 0.2743
 Epoch 15/16
 7/7 ————— 0s 29ms/step - accuracy: 0.9624 - loss: 0.2246 - val_accuracy: 0.9318 - val_loss: 0.2685
 Epoch 16/16
 7/7 ————— 0s 31ms/step - accuracy: 0.9624 - loss: 0.2185 - val_accuracy: 0.9337 - val_loss: 0.2632

So, what's the final verdict?

```
In [56]: model_B_on_A.evaluate(X_test_B, y_test_B)
```

63/63 ————— 0s 4ms/step - accuracy: 0.9247 - loss: 0.2592

```
Out[56]: [0.27674776315689087, 0.9204999804496765]
```

Great! We got a bit of transfer: the model's accuracy went up 2 percentage points, from 91.85% to 93.85%. This means the error rate dropped by almost 25%:

```
In [57]: 1 - (100 - 93.85) / (100 - 91.85)
```

```
Out[57]: 0.24539877300613477
```

Conclusion

In this little experiment we split Fashion-MNIST into two related tasks and tried to transfer some knowledge from the first to the second:

- task A – an 8-class problem made of all items **except** T-shirts and pullovers,
- task B – a binary Pullover vs. T-shirt/top classification with only 200 training examples.

We trained a small fully-connected network from scratch on task A, saved it, and then trained another network from scratch on task B. The “pure” model B reached about **91.8%** test accuracy.

Re-using the pretrained model A for task B (cloning it, slicing off the last layer, adding a new sigmoid output, freezing the old layers and then fine-tuning) gave a clear benefit: after the usual freeze-then-unfreeze schedule the transferred model attained **≈ 93.8%** accuracy. That’s a 2-point gain, i.e. roughly a 25% reduction in error rate, even though only dense layers were used (so only patterns at the exact same pixel locations could be shared – convolutional nets would transfer even better).

This illustrates the basic recipe for transfer learning with Keras: split the data, train a “base” model on a large related task, clone or load it, replace the final head, optionally freeze its layers, train on the new task, then unfreeze and fine-tune. Even when the secondary dataset is small, reusing pretrained layers can speed convergence and improve generalization.

(And of course, for vision problems you’ll usually transfer convolutional layers rather than dense ones, which allows learned features to be detected anywhere in the image.)

Faster Optimizers

Faster optimizers aim to converge in fewer epochs (and with less tuning) while still generalizing well. This section compares popular algorithms, summarizes their trade-offs, and gives a practical workflow for picking and tuning an optimizer.

Key algorithms (very briefly)

- SGD + momentum / Nesterov — simple, robust; often yields the best final generalization but can be slower to converge. Typical: momentum=0.9.

- RMSProp — adaptive per-parameter steps; good for noisy/ill-conditioned problems.
- Adam / AdamW — fast, well-behaved defaults for prototyping; AdamW decouples weight decay (preferred for regularization).
- Nadam / AMSGrad — small variants that can help stability in some cases.

Practical recipe

- Prototype with Adam ($\text{lr} \approx 1\text{e-}3$) to find a reasonable model and learning-rate range.
- For best final results, switch to SGD + momentum (momentum ≈ 0.9) and use a tuned LR schedule (step decay, cosine annealing, or cyclic).
- Use AdamW (not plain Adam with L2) when using decoupled weight decay.
- For large-batch training use learning-rate warmup; scale LR proportionally to batch size.
- Clip gradients (clipnorm ≈ 1.0) if you see unstable/very large updates.
- Always perform a small LR sweep (log scale) and run multiple seeds for robustness.

Typical starting hyperparameters

- SGD: lr 0.01–0.1, momentum 0.9
- Adam: lr $1\text{e-}3$, $\text{beta1}=0.9$, $\text{beta2}=0.999$, $\text{eps}=1\text{e-}7$ – $1\text{e-}8$
- AdamW: same as Adam, $\text{weight_decay} \approx 1\text{e-}2$
- RMSProp: lr $1\text{e-}3$, rho 0.9

What you will see next

- Automated comparisons (same architecture, same seed) to evaluate convergence speed and final validation accuracy on Fashion MNIST.
- Suggested experiments: LR sweeps, train longer with SGD+momentum, and compare learning curves and wall-clock time.

Here are the utility functions that will be used to build and train a deep neural network on the Fashion MNIST dataset, facilitating the comparison of different optimization algorithms.

```
In [11]: # extra code - a little function to test an optimizer on Fashion MNIST

def build_model(seed=42):
    """Builds a simple neural network model for Fashion MNIST.

    Args:
        seed (int, optional): Random seed for reproducibility. Defaults to 42.

    Returns:
        tf.keras.Sequential: A compiled Keras Sequential model with three dense
        """
    tf.random.set_seed(seed)
    return tf.keras.Sequential([
        tf.keras.layers.Input(shape=[28, 28]),
```

```

tf.keras.layers.Flatten(),
tf.keras.layers.Dense(100, activation="relu",
                        kernel_initializer="he_normal"),
tf.keras.layers.Dense(100, activation="relu",
                        kernel_initializer="he_normal"),
tf.keras.layers.Dense(100, activation="relu",
                        kernel_initializer="he_normal"),
tf.keras.layers.Dense(10, activation="softmax")
])

def build_and_train_model(optimizer):
    """Builds, compiles, and trains a model using the given optimizer.

    Args:
        optimizer: A TensorFlow optimizer instance to use for training.

    Returns:
        A History object containing the training history.
    """
    model = build_model()
    model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
                  metrics=["accuracy"])
    return model.fit(X_train, y_train, epochs=10,
                     validation_data=(X_valid, y_valid))

```

First, let's use SGD optimizer to train a deep neural network on the Fashion MNIST dataset as a baseline:

```
In [59]: optimizer = tf.keras.optimizers.SGD(learning_rate=0.001)
```

```
In [60]: history_sgd = build_and_train_model(optimizer) # extra code
```


Epoch 1/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.4694 - loss: 1.7054 - val_accuracy: 0.7316 - val_loss: 0.8711
Epoch 2/10
1719/1719 ————— 7s 4ms/step - accuracy: 0.7306 - loss: 0.8246 - val_accuracy: 0.7808 - val_loss: 0.6816
Epoch 3/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.7759 - loss: 0.6749 - val_accuracy: 0.7970 - val_loss: 0.6047
Epoch 4/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.7961 - loss: 0.6064 - val_accuracy: 0.8096 - val_loss: 0.5611
Epoch 5/10
1719/1719 ————— 7s 4ms/step - accuracy: 0.8095 - loss: 0.5651 - val_accuracy: 0.8176 - val_loss: 0.5325
Epoch 6/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8184 - loss: 0.5368 - val_accuracy: 0.8228 - val_loss: 0.5120
Epoch 7/10
1719/1719 ————— 7s 4ms/step - accuracy: 0.8254 - loss: 0.5160 - val_accuracy: 0.8258 - val_loss: 0.4967
Epoch 8/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8300 - loss: 0.4997 - val_accuracy: 0.8292 - val_loss: 0.4842
Epoch 9/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8343 - loss: 0.4864 - val_accuracy: 0.8310 - val_loss: 0.4742
Epoch 10/10
1719/1719 ————— 7s 4ms/step - accuracy: 0.8379 - loss: 0.4753 - val_accuracy: 0.8336 - val_loss: 0.4659

Momentum Optimization

Core Concept (What it does)

Momentum optimization is designed to accelerate Gradient Descent by simulating the physical behavior of a ball rolling down a hill. While standard Gradient Descent takes small, uniform steps based only on the local gradient, Momentum accumulates a "velocity" from previous steps. This allows the optimizer to gain speed in directions where gradients consistently point toward the minimum and dampen oscillations in "ravines" (narrow valleys where gradients point sharply toward the center rather than down the length). The result is significantly faster convergence and a better ability to escape shallow local minima.

Update Rule

The algorithm maintains a velocity vector $\mathbf{v}$ that is updated at each step, and this velocity is then used to update the model parameters θ :

1. Update Velocity:

$$\mathbf{v} \leftarrow \beta \mathbf{v} + \eta \nabla_{\theta} \mathcal{L}(\theta)$$

2. Update Parameters:

$$\theta \leftarrow \theta - \mathbf{v}$$

Notation:

- $\nabla_{\theta}\mathcal{L}(\theta)$: The gradient of the loss function at the current position.
- $\mathbf{v}$: The velocity (momentum) vector, representing the accumulated history of updates.
- β : The momentum coefficient (hyperparameter), typically set to 0.9. It acts as a fraction term, determining how much of the previous velocity is retained.
- η : The learning rate.
- θ : The model parameters being updated.

Practical Settings and Tips

• Common Hyperparameter Values:

- Momentum Coefficient (β): Usually set to 0.9 (retaining 90% of the previous velocity).
- Learning Rate (η): Generally requires careful tuning alongside β .

- **The "Overshooting" Risk:** If β is too close to 1.0 (e.g., 0.999), the optimizer may gain too much speed and overshoot the minimum, leading to prolonged oscillations before finally converging.

- **Implementation:** In Keras, Momentum is implemented as part of the SGD optimizer:

```
optimizer = tf.keras.optimizers.SGD(learning_rate=0.001,
momentum=0.9)
```

where `momentum=0.9` is the default value and it represents the fraction of the previous velocity that is retained in the current update.

- **Generalization:** SGD with Momentum is often the preferred choice for final production models because it tends to find flatter minima that generalize better than adaptive methods like Adam, provided the learning rate schedule is well-tuned.

```
In [61]: optimizer = tf.keras.optimizers.SGD(learning_rate=0.001, momentum=0.9)
```

where `learning_rate` is the step size η for updating the model parameters, and `momentum` is the coefficient β that determines how much of the previous update's velocity to retain. A common choice for momentum is around 0.9, which means that 90% of the previous velocity is retained in the current update.

```
In [62]: history_momentum = build_and_train_model(optimizer) # extra code
```

```

Epoch 1/10
1719/1719 ————— 10s 5ms/step - accuracy: 0.6685 - loss: 0.9823 - val_accuracy: 0.8150 - val_loss: 0.5061
Epoch 2/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8329 - loss: 0.4779 - val_accuracy: 0.8296 - val_loss: 0.4544
Epoch 3/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8514 - loss: 0.4241 - val_accuracy: 0.8458 - val_loss: 0.4163
Epoch 4/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8618 - loss: 0.3944 - val_accuracy: 0.8534 - val_loss: 0.3970
Epoch 5/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8675 - loss: 0.3738 - val_accuracy: 0.8558 - val_loss: 0.3897
Epoch 6/10
1719/1719 ————— 8s 4ms/step - accuracy: 0.8726 - loss: 0.3578 - val_accuracy: 0.8584 - val_loss: 0.3797
Epoch 7/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8760 - loss: 0.3448 - val_accuracy: 0.8588 - val_loss: 0.3763
Epoch 8/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8802 - loss: 0.3335 - val_accuracy: 0.8584 - val_loss: 0.3744
Epoch 9/10
1719/1719 ————— 8s 4ms/step - accuracy: 0.8828 - loss: 0.3236 - val_accuracy: 0.8578 - val_loss: 0.3710
Epoch 10/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8865 - loss: 0.3148 - val_accuracy: 0.8586 - val_loss: 0.3718

```

Nesterov Accelerated Gradient (NAG)

Core Concept (What it does)

Nesterov Accelerated Gradient is a powerful variant of **Momentum optimization** that "looks ahead" before making an update. While standard Momentum computes the gradient at the current position, NAG computes the gradient at an anticipated future position—slightly further in the direction of the momentum. This "look-ahead" mechanism allows the optimizer to adjust its course more intelligently, resulting in significantly faster convergence and reduced overshooting in many optimization landscapes.

Update Rule

The parameter update in NAG involves calculating the gradient at the predicted future position of the parameters ($\theta + \beta \mathbf{v}$):

1. **Compute Velocity (with Look-ahead Gradient):**

$$\mathbf{v} \leftarrow \beta \mathbf{v} - \eta \nabla_{\theta} \mathcal{L}(\theta + \beta \mathbf{v})$$

2. **Update Parameters:**

$$\theta \leftarrow \theta + \mathbf{v}$$

Notation:

- $\mathcal{L}(\theta + \beta \mathbf{v})$: The loss function evaluated at the "look-ahead" position.
- $\mathbf{v}$: The velocity (momentum) vector.
- β : The momentum coefficient (hyperparameter), typically set to 0.9.
- η : The learning rate.
- θ : The model parameters.

Practical Settings and Tips

- **Common Hyperparameter Values:**
 - Momentum Coefficient (β): Usually 0.9.
 - Learning Rate (η): Depends on the task, but often paired with a learning rate schedule.
- **Performance Advantage:** NAG almost always outperforms standard momentum optimization, especially in convex problems. It effectively "slows down" the optimizer as it approaches a minimum, preventing the heavy oscillations that often plague standard momentum.
- **Implementation:** In Keras, you can enable NAG by setting `nesterov=True` within the SGD optimizer:

```
optimizer = tf.keras.optimizers.SGD(learning_rate=0.001,  
momentum=0.9, nesterov=True)
```
- **Generalization:** Like SGD with Momentum, NAG is often favored for its strong final generalization performance when the learning rate is properly tuned.

```
In [63]: optimizer = tf.keras.optimizers.SGD(learning_rate=0.001, momentum=0.9,  
                                             nesterov=True) # use Nesterov Accelerated Gr
```

```
In [64]: history_nesterov = build_and_train_model(optimizer) # extra code
```

```

Epoch 1/10
1719/1719 ————— 10s 5ms/step - accuracy: 0.6698 - loss: 0.9871 - val_accuracy: 0.8280 - val_loss: 0.4821
Epoch 2/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8361 - loss: 0.4715 - val_accuracy: 0.8402 - val_loss: 0.4394
Epoch 3/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8520 - loss: 0.4234 - val_accuracy: 0.8464 - val_loss: 0.4179
Epoch 4/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8610 - loss: 0.3965 - val_accuracy: 0.8488 - val_loss: 0.4057
Epoch 5/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8671 - loss: 0.3774 - val_accuracy: 0.8542 - val_loss: 0.3949
Epoch 6/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8722 - loss: 0.3622 - val_accuracy: 0.8570 - val_loss: 0.3877
Epoch 7/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8756 - loss: 0.3493 - val_accuracy: 0.8594 - val_loss: 0.3797
Epoch 8/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8790 - loss: 0.3381 - val_accuracy: 0.8588 - val_loss: 0.3763
Epoch 9/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8826 - loss: 0.3280 - val_accuracy: 0.8598 - val_loss: 0.3705
Epoch 10/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8852 - loss: 0.3189 - val_accuracy: 0.8600 - val_loss: 0.3666

```

AdaGrad Optimization

Core Concept (What it does)

AdaGrad (Adaptive Gradient Algorithm) is an optimizer that automatically scales the learning rate for each parameter individually based on the history of gradients. It achieves this by tracking the sum of the squares of all past gradients. Parameters that receive large and frequent updates have their learning rates scaled down more aggressively, while parameters that receive small or infrequent updates maintain a higher learning rate. This makes AdaGrad particularly effective for handling **sparse data** (where most features are zero most of the time), as it ensures that rarely occurring but informative features still receive significant updates.

Update Rule

AdaGrad maintains an accumulator $\mathbf{s}_t$ that stores the sum of the squares of the gradients up to time step t :

1. Accumulate Squared Gradients:

$$\mathbf{s}_t = \mathbf{s}_{t-1} + \mathbf{g}_t^2$$

2. Update Parameters:

$$\theta \leftarrow \theta - \frac{\eta}{\sqrt{\mathbf{s}_t} + \epsilon} \mathbf{g}_t$$

Notation:

- $\mathbf{g}_t$: The gradient at time step t .
- $\mathbf{s}_t$: The accumulated sum of squared gradients (acts as a per-parameter learning rate scaler).
- η : The initial learning rate.
- ϵ : A small constant (e.g., $1e^{-7}$) to prevent division by zero.
- θ : The model parameters being updated.
- as a result, the effective learning rate for each parameter is $\eta/(\sqrt{\mathbf{s}_t} + \epsilon)$, which decreases over time for parameters with large accumulated gradients.

Practical Settings and Tips

- **Common Hyperparameter Values:**
 - Initial Learning Rate (η): Often set to 0.01, though it requires tuning for specific problems.
 - Numerical Stability Constant (ϵ): Typically $1e^{-7}$.
- **Strength in Sparsity:** AdaGrad excels in tasks with high-dimensional sparse inputs, such as **Natural Language Processing (NLP)** or recommendation systems, where some features are much rarer than others.
- **The "Vanishing Learning Rate" Limitation:** Because $\mathbf{s}_t$ accumulates squares (which are always positive), the denominator $\sqrt{\mathbf{s}_t}$ grows monotonically. In deep neural networks, this often causes the learning rate to decrease too quickly, effectively stopping the model from learning before it reaches the global optimum.
- **Successors:** Due to the vanishing learning rate issue, **RMSProp** and **Adam** are generally preferred for training deep neural networks, as they use a moving average to prevent the accumulator from growing indefinitely.

```
In [65]: optimizer = tf.keras.optimizers.Adagrad(learning_rate=0.001)
```

```
In [66]: history_adagrad = build_and_train_model(optimizer) # extra code
```

```

Epoch 1/10
1719/1719 ————— 10s 5ms/step - accuracy: 0.5587 - loss: 1.3803 - val_accuracy: 0.7778 - val_loss: 0.6837
Epoch 2/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.7803 - loss: 0.6681 - val_accuracy: 0.8046 - val_loss: 0.5854
Epoch 3/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8076 - loss: 0.5858 - val_accuracy: 0.8180 - val_loss: 0.5421
Epoch 4/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8191 - loss: 0.5453 - val_accuracy: 0.8260 - val_loss: 0.5162
Epoch 5/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8263 - loss: 0.5196 - val_accuracy: 0.8314 - val_loss: 0.4984
Epoch 6/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8313 - loss: 0.5013 - val_accuracy: 0.8350 - val_loss: 0.4851
Epoch 7/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8353 - loss: 0.4876 - val_accuracy: 0.8376 - val_loss: 0.4748
Epoch 8/10
1719/1719 ————— 8s 5ms/step - accuracy: 0.8382 - loss: 0.4766 - val_accuracy: 0.8402 - val_loss: 0.4664
Epoch 9/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8414 - loss: 0.4675 - val_accuracy: 0.8410 - val_loss: 0.4593
Epoch 10/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8434 - loss: 0.4597 - val_accuracy: 0.8430 - val_loss: 0.4531

```

RMSProp Optimization

Core Concept (What it does)

RMSProp (Root Mean Square Propagation) is an adaptive learning rate optimization algorithm designed to address the aggressive, monotonically decreasing learning rate of the AdaGrad algorithm. It achieves this by maintaining a moving average of squared gradients for each parameter. By dividing the gradient by the root mean square of these recent magnitudes, RMSProp scales the step size for each dimension, effectively reducing oscillations in "ravine-like" regions of the loss surface and handling non-stationary objectives efficiently.

Update Rule

RMSProp updates the model parameters by first calculating an exponentially weighted moving average of the squared gradients:

1. **Compute Accumulated Squared Gradient (Moving Average):**

$$\mathbf{s}_t = \rho \mathbf{s}_{t-1} + (1 - \rho) \mathbf{g}_t^2$$

2. **Update Parameters:**

$$\theta \leftarrow \theta - \frac{\eta}{\sqrt{\mathbf{s}_t} + \epsilon} \mathbf{g}_t$$

Notation:

- $\mathbf{g}_t$: The gradient at time step t .
- $\mathbf{s}_t$: The exponentially weighted moving average of squared gradients (the accumulated "history").
- ρ : The decay rate (hyperparameter), typically set around 0.9, which determines how much weight is given to recent vs. past gradients.
- η : The initial learning rate.
- ϵ : A small constant (e.g., $1e^{-7}$) added to prevent division by zero and maintain numerical stability.
- θ : The model parameters being updated.

Practical Settings and Tips

- **Common Hyperparameter Values:**
 - Decay Rate (ρ): Usually 0.9.
 - Learning Rate (η): Often starts at 0.001.
 - Numerical Stability Constant (ϵ): Typically $1e^{-7}$.
- **Best Use Cases:** RMSProp is known to perform exceptionally well on **Recurrent Neural Networks (RNNs)** and is effective for problems with noisy gradients or non-stationary objectives.
- **Comparison with Adam:** While Adam is often the first choice for modern deep learning, RMSProp remains a powerful alternative. Adam essentially adds momentum to the RMSProp approach; if Adam is overshooting or behaving erratically, RMSProp can sometimes provide more stable behavior.
- **Momentum Extension:** Although Keras provides a basic RMSProp, it can be combined with a momentum term for further improvements in convergence speed in some architectures.

```
In [67]: optimizer = tf.keras.optimizers.RMSprop(learning_rate=0.001, rho=0.9)
```

This code initializes the **RMSProp optimizer** with a `learning_rate` of 0.001 and a `rho` (decay rate) of 0.9. RMSProp adaptively adjusts the learning rate for each parameter by maintaining an exponentially weighted moving average of past squared gradients. The `learning_rate` controls the overall step size, while `rho` determines how quickly the optimizer "forgets" past gradients, allowing it to adapt to changing loss landscapes and overcome issues like the vanishing learning rate problem seen in AdaGrad.

```
In [68]: history_rmsprop = build_and_train_model(optimizer) # extra code
```


Epoch 1/10
1719/1719 ————— **10s** 5ms/step - accuracy: 0.7584 - loss: 0.6676 - val_accuracy: 0.8426 - val_loss: 0.4262

Epoch 2/10
1719/1719 ————— **9s** 5ms/step - accuracy: 0.8538 - loss: 0.4045 - val_accuracy: 0.8590 - val_loss: 0.3981

Epoch 3/10
1719/1719 ————— **9s** 5ms/step - accuracy: 0.8700 - loss: 0.3699 - val_accuracy: 0.8492 - val_loss: 0.4206

Epoch 4/10
1719/1719 ————— **9s** 5ms/step - accuracy: 0.8756 - loss: 0.3577 - val_accuracy: 0.8598 - val_loss: 0.4266

Epoch 5/10
1719/1719 ————— **8s** 5ms/step - accuracy: 0.8803 - loss: 0.3494 - val_accuracy: 0.8664 - val_loss: 0.4114

Epoch 6/10
1719/1719 ————— **9s** 5ms/step - accuracy: 0.8822 - loss: 0.3459 - val_accuracy: 0.8616 - val_loss: 0.4311

Epoch 7/10
1719/1719 ————— **9s** 5ms/step - accuracy: 0.8828 - loss: 0.3461 - val_accuracy: 0.8550 - val_loss: 0.4910

Epoch 8/10
1719/1719 ————— **10s** 5ms/step - accuracy: 0.8832 - loss: 0.3407 - val_accuracy: 0.8570 - val_loss: 0.4907

Epoch 9/10
1719/1719 ————— **8s** 5ms/step - accuracy: 0.8853 - loss: 0.3390 - val_accuracy: 0.8528 - val_loss: 0.5574

Epoch 10/10
1719/1719 ————— **10s** 6ms/step - accuracy: 0.8891 - loss: 0.3339 - val_accuracy: 0.8586 - val_loss: 0.6341

Adam Optimization

Core Concept (What it does)

Adam (Adaptive Moment Estimation) is a widely used optimization algorithm in deep learning that combines the strengths of two other popular methods: **Momentum** and **RMSProp**. Its core idea is to adapt the learning rate for each parameter individually by utilizing exponentially weighted averages of both past gradients (first moment) and past squared gradients (second moment). This adaptive approach makes Adam highly effective across a broad range of deep learning problems, often converging quickly and robustly.

Update Rule

Adam's parameter update process involves maintaining and correcting bias for both the first and second moment estimates of the gradients:

1. **Compute First Moment Estimate (Mean of Gradients):**

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t$$

2. **Compute Second Moment Estimate (Mean of Squared Gradients):**

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t^2$$

3. Bias-corrected First Moment:

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}$$

4. Bias-corrected Second Moment:

$$\hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}$$

5. Final Parameter Update:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \hat{\mathbf{m}}_t$$

Notation:

- $\mathbf{g}_t$: The gradient at time step t .
- $\mathbf{m}_t, \mathbf{v}_t$: The first (mean) and second (uncentered variance) moment estimates, respectively.
- β_1, β_2 : The decay rates (hyperparameters) for these moment estimates, controlling how much past information is retained.
- $\hat{\mathbf{m}}_t, \hat{\mathbf{v}}_t$: The bias-corrected moment estimates, which are crucial during the initial warm-up phase of training.
- η : The global learning rate.
- ϵ : A small constant (e.g., $1e^{-8}$) added to the denominator to prevent division by zero and ensure numerical stability.
- $\boldsymbol{\theta}$: The model parameters being updated.

Practical Settings and Tips

- **Common Hyperparameter Values:**
 - Learning Rate (η): Approximately 0.001.
 - First Moment Decay Rate (β_1): Approximately 0.9.
 - Second Moment Decay Rate (β_2): Approximately 0.999.
 - Numerical Stability Constant (ϵ): Approximately $1e^{-8}$.
- **Out-of-the-Box Performance:** Adam often works remarkably well with its default settings and is a solid choice for rapid prototyping due to its fast convergence.
- **Generalization vs. Convergence Speed:** While Adam typically converges quickly, it may sometimes achieve slightly worse generalization performance than fine-tuned SGD with momentum over very long training periods.
- **Weight Decay:** For scenarios requiring strong regularization, consider using **AdamW** instead of standard Adam, as it decouples weight decay from the adaptive learning rate mechanism, leading to more effective regularization.
- **Tuning:** Despite its robustness, careful tuning of hyperparameters (especially the learning rate) can still yield significant performance improvements for specific tasks.

```
In [69]: optimizer = tf.keras.optimizers.Adam(learning_rate=0.001, beta_1=0.9,
                                              beta_2=0.999)
```

```
In [70]: history_adam = build_and_train_model(optimizer) # extra code
```

```
Epoch 1/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.7753 - loss: 0.6442 - val_accuracy: 0.8366 - val_loss: 0.4262
Epoch 2/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8567 - loss: 0.3900 - val_accuracy: 0.8402 - val_loss: 0.4161
Epoch 3/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8735 - loss: 0.3460 - val_accuracy: 0.8404 - val_loss: 0.4165
Epoch 4/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8823 - loss: 0.3201 - val_accuracy: 0.8482 - val_loss: 0.4132
Epoch 5/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8893 - loss: 0.2992 - val_accuracy: 0.8674 - val_loss: 0.3568
Epoch 6/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8946 - loss: 0.2832 - val_accuracy: 0.8600 - val_loss: 0.3846
Epoch 7/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8999 - loss: 0.2681 - val_accuracy: 0.8620 - val_loss: 0.3954
Epoch 8/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.9049 - loss: 0.2557 - val_accuracy: 0.8544 - val_loss: 0.4681
Epoch 9/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.9075 - loss: 0.2451 - val_accuracy: 0.8682 - val_loss: 0.3921
Epoch 10/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.9096 - loss: 0.2380 - val_accuracy: 0.8694 - val_loss: 0.3817
```

Adamax Optimization

- What it does

A variant of Adam that uses the infinity norm for the second moment instead of the L2 norm, making it more robust to large gradients and potentially better for certain problems like natural language processing.

- Update rule

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t$$

$$\mathbf{v}_t = \max(\beta_2 \mathbf{v}_{t-1}, |\mathbf{g}_t|)$$

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}$$

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\eta}{\mathbf{v}_t} \hat{\mathbf{m}}_t$$

where

- $\mathbf{g}_t$ is the gradient at time step t
- $\mathbf{m}_t$ is the first moment estimate
- $\mathbf{v}_t$ is the infinity norm of past gradients
- β_1 and β_2 are decay rates
- $\hat{\mathbf{m}}_t$ is the bias-corrected first moment estimate
- η is the learning rate
- Typical settings and tips
 - Common values: $\eta \approx 0.001$, $\beta_1 \approx 0.9$, $\beta_2 \approx 0.999$.
 - Similar to Adam but can be more stable for sparse gradients; consider it as an alternative when Adam's L2 norm causes issues.
 - Often used in NLP tasks; tune learning rate and monitor convergence.

```
In [71]: optimizer = tf.keras.optimizers.Adamax(learning_rate=0.001, beta_1=0.9,
                                              beta_2=0.999)
```

```
In [72]: history_adamax = build_and_train_model(optimizer) # extra code
```

```
Epoch 1/10
1719/1719 ————— 12s 6ms/step - accuracy: 0.7597 - loss: 0.7048 - val_accuracy: 0.8416 - val_loss: 0.4345
Epoch 2/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8554 - loss: 0.4131 - val_accuracy: 0.8554 - val_loss: 0.3933
Epoch 3/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8697 - loss: 0.3657 - val_accuracy: 0.8580 - val_loss: 0.3728
Epoch 4/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8776 - loss: 0.3374 - val_accuracy: 0.8654 - val_loss: 0.3604
Epoch 5/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8845 - loss: 0.3164 - val_accuracy: 0.8640 - val_loss: 0.3615
Epoch 6/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8907 - loss: 0.3003 - val_accuracy: 0.8672 - val_loss: 0.3491
Epoch 7/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8962 - loss: 0.2861 - val_accuracy: 0.8684 - val_loss: 0.3477
Epoch 8/10
1719/1719 ————— 10s 5ms/step - accuracy: 0.9000 - loss: 0.2739 - val_accuracy: 0.8662 - val_loss: 0.3541
Epoch 9/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.9046 - loss: 0.2621 - val_accuracy: 0.8696 - val_loss: 0.3470
Epoch 10/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.9089 - loss: 0.2519 - val_accuracy: 0.8704 - val_loss: 0.3481
```

Nadam Optimization

- What it does
 - Combines Adam's adaptive learning rates with Nesterov's accelerated gradient,

computing gradients at the anticipated future position for more informed updates and faster convergence.

- Update rule

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t$$

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t^2$$

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}$$

$$\hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}$$

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} (\beta_1 \hat{\mathbf{m}}_t + \frac{(1 - \beta_1) \mathbf{g}_t}{1 - \beta_1^t})$$

where

- $\mathbf{g}_t$ is the gradient at time step t
 - $\mathbf{m}_t$ and $\mathbf{v}_t$ are the first and second moment estimates
 - β_1 and β_2 are decay rates
 - $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are bias-corrected estimates
 - η is the learning rate
 - ϵ is a small constant to prevent division by zero
- Typical settings and tips
 - Common values: $\eta \approx 0.001$, $\beta_1 \approx 0.9$, $\beta_2 \approx 0.999$, $\epsilon \approx 1e-8$.
 - Often converges faster than Adam due to Nesterov momentum; good for deep networks.
 - Use as a drop-in replacement for Adam; monitor for overfitting and tune learning rate.

```
In [73]: optimizer = tf.keras.optimizers.Nadam(learning_rate=0.001, beta_1=0.9,  
                                              beta_2=0.999)
```

```
In [74]: history_nadam = build_and_train_model(optimizer) # extra code
```

Epoch 1/10
1719/1719 ————— 13s 6ms/step - accuracy: 0.7786 - loss: 0.6328 - val_accuracy: 0.8446 - val_loss: 0.4238
Epoch 2/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.8613 - loss: 0.3789 - val_accuracy: 0.8520 - val_loss: 0.3912
Epoch 3/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8755 - loss: 0.3408 - val_accuracy: 0.8558 - val_loss: 0.3988
Epoch 4/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.8834 - loss: 0.3148 - val_accuracy: 0.8498 - val_loss: 0.4032
Epoch 5/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.8904 - loss: 0.2953 - val_accuracy: 0.8620 - val_loss: 0.3800
Epoch 6/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.8954 - loss: 0.2791 - val_accuracy: 0.8600 - val_loss: 0.3932
Epoch 7/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8998 - loss: 0.2661 - val_accuracy: 0.8514 - val_loss: 0.4351
Epoch 8/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.9034 - loss: 0.2572 - val_accuracy: 0.8626 - val_loss: 0.3953
Epoch 9/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.9087 - loss: 0.2446 - val_accuracy: 0.8692 - val_loss: 0.3887
Epoch 10/10
1719/1719 ————— 11s 6ms/step - accuracy: 0.9109 - loss: 0.2382 - val_accuracy: 0.8708 - val_loss: 0.3723

AdamW Optimization

Core Concept (What it does)

AdamW (Adam Weight Decay) is a variant of the Adam optimizer, and its primary innovation lies in **decoupling** weight decay from the gradient-based updates.

In the standard Adam optimizer, L2 regularization is typically handled as part of the loss function. This approach can lead to adaptive learning rates influencing the effectiveness of weight decay during the update process. AdamW, however, applies L2 regularization (i.e., weight decay) directly to the model's weights themselves, rather than to their gradients. This independent handling ensures that the regularization effect is not diluted by adaptive learning rates, often resulting in better model generalization and more stable training performance.

Update Rule

AdamW's parameter update combines Adam's momentum estimates and adaptive learning rates, while an additional and independent weight decay term is added during the parameter update step:

1. **Compute First Moment Estimate:**

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t$$

2. Compute Second Moment Estimate:

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t^2$$

3. Bias-corrected First Moment:

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}$$

4. Bias-corrected Second Moment:

$$\hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}$$

5. Final Parameter Update:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \hat{\mathbf{m}}_t - \eta \lambda \boldsymbol{\theta}$$

Notation:

- $\mathbf{g}_t$: The gradient at time step t .
- $\mathbf{m}_t, \mathbf{v}_t$: The first and second moment estimates, respectively.
- β_1, β_2 : The decay rates for these moment estimates.
- $\hat{\mathbf{m}}_t, \hat{\mathbf{v}}_t$: The bias-corrected moment estimates.
- η : The learning rate.
- ϵ : A small constant to prevent division by zero, ensuring numerical stability.
- λ : The weight decay coefficient, applied directly to the model parameters $\boldsymbol{\theta}$.
- the decoupled weight decay term ($\eta \lambda \boldsymbol{\theta}$) is subtracted from the parameters independently of the adaptive learning rate, ensuring that regularization is effective regardless of the gradient magnitudes.

Practical Settings and Tips

- **Common Hyperparameter Values:**
 - Learning Rate (η): Approximately 0.001.
 - First Moment Decay Rate (β_1): Approximately 0.9.
 - Second Moment Decay Rate (β_2): Approximately 0.999.
 - Numerical Stability Constant (ϵ): Approximately $1e^{-8}$.
 - Weight Decay Coefficient (λ): Commonly set to 0.01.
- **Regularization Advantage:** AdamW typically outperforms standard Adam in scenarios requiring strong regularization, especially on large datasets, by providing superior generalization capabilities.
- **Replacement Strategy:** When your training pipeline requires weight decay, it is highly recommended to use AdamW instead of the Adam optimizer.
- **Hyperparameter Tuning:** Always carefully tune the weight decay coefficient λ and continuously monitor the model's performance on the validation set to prevent overfitting.

Note: Since TF 1.12, `AdamW` is no longer experimental. It is available at `tf.keras.optimizers.AdamW` instead of `tf.keras.optimizers.experimental.AdamW`.

```
In [75]: optimizer = tf.keras.optimizers.AdamW(weight_decay=1e-5, learning_rate=0.001,
                                              beta_1=0.9, beta_2=0.999)
```

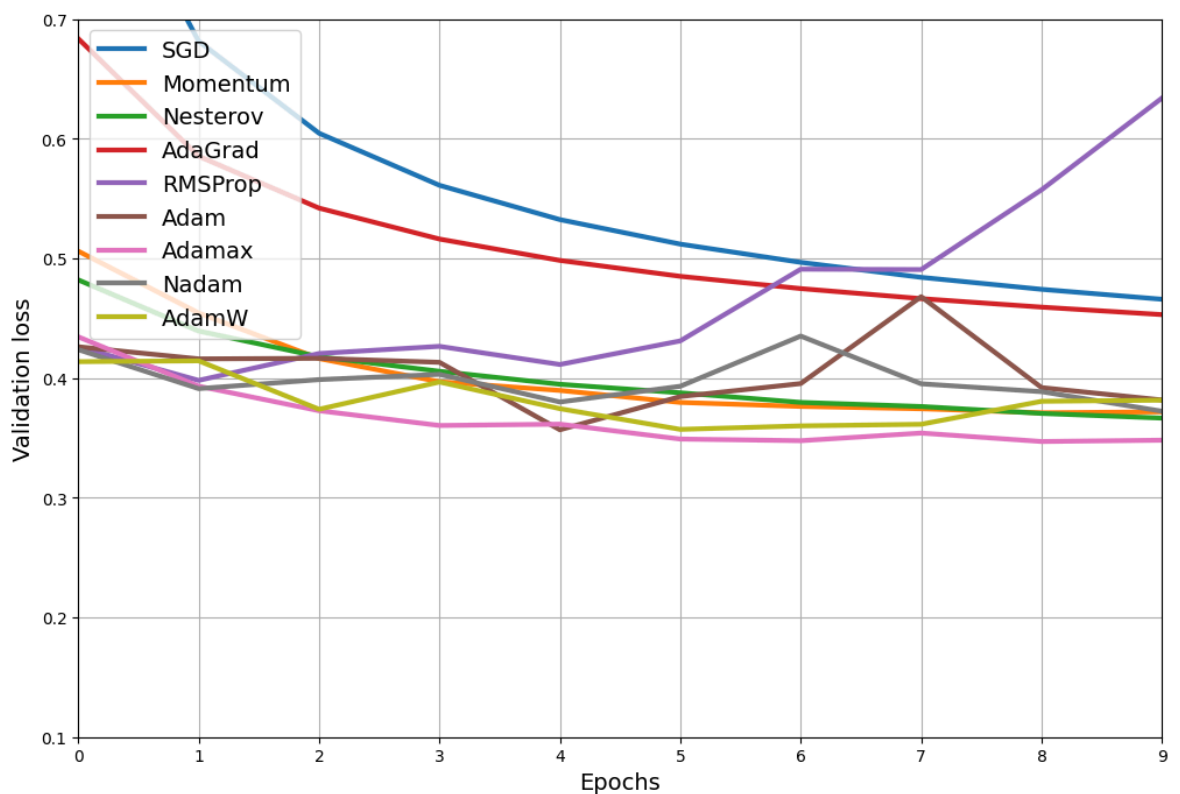
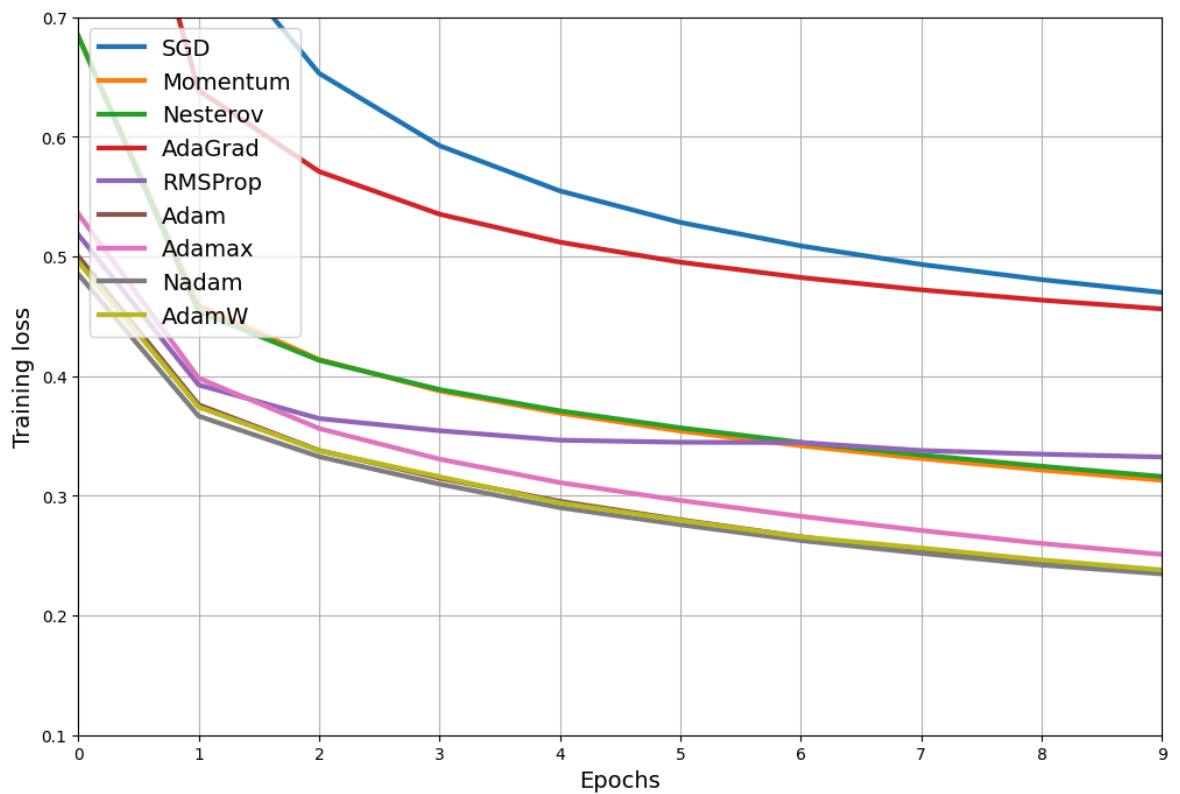
```
In [76]: history_adamw = build_and_train_model(optimizer) # extra code
```

```
Epoch 1/10
1719/1719 ————— 12s 6ms/step - accuracy: 0.7771 - loss: 0.6324 - val_accuracy: 0.8390 - val_loss: 0.4137
Epoch 2/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8582 - loss: 0.3864 - val_accuracy: 0.8382 - val_loss: 0.4145
Epoch 3/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8723 - loss: 0.3456 - val_accuracy: 0.8608 - val_loss: 0.3740
Epoch 4/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8792 - loss: 0.3216 - val_accuracy: 0.8518 - val_loss: 0.3968
Epoch 5/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8893 - loss: 0.2996 - val_accuracy: 0.8590 - val_loss: 0.3745
Epoch 6/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.8951 - loss: 0.2814 - val_accuracy: 0.8700 - val_loss: 0.3572
Epoch 7/10
1719/1719 ————— 9s 5ms/step - accuracy: 0.8981 - loss: 0.2674 - val_accuracy: 0.8720 - val_loss: 0.3601
Epoch 8/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.9036 - loss: 0.2558 - val_accuracy: 0.8722 - val_loss: 0.3614
Epoch 9/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.9079 - loss: 0.2441 - val_accuracy: 0.8700 - val_loss: 0.3806
Epoch 10/10
1719/1719 ————— 10s 6ms/step - accuracy: 0.9121 - loss: 0.2362 - val_accuracy: 0.8764 - val_loss: 0.3816
```

```
In [77]: # extra code - visualize the learning curves of all the optimizers

for loss in ("loss", "val_loss"):
    plt.figure(figsize=(12, 8))
    opt_names = "SGD Momentum Nesterov AdaGrad RMSProp Adam Adamax Nadam AdamW"
    for history, opt_name in zip((history_sgd, history_momentum, history_nesterov,
                                history_adagrad, history_rmsprop, history_adam,
                                history_adamax, history_nadam, history_adamw),
                                opt_names.split()):
        plt.plot(history.history[loss], label=f"{opt_name}", linewidth=3)

    plt.grid()
    plt.xlabel("Epochs")
    plt.ylabel(f"{loss}: \"Training loss\", \"val_loss\": \"Validation loss\"}[loss]")
    plt.legend(loc="upper left")
    plt.axis([0, 9, 0.1, 0.7])
    plt.show()
```

Conclusion: Faster Optimizers

Standard Stochastic Gradient Descent (SGD) is often too slow for training deep neural networks and struggles with complex loss landscapes, such as narrow "ravines." To accelerate convergence and boost performance, researchers have developed **Faster Optimizers**. These are generally categorized into momentum-based methods, adaptive learning rate methods, and advanced hybrids.

1. Momentum-Based Optimizers

These algorithms simulate physical inertia to speed up training.

- **Momentum Optimization:** Simulates a ball rolling down a slope. Unlike vanilla SGD which only considers the local gradient, momentum accumulates a "velocity" from past gradients. This allows it to accelerate through plateaus and dampen oscillations in steep valleys, often converging up to 10 times faster than SGD.
- **Nesterov Accelerated Gradient (NAG):** A variant of momentum that calculates the gradient at a "look-ahead" position (slightly ahead in the direction of the momentum). This mechanism makes updates more precise and significantly reduces overshooting, making it almost always faster and more stable than standard momentum.

2. Adaptive Learning Rate Optimizers

These algorithms individually tune the learning rate for every parameter, which is highly effective for data with varying scales or sparsity.

- **AdaGrad:** Scales the learning rate of each parameter based on the sum of past squared gradients. While useful for sparse data, its learning rate decays monotonically and often causes training to **stop prematurely** in deep networks.
- **RMSProp:** Fixes AdaGrad's flaw by using an "exponentially decaying moving average" to accumulate recent squared gradients. It ensures the learning rate doesn't vanish and is a preferred choice for **Recurrent Neural Networks (RNNs)**.

3. Advanced and Hybrid Optimizers

The industry standards that combine momentum stability with adaptive scaling.

- **Adam (Adaptive Moment Estimation):** Combines the ideas of both Momentum (tracking the mean of gradients) and RMSProp (tracking the variance). It is robust to hyperparameters and is the **top choice for rapid prototyping**.
- **Nadam:** Incorporates the **Nesterov trick** into the Adam optimizer, typically resulting in even faster convergence than standard Adam.
- **AdamW:** A crucial improvement that **decouples** weight decay from the gradient updates. It is the default optimizer for modern large-scale models (like **Transformers** and **LLaMA**) because it offers much stronger generalization capabilities.

Practical Guidelines and Recommendations

- **Initial Development:** Prioritize **Adam** or **Nadam** for their ease of use and exceptionally fast initial convergence.
- **Peak Generalization:** If you have the resources for extensive hyperparameter tuning, **SGD with Momentum (and NAG)** paired with a well-designed **Learning**

Rate Schedule (such as the **1 cycle policy**) often finds better global optima than Adam.

- **Large-Batch Training:** For distributed training with very large batches, consider specialized optimizers like **LARS** to maintain training stability.

This concludes our exploration of optimization algorithms—next, we will dive into advanced learning rate scheduling and other sophisticated training techniques!

Learning Rate Scheduling

Learning-rate (LR) scheduling controls **how the step size changes during training** — a small, well-chosen schedule often speeds convergence and improves final generalization more than many other hyperparameter tweaks.

Practical overview

- Conceptual Refinement: The Learning Rate Trilemma
 - In deep learning, you face a trade-off:
 - High Learning Rate: Risks diverging or overshooting the global minimum.
 - Low Learning Rate: Takes too long to converge and may get stuck in poor local minima.
 - Decaying Learning Rate: Starts high to make rapid progress and decreases to "settle" into the minimum.generalization.
- Common schedules:
 - Step decay — drop LR by a factor at fixed epochs (simple, robust).
 - Exponential decay — smooth multiplicative decay.
 - Cosine annealing (with/without restarts) — smooth decay that often yields better final results.
 - One-cycle / cyclical LR — helps escape sharp minima and speeds training.
 - Warmup — ramp LR up for a few epochs/steps (essential for large-batch training).
 - ReduceLROnPlateau — lower LR when a monitored metric stops improving.
- Keras tools: `tf.keras.optimizers.schedules.*`, `tf.keras.callbacks.LearningRateScheduler`, `tf.keras.callbacks.ReduceLROnPlateau`.
 - `tf.keras.optimizers.schedules.*`: Optimizer schedules are stateless functions of the current step
 - `tf.keras.callbacks.*`: Callbacks can implement more complex logic based on training progress and metrics.
- Recipe:
 - Do a short LR sweep to find a good max LR.
 - Prototype with `Adam` ($lr \approx 1e-3$); for best final results switch to `SGD` +momentum ($lr \approx 0.01-0.1$) and use a decay (cosine or one-cycle).
 - Use 5–10% of total steps for warmup when training with large batches.

- Monitor validation metrics and save the best weights (early stopping + LR reduction).
- Typical defaults to try: Adam lr=1e-3, SGD + momentum lr=0.01 (scale with batch size), final LR floor $\approx 1e-6$.

The cells that follow show concrete Keras examples and recipes (LR schedules, callbacks, and a one-cycle implementation).

Power Scheduling

Power scheduling decays the learning rate using a polynomial function, where the learning rate decreases as training progresses. The general formula is:

$$\eta_t = \frac{\eta_0}{(1 + \text{decay_rate} \cdot t/s)^{\text{power}}}$$

where

- Keras uses `power = 1`.
- `decay_rate` controls how quickly the learning rate decreases
- `s` is a scaling factor (decay steps often set to the total number of steps or epochs).
- `t` is the current step or epoch.

Note: Optimizers used to have a `decay` argument for this, but it was deprecated. The legacy method of passing a `decay` argument directly to an optimizer (e.g., `tf.keras.optimizers.Adam(learning_rate=0.001, decay=1e-6)`) is no longer recommended. You must use the schedulers in `tf.keras.optimizers.schedules` instead.

```
In [99]: dir(tf.keras.optimizers.schedules) # extra code - list all learning rate schedul
```

```
Out[99]: ['CosineDecay',
          'CosineDecayRestarts',
          'ExponentialDecay',
          'InverseTimeDecay',
          'LearningRateSchedule',
          'PiecewiseConstantDecay',
          'PolynomialDecay',
          '__builtins__',
          '__cached__',
          '__doc__',
          '__file__',
          '__loader__',
          '__name__',
          '__package__',
          '__path__',
          '__spec__',
          'deserialize',
          'serialize']
```

In the following, we will use `tf.keras.optimizers.schedules.InverseTimeDecay` to implement power scheduling.

```
In [100... power_lr_schedule = tf.keras.optimizers.schedules.InverseTimeDecay(  
    initial_learning_rate=0.01,  
    decay_steps=10_000,  
    decay_rate=1.0,  
    staircase=False  
)  
optimizer = tf.keras.optimizers.SGD(learning_rate=power_lr_schedule)
```

The `InverseTimeDecay` scheduler calculates the learning rate η_t at step t using the following formula:

$$\eta_t = \frac{\eta_0}{1 + c \cdot \frac{t}{s}}$$

If you set `staircase=True`, Keras replaces the continuous ratio t/s with its floor value $\lfloor t/s \rfloor$:

$$\eta_t = \frac{\eta_0}{1 + c \cdot \lfloor t/s \rfloor}$$

In this implementation, `InverseTimeDecay` effectively performs **power scheduling** with the power coefficient set to 1. Specifically:

- η_0 represents the `initial_learning_rate`.
- c represents the `decay_rate`.
- s represents the `decay_steps` (the scaling factor).

```
In [101... history_power_scheduling = build_and_train_model(optimizer) # extra code
```

```

Epoch 1/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.7754 - loss: 0.6622 - va
l_accuracy: 0.8280 - val_loss: 0.4802
Epoch 2/10
1719/1719 ————— 3s 1ms/step - accuracy: 0.8363 - loss: 0.4621 - va
l_accuracy: 0.8330 - val_loss: 0.4424
Epoch 3/10
1719/1719 ————— 2s 1ms/step - accuracy: 0.8513 - loss: 0.4198 - va
l_accuracy: 0.8424 - val_loss: 0.4230
Epoch 4/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8598 - loss: 0.3953 - va
l_accuracy: 0.8490 - val_loss: 0.4116
Epoch 5/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8657 - loss: 0.3783 - va
l_accuracy: 0.8540 - val_loss: 0.4007
Epoch 6/10
1719/1719 ————— 2s 1ms/step - accuracy: 0.8700 - loss: 0.3653 - va
l_accuracy: 0.8592 - val_loss: 0.3918
Epoch 7/10
1719/1719 ————— 2s 1ms/step - accuracy: 0.8734 - loss: 0.3549 - va
l_accuracy: 0.8614 - val_loss: 0.3845
Epoch 8/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8760 - loss: 0.3462 - va
l_accuracy: 0.8638 - val_loss: 0.3774
Epoch 9/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8781 - loss: 0.3388 - va
l_accuracy: 0.8652 - val_loss: 0.3725
Epoch 10/10
1719/1719 ————— 2s 1ms/step - accuracy: 0.8803 - loss: 0.3324 - va
l_accuracy: 0.8670 - val_loss: 0.3679

```

In [102...

```

import matplotlib.pyplot as plt
import tensorflow as tf
import numpy as np # Import numpy if not already present

# Calculate steps per epoch based on your training data and batch size
import math
batch_size = 32 # default used in model.fit if not specified
steps_per_epoch = math.ceil(len(X_train) / batch_size)

def plot_learning_rate_schedule(
    history: tf.keras.callbacks.History,
    lr_schedule_fn: callable,
    plot_label: str = "Learning rate",
    plot_title: str = "Learning Rate Schedule"
) -> None:
    """
    Visualizes a learning rate schedule from a Keras History object.

    Args:
        history: The Keras History object obtained from model.fit().
        lr_schedule_fn: The tf.keras.optimizers.schedules.LearningRateSchedule or
            a callable that takes a step (int) and returns the le
        plot_label: Label for the plotted line in the legend.
        plot_title: Title for the plot.
    """
    plt.figure(figsize=(12, 8))

    # Calculate learning rates for each epoch using the provided schedule functi
    # We cast the epoch number to a tf.int64 tensor as required by tf.keras.opti

```

```

learning_rates_at_epochs = [
    lr_schedule_fn(tf.constant(epoch, dtype=tf.int64)).numpy()
    for epoch in history.epoch
]

# Learning rate
plt.plot(history.epoch, learning_rates_at_epochs, label=plot_label, linewidth=2)
plt.grid(True)
plt.xlabel("Epochs")
plt.ylabel("Learning rate", color='b') # Modified to set color for primary y-axis
plt.tick_params('y', colors='b') # Added for primary y-axis
plt.legend()

# validation loss
ax2 = plt.gca().twinx()
ax2.plot(history.epoch, history.history["val_loss"], "r^-", label="Validation Loss")
ax2.set_ylabel('Validation Loss', color='r')
ax2.tick_params('y', colors='r')

plt.title(plot_title)
plt.legend(loc="lower left")
plt.show()

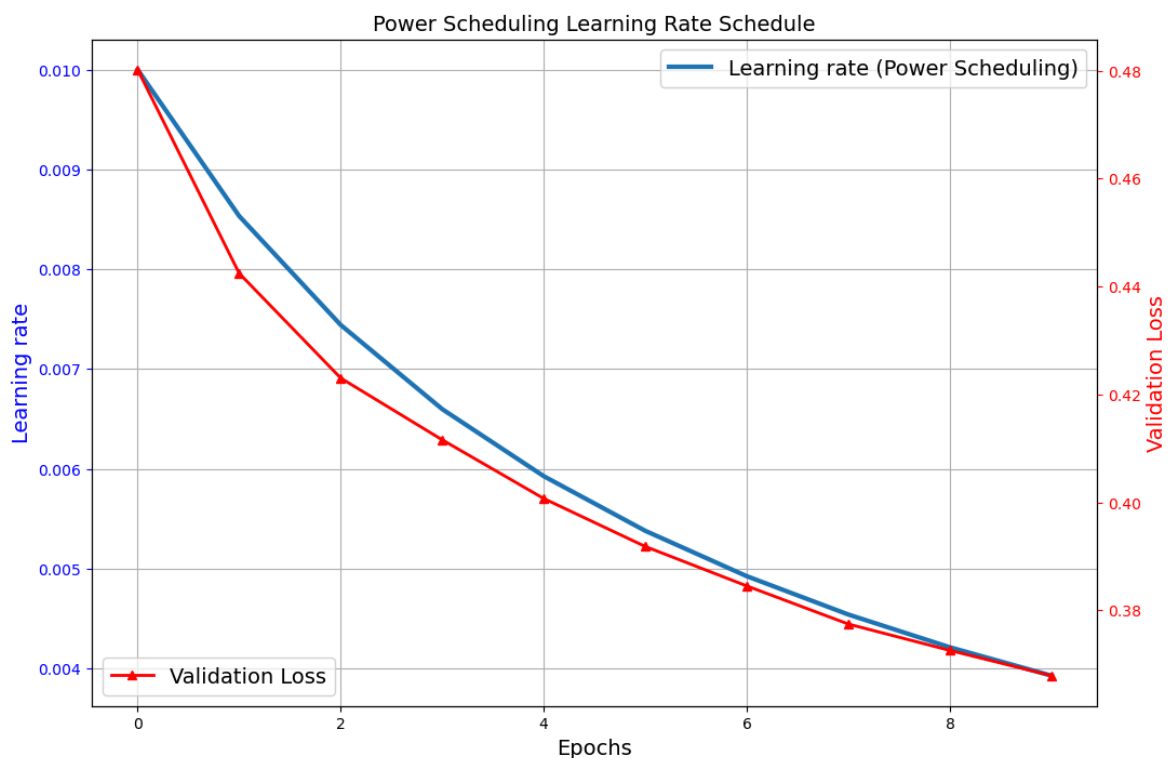
```

In [103... *# visualize the learning rate schedule of the power scheduling optimizer*

```

# We need to multiply the epoch number by steps_per_epoch to get the correct steps
plot_learning_rate_schedule(
    history=history_power_scheduling,
    lr_schedule_fn=lambda epoch: power_lr_schedule(epoch * steps_per_epoch),
    plot_label="Learning rate (Power Scheduling)",
    plot_title="Power Scheduling Learning Rate Schedule"
)

```



In [104... *# extra code - this cell plots power scheduling with staircase=True or False*

```

initial_learning_rate = 0.01

```

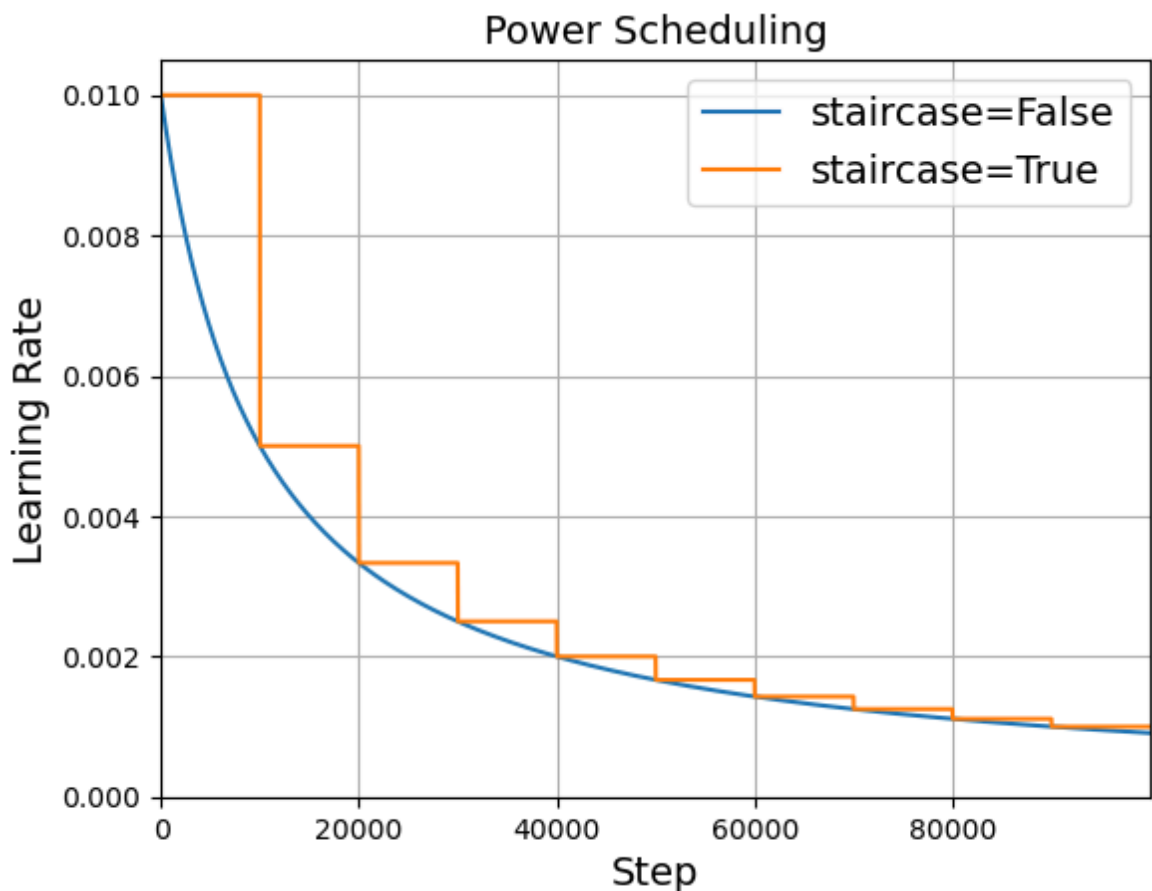
```

decay_rate = 1.0
decay_steps = 10_000

steps = np.arange(100_000)
lrs = initial_learning_rate / (1 + decay_rate * steps / decay_steps)
lrs2 = initial_learning_rate / (1 + decay_rate * np.floor(steps / decay_steps))

plt.plot(steps, lrs, "-", label="staircase=False")
plt.plot(steps, lrs2, "-", label="staircase=True")
plt.axis((0, steps.max(), 0, 0.0105))
plt.xlabel("Step")
plt.ylabel("Learning Rate")
plt.title("Power Scheduling", fontsize=14)
plt.legend()
plt.grid(True)
plt.show()

```



Exponential Scheduling

$$\eta_t = \eta_0 \cdot \text{decay_rate}^{t/s}$$

where

- `decay_rate` is the base of the exponential decay (e.g., 0.96).
- `s` is a scaling factor (decay steps often set to the total number of steps or epochs).
- `t` is the current step or epoch.

Next, the `ExponentialDecay` scheduler calculates the learning rate η_t at step t using the following formula:

$$\eta_t = \eta_0 \cdot \gamma^{t/s}$$

If you set `staircase=True`, Keras replaces the continuous ratio t/s with its floor value $\lfloor t/s \rfloor$, resulting in a step-like decay:

$$\eta_t = \eta_0 \cdot \gamma^{\lfloor t/s \rfloor}$$

In this implementation:

- η_0 corresponds to the `initial_learning_rate`.
- γ corresponds to the `decay_rate` (the base of the exponential).
- s corresponds to the `decay_steps` (the interval over which the decay is applied).

This ensures the learning rate is multiplied by the decay rate every s steps, allowing for a smooth or discrete reduction in the step size as training progresses.

```
In [105... exponential_lr_schedule = tf.keras.optimizers.schedules.ExponentialDecay(  
    initial_learning_rate=0.01,  
    decay_steps=20_000,  
    decay_rate=0.1,  
    staircase=False  
)  
optimizer = tf.keras.optimizers.SGD(learning_rate=exponential_lr_schedule)
```

```
In [106... history_exponential_scheduling = build_and_train_model(optimizer) # extra code
```

```

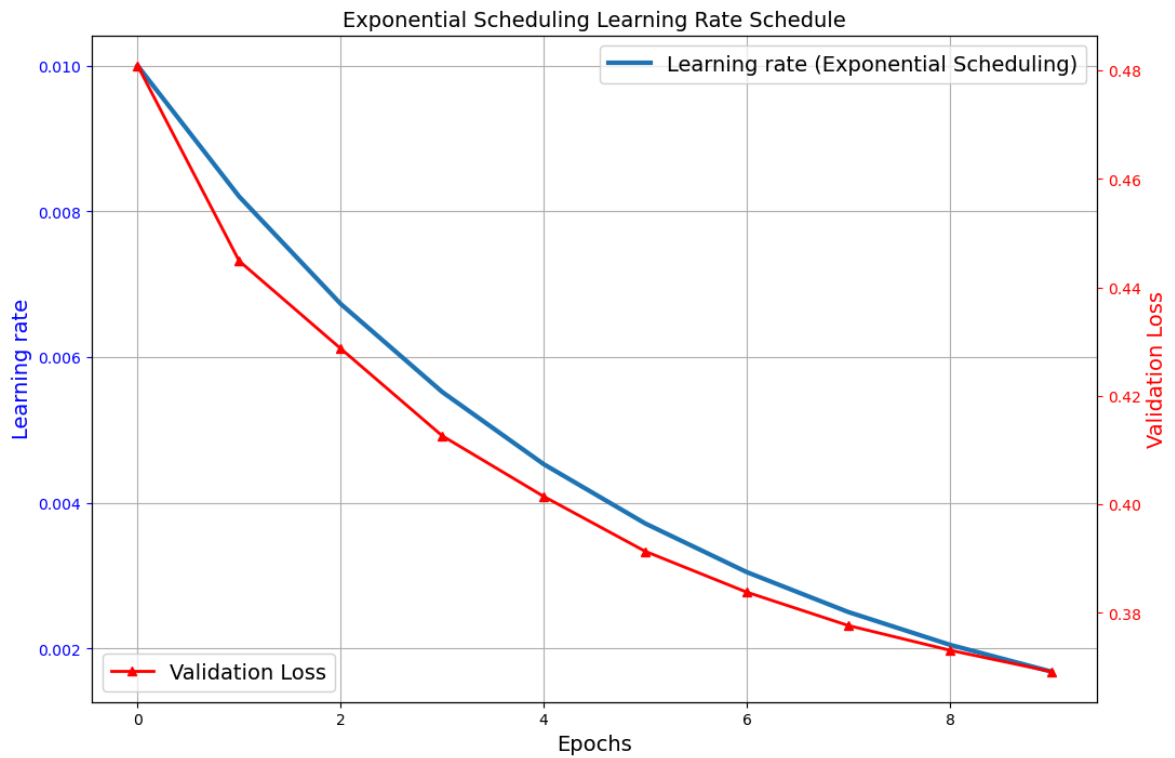
Epoch 1/10
1719/1719 ————— 4s 2ms/step - accuracy: 0.7688 - loss: 0.6799 - va
l_accuracy: 0.8348 - val_loss: 0.4809
Epoch 2/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8375 - loss: 0.4671 - va
l_accuracy: 0.8428 - val_loss: 0.4449
Epoch 3/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8529 - loss: 0.4238 - va
l_accuracy: 0.8470 - val_loss: 0.4287
Epoch 4/10
1719/1719 ————— 5s 3ms/step - accuracy: 0.8607 - loss: 0.3993 - va
l_accuracy: 0.8530 - val_loss: 0.4127
Epoch 5/10
1719/1719 ————— 4s 3ms/step - accuracy: 0.8667 - loss: 0.3828 - va
l_accuracy: 0.8556 - val_loss: 0.4015
Epoch 6/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8708 - loss: 0.3710 - va
l_accuracy: 0.8596 - val_loss: 0.3914
Epoch 7/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8738 - loss: 0.3620 - va
l_accuracy: 0.8596 - val_loss: 0.3838
Epoch 8/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8760 - loss: 0.3550 - va
l_accuracy: 0.8624 - val_loss: 0.3777
Epoch 9/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8775 - loss: 0.3495 - va
l_accuracy: 0.8662 - val_loss: 0.3731
Epoch 10/10
1719/1719 ————— 4s 2ms/step - accuracy: 0.8787 - loss: 0.3450 - va
l_accuracy: 0.8678 - val_loss: 0.3692

```

```

In [107... # visualize the learning rate schedule of the exponential scheduling optimizer
plot_learning_rate_schedule(
    history=history_exponential_scheduling,
    lr_schedule_fn=lambda epoch: exponential_lr_schedule(epoch * steps_per_epoch
    plot_label="Learning rate (Exponential Scheduling)",
    plot_title="Exponential Scheduling Learning Rate Schedule"
)

```



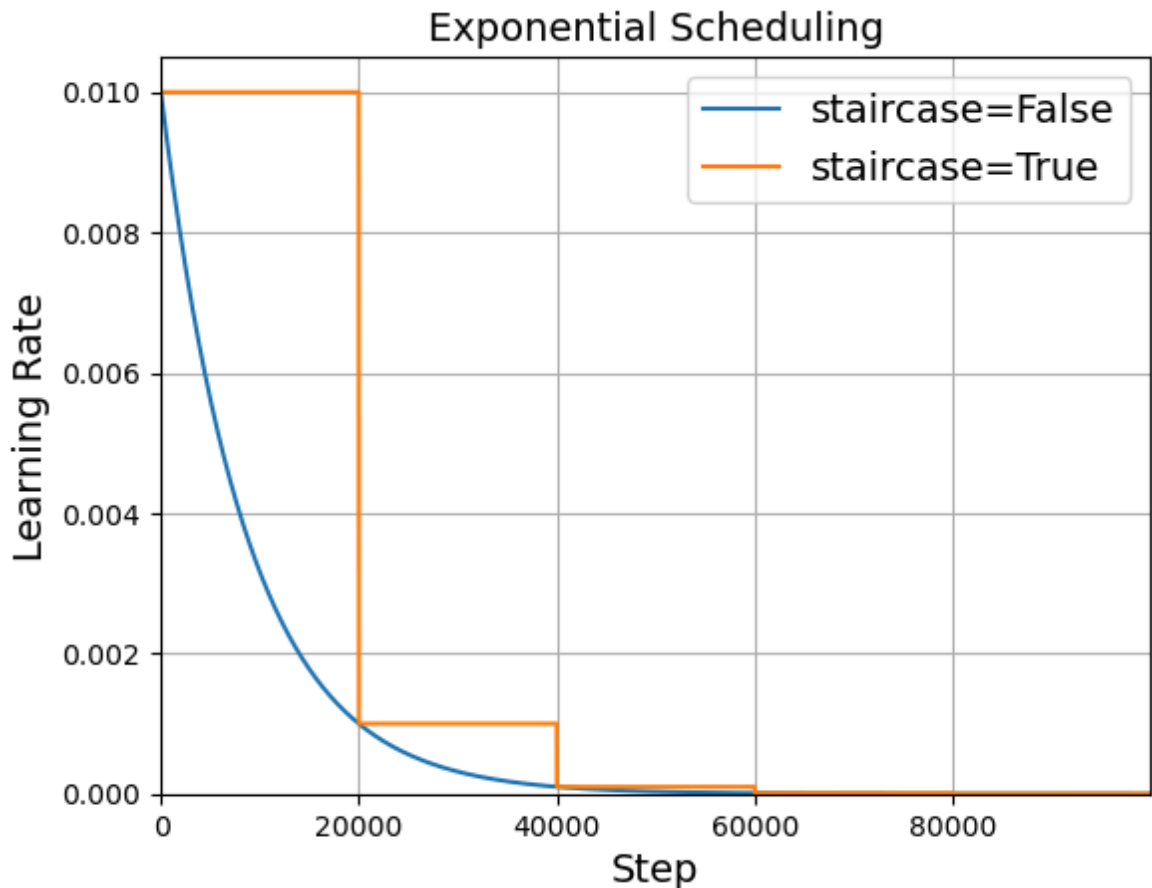
In [108...

extra code - this cell plots exponential scheduling

```
initial_learning_rate = 0.01
decay_rate = 0.1
decay_steps = 20_000

steps = np.arange(100_000)
lrs = initial_learning_rate * decay_rate ** (steps / decay_steps)
lrs2 = initial_learning_rate * decay_rate ** np.floor(steps / decay_steps)

plt.plot(steps, lrs, "-", label="staircase=False")
plt.plot(steps, lrs2, "-", label="staircase=True")
plt.axis([0, steps.max(), 0, 0.0105])
plt.xlabel("Step")
plt.ylabel("Learning Rate")
plt.title("Exponential Scheduling", fontsize=14)
plt.legend()
plt.grid(True)
plt.show()
```



Using LearningRateScheduler Callback

Keras also provides a `LearningRateScheduler` callback class that lets you define your own scheduling function. Let's see how you could use it to implement exponential decay. Note that in this case the learning rate only changes at each epoch, not at each step:

```
In [109... def exponential_decay_fn(epoch):
    return 0.01 * 0.1 ** (epoch / 20)
```

```
In [110... def exponential_decay(lr0, s):
    """Returns a function that computes the exponential decay of the learning ra
    Args:
        lr0: initial learning rate (float).
        s: decay speed (float). The larger `s` is, the slower the learning rate
            decreases."""
    def exponential_decay_fn(epoch):
        """Computes the exponential decay of the learning rate for a given epoch
        Args:
            epoch: current epoch number (int).
        Returns:
            The exponentially decayed learning rate (float) for the given epoch.
        """
        return lr0 * 0.1 ** (epoch / s)
    return exponential_decay_fn

exponential_decay_fn = exponential_decay(lr0=0.01, s=20)
```

Then, integrate a dynamic learning rate schedule using the `LearningRateScheduler` callback. The `LearningRateScheduler` constructor accepts a callable, in this case `exponential_decay_fn`, which is expected to take the current epoch index as an input and return the desired learning rate. This callback is executed at the onset of each epoch within the `model.fit()` method. By including it in the callbacks list, the optimizer's learning rate is programmatically updated before the next set of weight updates begins.

```
In [111... # extra code - build and compile a model for Fashion MNIST

tf.random.set_seed(42)
model = build_model()
optimizer = tf.keras.optimizers.SGD(learning_rate=0.001)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
              metrics=["accuracy"])

In [112... n_epochs = 20

lr_scheduler = tf.keras.callbacks.LearningRateScheduler(exponential_decay_fn)
history = model.fit(X_train, y_train, epochs=n_epochs,
                  validation_data=(X_valid, y_valid),
                  callbacks=[lr_scheduler])
```

Epoch 1/20
1719/1719 ————— 4s 2ms/step - accuracy: 0.7699 - loss: 0.6764 - val_accuracy: 0.8258 - val_loss: 0.4846 - learning_rate: 0.0100

Epoch 2/20
1719/1719 ————— 3s 1ms/step - accuracy: 0.8381 - loss: 0.4606 - val_accuracy: 0.8364 - val_loss: 0.4447 - learning_rate: 0.0089

Epoch 3/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8529 - loss: 0.4163 - val_accuracy: 0.8488 - val_loss: 0.4166 - learning_rate: 0.0079

Epoch 4/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8610 - loss: 0.3901 - val_accuracy: 0.8540 - val_loss: 0.4014 - learning_rate: 0.0071

Epoch 5/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8680 - loss: 0.3715 - val_accuracy: 0.8570 - val_loss: 0.3896 - learning_rate: 0.0063

Epoch 6/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8719 - loss: 0.3575 - val_accuracy: 0.8616 - val_loss: 0.3785 - learning_rate: 0.0056

Epoch 7/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8754 - loss: 0.3463 - val_accuracy: 0.8646 - val_loss: 0.3703 - learning_rate: 0.0050

Epoch 8/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8788 - loss: 0.3372 - val_accuracy: 0.8676 - val_loss: 0.3633 - learning_rate: 0.0045

Epoch 9/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8813 - loss: 0.3295 - val_accuracy: 0.8690 - val_loss: 0.3572 - learning_rate: 0.0040

Epoch 10/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8835 - loss: 0.3230 - val_accuracy: 0.8706 - val_loss: 0.3536 - learning_rate: 0.0035

Epoch 11/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8858 - loss: 0.3175 - val_accuracy: 0.8716 - val_loss: 0.3492 - learning_rate: 0.0032

Epoch 12/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8874 - loss: 0.3127 - val_accuracy: 0.8730 - val_loss: 0.3459 - learning_rate: 0.0028

Epoch 13/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8888 - loss: 0.3085 - val_accuracy: 0.8748 - val_loss: 0.3432 - learning_rate: 0.0025

Epoch 14/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8903 - loss: 0.3049 - val_accuracy: 0.8758 - val_loss: 0.3406 - learning_rate: 0.0022

Epoch 15/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8915 - loss: 0.3017 - val_accuracy: 0.8758 - val_loss: 0.3390 - learning_rate: 0.0020

Epoch 16/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8925 - loss: 0.2989 - val_accuracy: 0.8764 - val_loss: 0.3370 - learning_rate: 0.0018

Epoch 17/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8935 - loss: 0.2964 - val_accuracy: 0.8760 - val_loss: 0.3356 - learning_rate: 0.0016

Epoch 18/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8943 - loss: 0.2942 - val_accuracy: 0.8776 - val_loss: 0.3342 - learning_rate: 0.0014

Epoch 19/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8951 - loss: 0.2922 - val_accuracy: 0.8786 - val_loss: 0.3331 - learning_rate: 0.0013

Epoch 20/20
1719/1719 ————— 2s 1ms/step - accuracy: 0.8960 - loss: 0.2905 - val_accuracy: 0.8788 - val_loss: 0.3319 - learning_rate: 0.0011

Therefore, `tf.keras.callbacks.LearningRateScheduler` is a Keras callback that allows you to implement custom learning rate scheduling during training.

- **Core Functionality:** It takes a single argument: a Python callable (a function) that determines the learning rate for a given epoch.
- **Execution:** This function (e.g., `exponential_decay_fn`) is invoked at the beginning of each epoch, receiving the current epoch number as its sole argument. It then returns the new learning rate that the optimizer will use for that epoch.
- **Flexibility:** This mechanism provides a flexible and programmatic way to implement various custom learning rate decay strategies, including:
 - Exponential decay
 - Step decay
 - Polynomial decay
- **Integration:** The created `lr_scheduler` object is passed to the `callbacks` argument of the `model.fit()` method to seamlessly apply the defined schedule throughout the training process.

```
In [113... history.history.keys()
```

```
Out[113... dict_keys(['accuracy', 'loss', 'val_accuracy', 'val_loss', 'learning_rate'])
```

```
In [114... # visualize the learning rate schedule of the exponential scheduling optimizer w
def plot_lr_and_loss_from_history(history: tf.keras.callbacks.History, n_epochs:
    """
    Visualizes the learning rate schedule and validation loss from a Keras Histo

    This function plots the learning rate used at each epoch (retrieved from `hi
    against the validation loss, allowing for easy comparison of their trends ov

    Args:
        history (tf.keras.callbacks.History): The Keras History object obtained
            which contains training metrics including 'learning_rate' or 'lr' an
        n_epochs (int): The total number of epochs for which the model was train
            Used for setting appropriate x-axis ticks.
        title (str, optional): The title to display above the plot. Defaults to
            "Learning Rate and Validation Loss".
    """
    plt.figure(figsize=(12, 8))

    # Try to retrieve learning rate from history, checking both 'learning_rate'
    learning_rates_data = history.history.get("learning_rate")
    if learning_rates_data is None:
        learning_rates_data = history.history.get("lr")

    if learning_rates_data:
        plt.plot(history.epoch, learning_rates_data, 'bo-', label="Learning Rate
        plt.ylabel("Learning Rate", color='b')
        plt.tick_params('y', colors='b')
        plt.legend(loc="upper right")
    else:
        print("Warning: Learning rate data ('learning_rate' or 'lr') not found i
        plt.ylabel("Primary Y-axis (LR data missing)", color='b')
        plt.tick_params('y', colors='b')
```

```

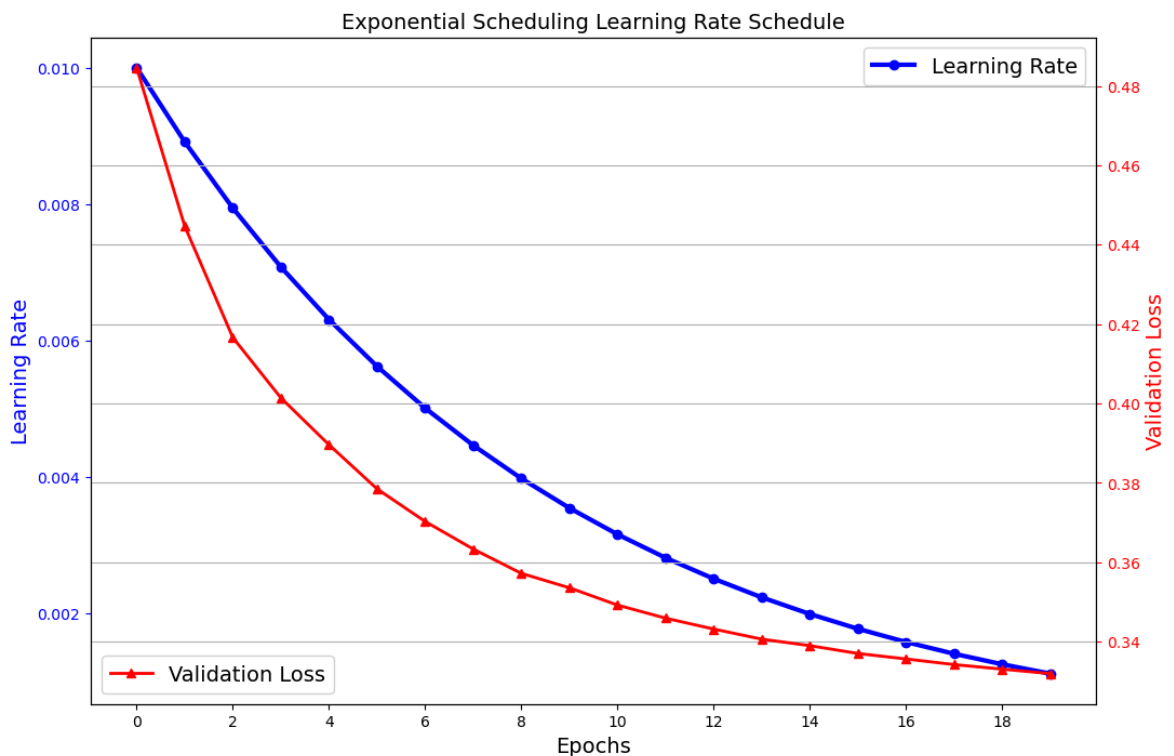
plt.xlabel("Epochs")
plt.title(title)
plt.xticks(range(0, n_epochs + 1, max(1, n_epochs // 10))) # Adjusted xticks

ax2 = plt.gca().twinx()
ax2.plot(history.epoch, history.history["val_loss"], "r^-", label="Validation Loss")
ax2.set_ylabel('Validation Loss', color='r')
ax2.tick_params('y', colors='r')
ax2.legend(loc="lower left")

plt.grid(True)
plt.show()

```

In [115... plot_lr_and_loss_from_history(history, n_epochs, title="Exponential Scheduling Learning L



Alternatively, the schedule function can take the current learning rate as a second argument:

In [116...

```
def exponential_decay_fn(epoch, lr):
    return lr * 0.1 ** (1 / 20)
```

Extra material: if you want to update the learning rate at each iteration rather than at each epoch, you can write a custom callback. Since the `batch` argument passed to `on_batch_begin()` is reset at each epoch, you should use the `self.model.optimizer.iterations` variable to track the total number of steps:

In [117...

```
K = tf.keras.backend

class ExponentialDecay(tf.keras.callbacks.Callback):
    """Custom Keras callback to implement exponential learning rate decay per batch

    This callback updates the learning rate at the beginning of each batch
```


rather than at the beginning of each epoch.

Args:

`n_steps` (int): The number of steps over which the learning rate decays by a factor of 0.1.

"""

```
def __init__(self, n_steps=40_000):
    super().__init__()
    self.n_steps = n_steps

    def on_batch_begin(self, batch, logs=None):
        # Note: the `batch` argument is reset at each epoch
        lr = self.model.optimizer.learning_rate.numpy()
        new_learning_rate = lr * 0.1 ** (1 / self.n_steps)
        self.model.optimizer.learning_rate = new_learning_rate

    def on_epoch_end(self, epoch, logs=None):
        logs = logs or {}
        logs['lr'] = self.model.optimizer.learning_rate.numpy()
```

```
In [118... lr0 = 0.01
model = build_model()
optimizer = tf.keras.optimizers.SGD(learning_rate=lr0)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
               metrics=["accuracy"])
```

```
In [119... import math

batch_size = 32
n_steps = n_epochs * math.ceil(len(X_train) / batch_size)
exp_decay = ExponentialDecay(n_steps)
history = model.fit(X_train, y_train, epochs=n_epochs,
                    validation_data=(X_valid, y_valid),
                    callbacks=[exp_decay])
```

Epoch 1/20
1719/1719 ————— 4s 2ms/step - accuracy: 0.7601 - loss: 0.6936 - val_accuracy: 0.8282 - val_loss: 0.4892 - lr: 0.0089

Epoch 2/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8323 - loss: 0.4749 - val_accuracy: 0.8382 - val_loss: 0.4482 - lr: 0.0079

Epoch 3/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8481 - loss: 0.4312 - val_accuracy: 0.8430 - val_loss: 0.4267 - lr: 0.0071

Epoch 4/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8568 - loss: 0.4053 - val_accuracy: 0.8514 - val_loss: 0.4094 - lr: 0.0063

Epoch 5/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8635 - loss: 0.3874 - val_accuracy: 0.8548 - val_loss: 0.3966 - lr: 0.0056

Epoch 6/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8678 - loss: 0.3736 - val_accuracy: 0.8586 - val_loss: 0.3873 - lr: 0.0050

Epoch 7/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8716 - loss: 0.3629 - val_accuracy: 0.8620 - val_loss: 0.3797 - lr: 0.0045

Epoch 8/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8750 - loss: 0.3538 - val_accuracy: 0.8648 - val_loss: 0.3729 - lr: 0.0040

Epoch 9/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8774 - loss: 0.3463 - val_accuracy: 0.8656 - val_loss: 0.3674 - lr: 0.0035

Epoch 10/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8797 - loss: 0.3399 - val_accuracy: 0.8674 - val_loss: 0.3635 - lr: 0.0032

Epoch 11/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8817 - loss: 0.3343 - val_accuracy: 0.8684 - val_loss: 0.3605 - lr: 0.0028

Epoch 12/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8831 - loss: 0.3294 - val_accuracy: 0.8696 - val_loss: 0.3579 - lr: 0.0025

Epoch 13/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8844 - loss: 0.3252 - val_accuracy: 0.8696 - val_loss: 0.3554 - lr: 0.0022

Epoch 14/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8856 - loss: 0.3215 - val_accuracy: 0.8700 - val_loss: 0.3532 - lr: 0.0020

Epoch 15/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8867 - loss: 0.3182 - val_accuracy: 0.8710 - val_loss: 0.3517 - lr: 0.0018

Epoch 16/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8876 - loss: 0.3153 - val_accuracy: 0.8716 - val_loss: 0.3500 - lr: 0.0016

Epoch 17/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8886 - loss: 0.3128 - val_accuracy: 0.8712 - val_loss: 0.3485 - lr: 0.0014

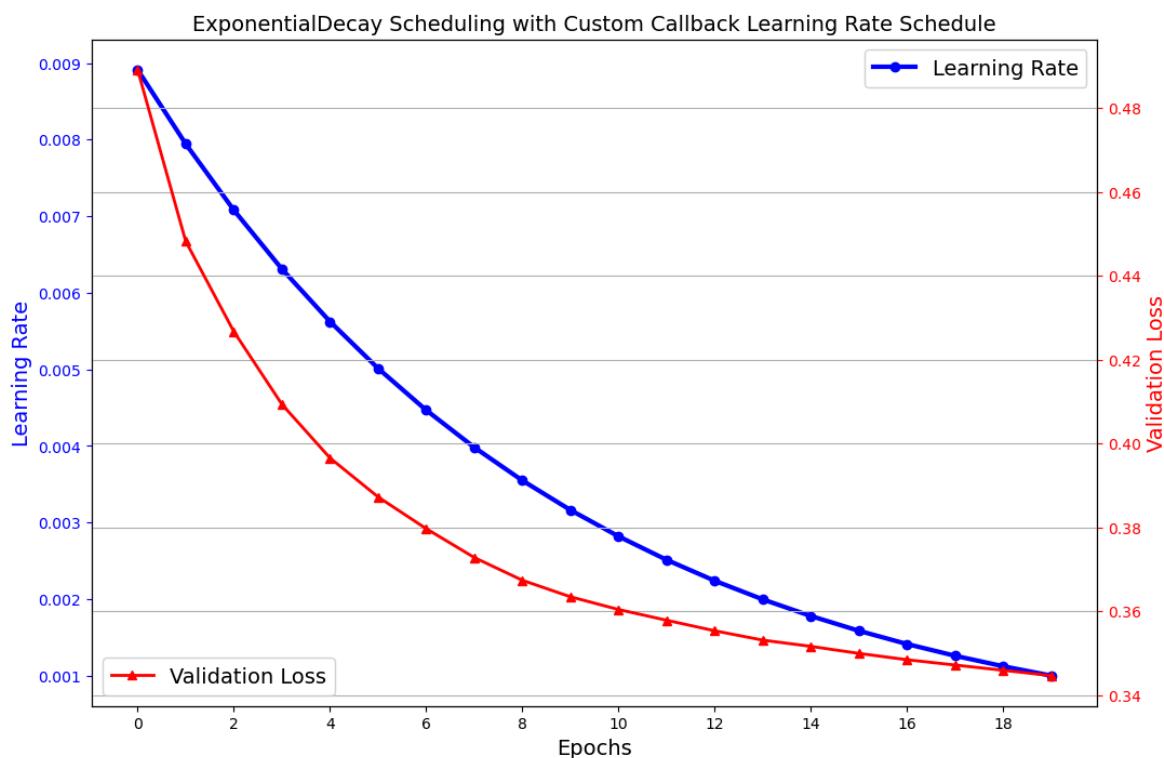
Epoch 18/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8892 - loss: 0.3105 - val_accuracy: 0.8718 - val_loss: 0.3472 - lr: 0.0013

Epoch 19/20
1719/1719 ————— 4s 2ms/step - accuracy: 0.8900 - loss: 0.3085 - val_accuracy: 0.8710 - val_loss: 0.3460 - lr: 0.0011

Epoch 20/20
1719/1719 ————— 3s 2ms/step - accuracy: 0.8908 - loss: 0.3067 - val_accuracy: 0.8718 - val_loss: 0.3447 - lr: 1.0000e-03

In [120...

```
plot_lr_and_loss_from_history(history, n_epochs, title="ExponentialDecay Schedul
```



In [121...

```
# investigate the methods in tf.keras.callbacks.Callback
for method in dir(tf.keras.callbacks.Callback):
    if not method.startswith("_"):
        print(method)
```

```
model
on_batch_begin
on_batch_end
on_epoch_begin
on_epoch_end
on_predict_batch_begin
on_predict_batch_end
on_predict_begin
on_predict_end
on_test_batch_begin
on_test_batch_end
on_test_begin
on_test_end
on_train_batch_begin
on_train_batch_end
on_train_begin
on_train_end
set_model
set_params
```

Piecewise Constant Scheduling

You manually define different learning rates for different phases of training (e.g., 0.1 for 5 epochs, then 0.01 for 10 epochs).

```
In [122... lr_schedule = tf.keras.optimizers.schedules.PiecewiseConstantDecay(
    boundaries=[50_000, 80_000],
    values=[0.01, 0.005, 0.001]
)
optimizer = tf.keras.optimizers.SGD(learning_rate=lr_schedule)
```

In this example, we will use

`tf.keras.optimizers.schedules.PiecewiseConstantDecay` to implement piecewise constant scheduling, where the learning rate changes at specified boundaries.

- `boundaries` is a list of epoch indices where the learning rate changes.
- `values` is a list of learning rates corresponding to each interval defined by the boundaries.

```
In [123... history_pieewise_scheduling = build_and_train_model(optimizer) # extra code
```

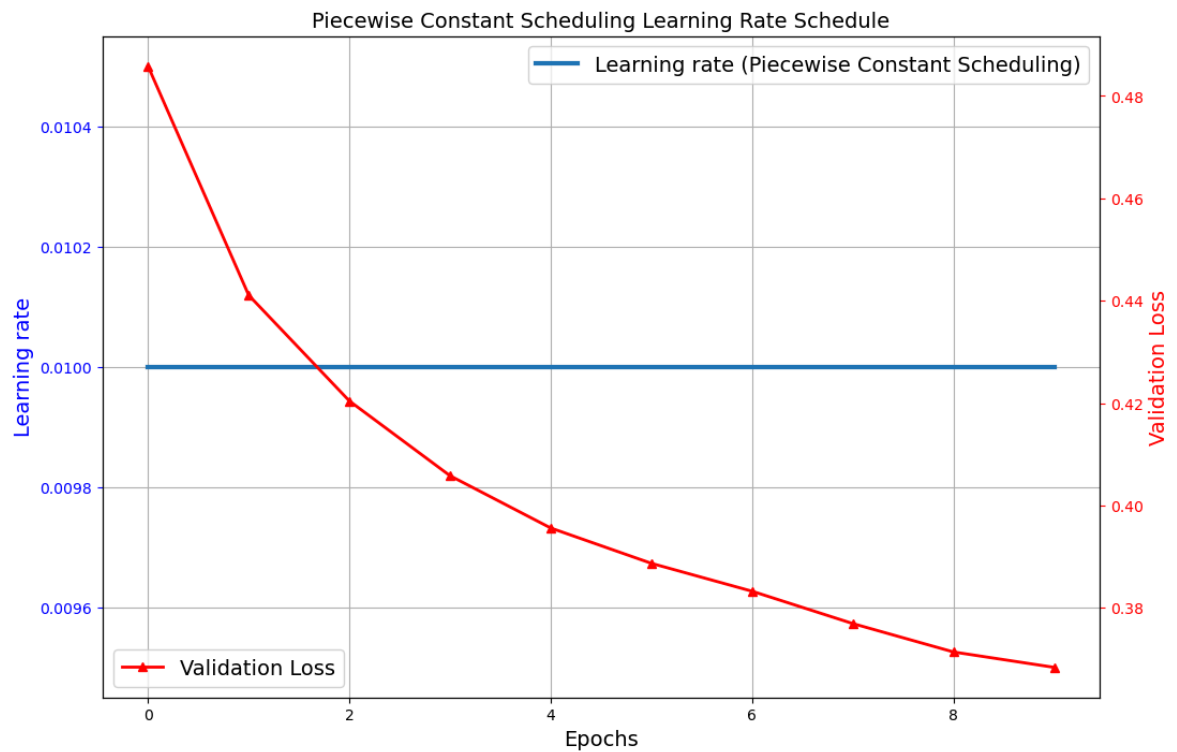
```
Epoch 1/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.7506 - loss: 0.7136 - va
l_accuracy: 0.8314 - val_loss: 0.4858
Epoch 2/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8336 - loss: 0.4709 - va
l_accuracy: 0.8398 - val_loss: 0.4411
Epoch 3/10
1719/1719 ————— 3s 1ms/step - accuracy: 0.8491 - loss: 0.4233 - va
l_accuracy: 0.8462 - val_loss: 0.4204
Epoch 4/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8591 - loss: 0.3951 - va
l_accuracy: 0.8504 - val_loss: 0.4058
Epoch 5/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8646 - loss: 0.3746 - va
l_accuracy: 0.8542 - val_loss: 0.3956
Epoch 6/10
1719/1719 ————— 4s 2ms/step - accuracy: 0.8699 - loss: 0.3588 - va
l_accuracy: 0.8594 - val_loss: 0.3887
Epoch 7/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8747 - loss: 0.3453 - va
l_accuracy: 0.8604 - val_loss: 0.3832
Epoch 8/10
1719/1719 ————— 2s 1ms/step - accuracy: 0.8786 - loss: 0.3339 - va
l_accuracy: 0.8624 - val_loss: 0.3769
Epoch 9/10
1719/1719 ————— 2s 1ms/step - accuracy: 0.8818 - loss: 0.3241 - va
l_accuracy: 0.8654 - val_loss: 0.3714
Epoch 10/10
1719/1719 ————— 3s 2ms/step - accuracy: 0.8844 - loss: 0.3150 - va
l_accuracy: 0.8674 - val_loss: 0.3684
```

```
In [124... print(steps_per_epoch)
```

```
1719
```

```
In [125... plot_learning_rate_schedule(
    history=history_pieewise_scheduling,
    lr_schedule_fn=lambda epoch: lr_schedule(epoch * steps_per_epoch),
    plot_label="Learning rate (Piecewise Constant Scheduling)",
```

```
plot_title="Piecewise Constant Scheduling Learning Rate Schedule"
)
```



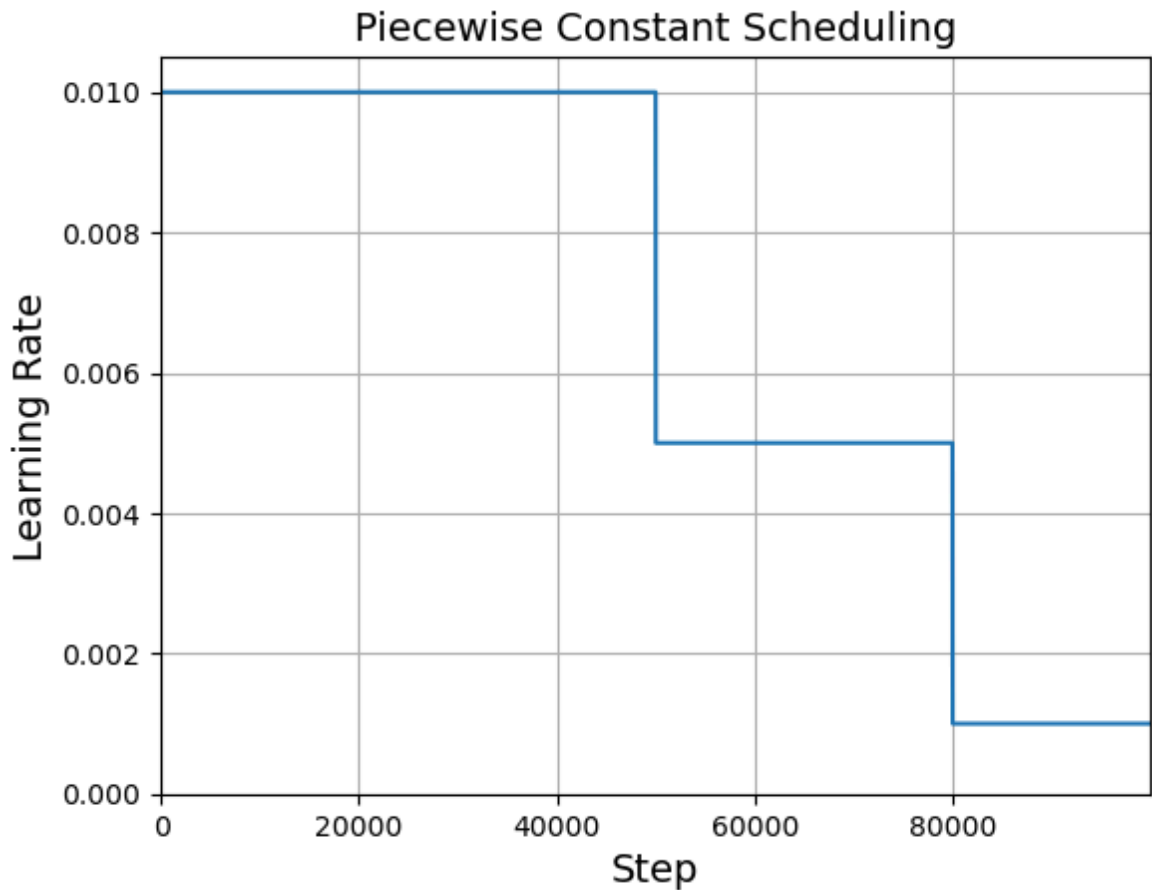
In [126... *# extra code - this cell plots piecewise constant scheduling*

```
boundaries = [50_000, 80_000]
values = [0.01, 0.005, 0.001]

steps = np.arange(100_000)

lrs = np.full(len(steps), values[0])
for boundary, value in zip(boundaries, values[1:]):
    lrs[boundary:] = value

plt.plot(steps, lrs, "-")
plt.axis([0, steps.max(), 0, 0.0105])
plt.xlabel("Step")
plt.ylabel("Learning Rate")
plt.title("Piecewise Constant Scheduling", fontsize=14)
plt.grid(True)
plt.show()
```



Just like we did with exponential scheduling, we could also implement piecewise constant scheduling manually:

```
In [127... def piecewise_constant_fn(epoch):
    if epoch < 5:
        return 0.01
    elif epoch < 15:
        return 0.005
    else:
        return 0.001
```

```
In [128... # extra code - this cell demonstrates a more general way to define
# piecewise constant scheduling.

def piecewise_constant(boundaries, values):
    boundaries = np.array([0] + boundaries)
    values = np.array(values)
    def piecewise_constant_fn(epoch):
        return values[(boundaries > epoch).argmax() - 1]
    return piecewise_constant_fn

piecewise_constant_fn = piecewise_constant([5, 15], [0.01, 0.005, 0.001])
```

```
In [129... # extra code - use a tf.keras.callbacks.LearningRateScheduler like earlier

n_epochs = 25

lr_scheduler = tf.keras.callbacks.LearningRateScheduler(piecewise_constant_fn)

model = build_model()
```

```
optimizer = tf.keras.optimizers.Nadam(learning_rate=lr0)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
              metrics=["accuracy"])
history = model.fit(X_train, y_train, epochs=n_epochs,
                   validation_data=(X_valid, y_valid),
                   callbacks=[lr_scheduler])
```

Epoch 1/25
1719/1719 ————— 4s 2ms/step - accuracy: 0.8087 - loss: 0.5422 - val_accuracy: 0.8310 - val_loss: 0.4606 - learning_rate: 0.0100

Epoch 2/25
1719/1719 ————— 4s 2ms/step - accuracy: 0.8435 - loss: 0.4459 - val_accuracy: 0.8410 - val_loss: 0.4544 - learning_rate: 0.0100

Epoch 3/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8495 - loss: 0.4311 - val_accuracy: 0.8482 - val_loss: 0.4392 - learning_rate: 0.0100

Epoch 4/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8554 - loss: 0.4177 - val_accuracy: 0.8456 - val_loss: 0.4173 - learning_rate: 0.0100

Epoch 5/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8577 - loss: 0.4057 - val_accuracy: 0.8434 - val_loss: 0.4337 - learning_rate: 0.0100

Epoch 6/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8739 - loss: 0.3524 - val_accuracy: 0.8486 - val_loss: 0.4032 - learning_rate: 0.0050

Epoch 7/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8770 - loss: 0.3461 - val_accuracy: 0.8538 - val_loss: 0.4024 - learning_rate: 0.0050

Epoch 8/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8794 - loss: 0.3375 - val_accuracy: 0.8424 - val_loss: 0.4326 - learning_rate: 0.0050

Epoch 9/25
1719/1719 ————— 4s 2ms/step - accuracy: 0.8792 - loss: 0.3343 - val_accuracy: 0.8474 - val_loss: 0.4241 - learning_rate: 0.0050

Epoch 10/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8832 - loss: 0.3309 - val_accuracy: 0.8518 - val_loss: 0.4073 - learning_rate: 0.0050

Epoch 11/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8834 - loss: 0.3254 - val_accuracy: 0.8558 - val_loss: 0.4219 - learning_rate: 0.0050

Epoch 12/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8840 - loss: 0.3264 - val_accuracy: 0.8388 - val_loss: 0.4504 - learning_rate: 0.0050

Epoch 13/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8853 - loss: 0.3226 - val_accuracy: 0.8430 - val_loss: 0.4380 - learning_rate: 0.0050

Epoch 14/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8872 - loss: 0.3146 - val_accuracy: 0.8504 - val_loss: 0.4425 - learning_rate: 0.0050

Epoch 15/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8883 - loss: 0.3148 - val_accuracy: 0.8560 - val_loss: 0.4001 - learning_rate: 0.0050

Epoch 16/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8982 - loss: 0.2789 - val_accuracy: 0.8744 - val_loss: 0.3713 - learning_rate: 0.0010

Epoch 17/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.9001 - loss: 0.2709 - val_accuracy: 0.8732 - val_loss: 0.3699 - learning_rate: 0.0010

Epoch 18/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.9015 - loss: 0.2663 - val_accuracy: 0.8728 - val_loss: 0.3772 - learning_rate: 0.0010

Epoch 19/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.9025 - loss: 0.2633 - val_accuracy: 0.8740 - val_loss: 0.3855 - learning_rate: 0.0010

Epoch 20/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.9037 - loss: 0.2607 - val_accuracy: 0.8748 - val_loss: 0.3953 - learning_rate: 0.0010

Epoch 21/25

1719/1719 ————— 3s 2ms/step - accuracy: 0.9045 - loss: 0.2575 - val_accuracy: 0.8728 - val_loss: 0.4059 - learning_rate: 0.0010

Epoch 22/25

1719/1719 ————— 3s 2ms/step - accuracy: 0.9047 - loss: 0.2554 - val_accuracy: 0.8720 - val_loss: 0.4179 - learning_rate: 0.0010

Epoch 23/25

1719/1719 ————— 3s 2ms/step - accuracy: 0.9059 - loss: 0.2529 - val_accuracy: 0.8722 - val_loss: 0.4285 - learning_rate: 0.0010

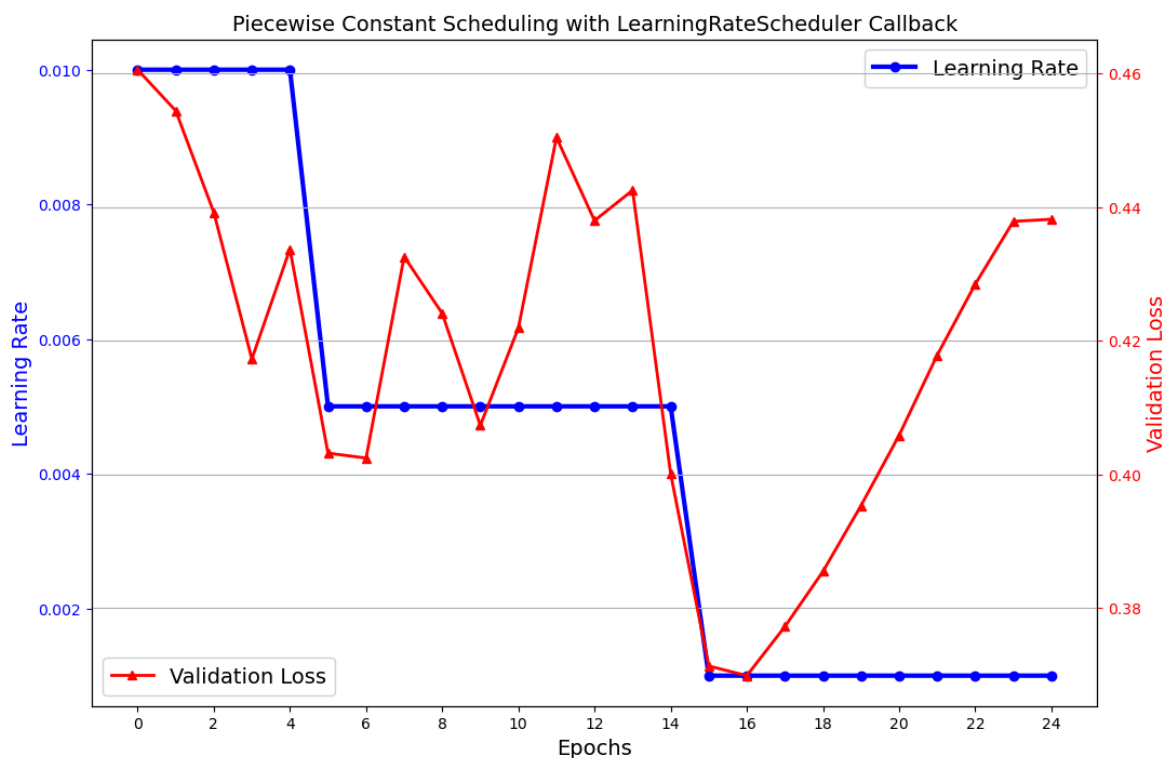
Epoch 24/25

1719/1719 ————— 3s 2ms/step - accuracy: 0.9069 - loss: 0.2508 - val_accuracy: 0.8726 - val_loss: 0.4379 - learning_rate: 0.0010

Epoch 25/25

1719/1719 ————— 3s 2ms/step - accuracy: 0.9073 - loss: 0.2495 - val_accuracy: 0.8746 - val_loss: 0.4382 - learning_rate: 0.0010

In [130... `plot_lr_and_loss_from_history(history, n_epochs, title="Piecewise Constant Sched`



We've looked at `InverseTimeDecay`, `ExponentialDecay`, and `PiecewiseConstantDecay`. A few more schedulers are available in `tf.keras.optimizers.schedules`, here is the full list:

In [131...

```
for name in sorted(dir(tf.keras.optimizers.schedules)):  
    if name[0] == name[0].lower(): # must start with capital letter  
        continue  
    scheduler_class = getattr(tf.keras.optimizers.schedules, name)  
    print(f"• {name} - {scheduler_class.__doc__.splitlines()[0]}")
```

- `CosineDecay` – A `LearningRateSchedule` that uses a cosine decay with optional warmup.
- `CosineDecayRestarts` – A `LearningRateSchedule` that uses a cosine decay schedule with restarts.
- `ExponentialDecay` – A `LearningRateSchedule` that uses an exponential decay schedule.
- `InverseTimeDecay` – A `LearningRateSchedule` that uses an inverse time decay schedule.
- `LearningRateSchedule` – The learning rate schedule base class.
- `PiecewiseConstantDecay` – A `LearningRateSchedule` that uses a piecewise constant decay schedule.
- `PolynomialDecay` – A `LearningRateSchedule` that uses a polynomial decay schedule.

Performance Scheduling (ReduceLROnPlateau)

Performance scheduling adjusts the learning rate based on the model's performance metrics, such as validation loss or accuracy. Unlike time-based schedules, it dynamically reduces the learning rate when a monitored metric stops improving, helping to fine-tune convergence and prevent overfitting. This is particularly useful for adaptive training where the optimal learning rate changes over time.

In [132...

```
# extra code - build and compile the model

model = build_model()
optimizer = tf.keras.optimizers.SGD(learning_rate=lr0)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
              metrics=["accuracy"])
```

In Keras, this is implemented using the `ReduceLROnPlateau` callback, which monitors a metric (e.g., validation loss) and reduces the learning rate by a factor when the metric hasn't improved for a specified number of epochs (patience). It can also stop training early if desired.

Key parameters:

- `monitor` : The metric to monitor (e.g., 'val_loss').
- `factor` : Factor by which the learning rate will be reduced (e.g., 0.5).
- `patience` : Number of epochs with no improvement after which learning rate will be reduced.
- `min_lr` : Minimum learning rate.
- `mode` : 'auto', 'min', or 'max' depending on whether the metric should be minimized or maximized.

This approach is robust for datasets where the optimal learning rate isn't known in advance and can lead to better generalization by allowing the model to settle into a minimum more gradually. Always combine with early stopping for efficiency.

In [133...

```
lr_scheduler = tf.keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.5,
                                                    patience=10,
                                                    min_lr=1e-6)
history = model.fit(X_train, y_train, epochs=n_epochs,
```

```
validation_data=(X_valid, y_valid),  
callbacks=[lr_scheduler])
```

Epoch 1/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.7671 - loss: 0.6767 - val_accuracy: 0.8340 - val_loss: 0.4850 - learning_rate: 0.0100

Epoch 2/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8335 - loss: 0.4718 - val_accuracy: 0.8410 - val_loss: 0.4524 - learning_rate: 0.0100

Epoch 3/25
1719/1719 ————— 3s 1ms/step - accuracy: 0.8496 - loss: 0.4249 - val_accuracy: 0.8504 - val_loss: 0.4248 - learning_rate: 0.0100

Epoch 4/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8577 - loss: 0.3961 - val_accuracy: 0.8526 - val_loss: 0.4150 - learning_rate: 0.0100

Epoch 5/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8647 - loss: 0.3760 - val_accuracy: 0.8538 - val_loss: 0.4067 - learning_rate: 0.0100

Epoch 6/25
1719/1719 ————— 3s 1ms/step - accuracy: 0.8698 - loss: 0.3603 - val_accuracy: 0.8582 - val_loss: 0.3909 - learning_rate: 0.0100

Epoch 7/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8745 - loss: 0.3467 - val_accuracy: 0.8608 - val_loss: 0.3864 - learning_rate: 0.0100

Epoch 8/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.8782 - loss: 0.3352 - val_accuracy: 0.8612 - val_loss: 0.3815 - learning_rate: 0.0100

Epoch 9/25
1719/1719 ————— 3s 1ms/step - accuracy: 0.8812 - loss: 0.3253 - val_accuracy: 0.8626 - val_loss: 0.3780 - learning_rate: 0.0100

Epoch 10/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8844 - loss: 0.3164 - val_accuracy: 0.8630 - val_loss: 0.3760 - learning_rate: 0.0100

Epoch 11/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.8876 - loss: 0.3083 - val_accuracy: 0.8634 - val_loss: 0.3717 - learning_rate: 0.0100

Epoch 12/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.8904 - loss: 0.3012 - val_accuracy: 0.8612 - val_loss: 0.3748 - learning_rate: 0.0100

Epoch 13/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.8926 - loss: 0.2943 - val_accuracy: 0.8642 - val_loss: 0.3716 - learning_rate: 0.0100

Epoch 14/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.8956 - loss: 0.2879 - val_accuracy: 0.8624 - val_loss: 0.3723 - learning_rate: 0.0100

Epoch 15/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.8979 - loss: 0.2819 - val_accuracy: 0.8648 - val_loss: 0.3728 - learning_rate: 0.0100

Epoch 16/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9003 - loss: 0.2763 - val_accuracy: 0.8618 - val_loss: 0.3763 - learning_rate: 0.0100

Epoch 17/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9057 - loss: 0.2596 - val_accuracy: 0.8690 - val_loss: 0.3482 - learning_rate: 0.0050

Epoch 18/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9072 - loss: 0.2559 - val_accuracy: 0.8700 - val_loss: 0.3490 - learning_rate: 0.0050

Epoch 19/25
1719/1719 ————— 3s 1ms/step - accuracy: 0.9087 - loss: 0.2530 - val_accuracy: 0.8692 - val_loss: 0.3502 - learning_rate: 0.0050

Epoch 20/25
1719/1719 ————— 3s 2ms/step - accuracy: 0.9097 - loss: 0.2501 - val_accuracy: 0.8696 - val_loss: 0.3510 - learning_rate: 0.0050

Epoch 21/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9108 - loss: 0.2473 - val_accuracy: 0.8716 - val_loss: 0.3509 - learning_rate: 0.0050
Epoch 22/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9117 - loss: 0.2447 - val_accuracy: 0.8720 - val_loss: 0.3520 - learning_rate: 0.0050
Epoch 23/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9149 - loss: 0.2371 - val_accuracy: 0.8780 - val_loss: 0.3338 - learning_rate: 0.0025
Epoch 24/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9161 - loss: 0.2351 - val_accuracy: 0.8774 - val_loss: 0.3343 - learning_rate: 0.0025
Epoch 25/25
1719/1719 ————— 2s 1ms/step - accuracy: 0.9164 - loss: 0.2335 - val_accuracy: 0.8776 - val_loss: 0.3343 - learning_rate: 0.0025

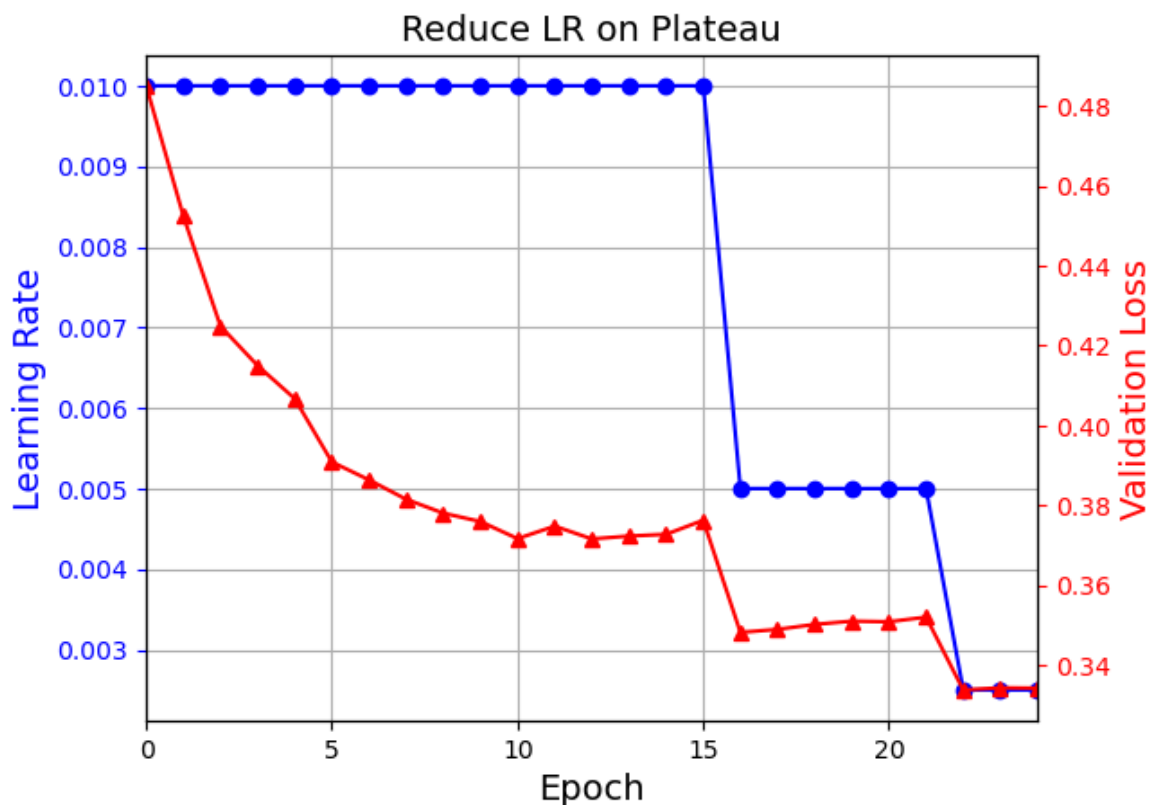
In [134...

```
# extra code - this cell plots performance scheduling

plt.plot(history.epoch, history.history["learning_rate"], "bo-")
plt.xlabel("Epoch")
plt.ylabel("Learning Rate", color='b')
plt.tick_params('y', colors='b')
plt.gca().set_xlim(0, n_epochs - 1)
plt.grid(True)

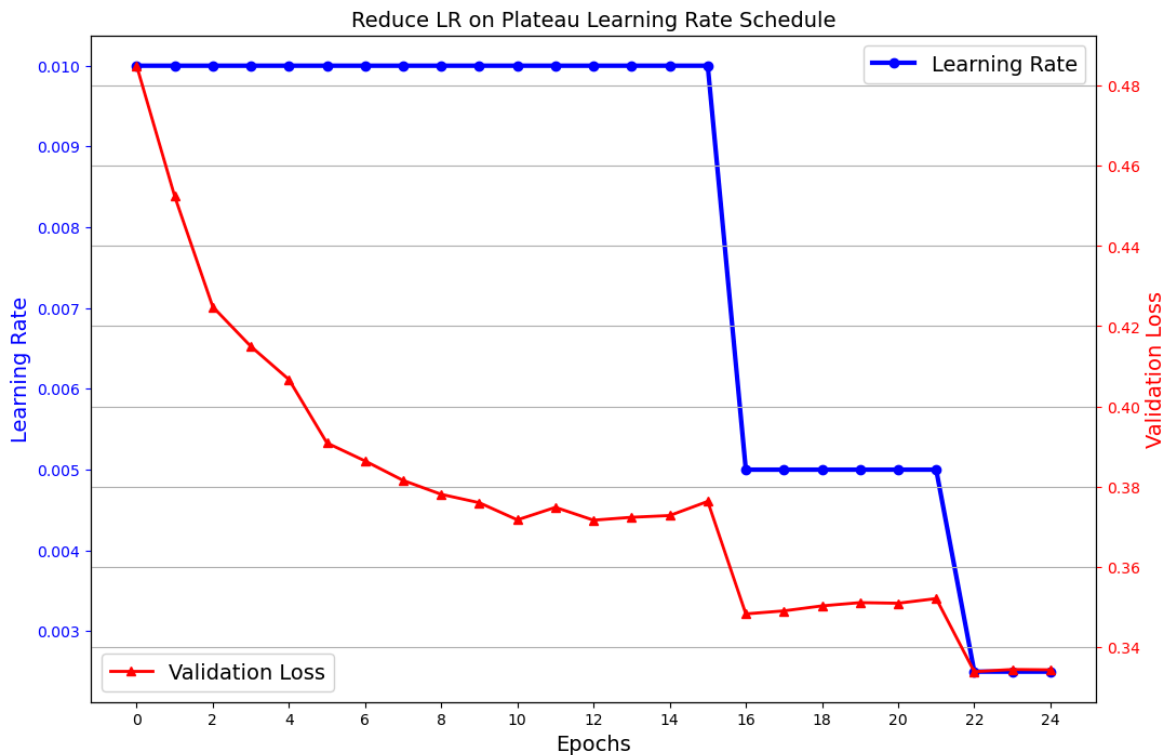
ax2 = plt.gca().twinx()
ax2.plot(history.epoch, history.history["val_loss"], "r^-")
ax2.set_ylabel('Validation Loss', color='r')
ax2.tick_params('y', colors='r')

plt.title("Reduce LR on Plateau", fontsize=14)
plt.show()
```



In [135...

```
plot_lr_and_loss_from_history(history, n_epochs, title="Reduce LR on Plateau Lea
```



This figure visualizes the effects of a **ReduceLRonPlateau** learning rate schedule during model training, correlating the learning rate (blue line) with the validation loss (red line) over epochs.

Key Insights:

1. Initial Learning and Plateau (Epochs 0-15):

- The learning rate starts at 0.01 and remains constant for the first 15 epochs.
- During this phase, the validation loss decreases significantly from approximately 0.48 to around 0.375, indicating that the model is learning effectively.
- Around epoch 10-15, the validation loss starts to flatten out, suggesting that the model's progress has slowed with the current learning rate.

2. First Learning Rate Reduction (Epoch 15):

- At epoch 15, the **ReduceLRonPlateau** callback detects that the validation loss has stopped improving (or improved insufficiently for a specified **patience** period).
- Consequently, the learning rate is sharply reduced from 0.01 to 0.005 (a factor of 0.5).
- This reduction reinvigorates training, as the validation loss continues to drop further, reaching a new low of approximately 0.35 around epoch 16-17.

3. Second Plateau and Reduction (Epochs 16-21):

- After the first reduction, the learning rate remains constant at 0.005.
- From epoch 16 to 21, the validation loss again stabilizes and shows minimal improvement, hovering around 0.35.

- This signals another plateau, and the `ReduceLROnPlateau` callback prepares for a second adjustment.

4. Second Learning Rate Reduction (Epoch 21-22):

- Between epoch 21 and 22, the learning rate is again halved, dropping from 0.005 to 0.0025.
- This second reduction proves beneficial, as the validation loss experiences another clear drop, reaching its lowest point in the plot at approximately 0.33.

5. Final Convergence (Epochs 22-24):

- With the learning rate at 0.0025, the validation loss stabilizes at its minimum observed value, indicating good convergence with the adjusted learning rate.

Conclusion:

The `ReduceLROnPlateau` callback effectively manages the learning rate, dynamically adjusting it downwards when the model's performance on the validation set plateaus. Each reduction successfully helps the model escape local plateaus and achieve lower validation loss, leading to better overall convergence. This demonstrates the power of adaptive learning rate scheduling in fine-tuning model training and improving generalization.

1Cycle scheduling

The `ExponentialLearningRate` custom callback updates the learning rate during training, at the end of each batch. It multiplies it by a constant `factor`. It also saves the learning rate and loss at each batch. Since `logs["loss"]` is actually the mean loss since the start of the epoch, and we want to save the batch loss instead, we must compute the mean times the number of batches since the beginning of the epoch to get the total loss so far, then we subtract the total loss at the previous batch to get the current batch's loss.

1. Finding the Optimal `max_lr` (`find_learning_rate` and `plot_lr_vs_loss`)

Before applying 1Cycle scheduling, it's common practice to perform a "learning rate range test" to identify an optimal `max_lr`.

`ExponentialLearningRate` Callback

This custom Keras callback is designed specifically for the LR range test:

- In its `on_batch_end` method, it records the current learning rate and the batch loss.
- It then exponentially increases the learning rate for the next batch.

- By continuously increasing the LR, it helps identify the "sweet spot" where the loss starts to drop rapidly but hasn't yet exploded.

In []: `K = tf.keras.backend`

```
class ExponentialLearningRate(tf.keras.callbacks.Callback):
    """A callback that exponentially increases the learning rate at each batch.

    Attributes:
        factor: The factor by which the learning rate is multiplied at each batch.
        rates: A list of learning rates recorded at each batch.
        losses: A list of losses recorded at each batch.
    """
    def __init__(self, factor):
        self.factor = factor # the factor by which the learning rate is multiplied
        self.rates = []
        self.losses = []

    def on_epoch_begin(self, epoch, logs=None):
        """ Resets the sum of epoch losses at the beginning of each epoch. """
        self.sum_of_epoch_losses = 0

    def on_batch_end(self, batch, logs=None):
        """ Updates the learning rate exponentially and records the learning rate.
        # Calculate the batch loss from the mean epoch loss and the sum of epoch losses
        mean_epoch_loss = logs["loss"] # the epoch's mean loss so far
        new_sum_of_epoch_losses = mean_epoch_loss * (batch + 1)
        batch_loss = new_sum_of_epoch_losses - self.sum_of_epoch_losses
        self.sum_of_epoch_losses = new_sum_of_epoch_losses

        # Get the current learning rate, record it along with the batch loss
        lr = self.model.optimizer.learning_rate.numpy()
        self.rates.append(lr)
        self.losses.append(batch_loss)

        # Update the learning rate by multiplying it with the factor.
        self.model.optimizer.learning_rate = lr * self.factor
```

In the context of Keras callbacks like `ExponentialLearningRate`, the `logs` parameter is a dictionary that contains various metrics and information available at the specific point in the training process when the callback method is invoked.

Here's a breakdown of what `logs` typically contains and its usage in the provided code:

1. What `logs` is:

- It's a dictionary passed by the Keras training loop to callback methods such as `on_epoch_begin`, `on_epoch_end`, `on_batch_begin`, and `on_batch_end`.
- The contents of this dictionary vary slightly depending on which method is called.

2. Contents of `logs` (common examples):

- For `on_batch_end` (as seen in your code):

- `logs["loss"]` : The current mean training loss for the epoch up to that batch. This is a running average.
- `logs["accuracy"]` (if applicable): The current mean training accuracy for the epoch up to that batch.
- Other metrics configured in `model.compile()`, such as `val_loss`, `val_accuracy`, etc., are typically present in `on_epoch_end` after validation has run.
- For `on_epoch_end` :
 - `logs["loss"]` : The average training loss for the entire epoch.
 - `logs["accuracy"]` (if applicable): The average training accuracy for the entire epoch.
 - `logs["val_loss"]` : The validation loss for the entire epoch.
 - `logs["val_accuracy"]` (if applicable): The validation accuracy for the entire epoch.

3. Usage in `ExponentialLearningRate`'s `on_batch_end` method:

- `mean_epoch_loss = logs["loss"]` : Here, the callback accesses the `loss` key from the `logs` dictionary. This `logs["loss"]` value represents the *mean loss of the current epoch up to the current batch*.
- The code then uses this `mean_epoch_loss` along with the current `batch` number to calculate the `batch_loss` (the loss for the specific batch that just finished), which is then recorded in `self.losses`.

In summary, `logs` serves as a vital communication channel from the Keras training system to your custom callbacks, providing real-time data about the model's performance and state, which you can then use to implement custom logic.

`find_learning_rate(model, X, y, ...)` Function

This function orchestrates the LR range test:

- It temporarily saves the model's initial weights and the optimizer's initial learning rate (`init_weights`, `init_lr`).
- It calculates a factor to exponentially increase the learning rate from `min_rate` to `max_rate` over the specified number of iterations.
- It sets the optimizer's learning rate to `min_rate`.
- It trains the model for a short period (e.g., 1 epoch) using the `ExponentialLearningRate` callback.
- Crucially, after the test, it resets the model's weights and the optimizer's learning rate to their initial states to avoid polluting the actual training run.
- It returns the recorded rates and losses. This process allows you to empirically determine a suitable `max_lr` for the 1Cycle policy, which can significantly enhance training speed and convergence.

```
In [1]: def find_learning_rate(model, X, y, epochs=1, batch_size=32, min_rate=1e-4,
                                max_rate=1):
        """Finds the optimal learning rate by exponentially increasing it and record
```

This function trains the model for a specified number of epochs and batch size while exponentially increasing the learning rate from `min\_rate` to `max\_rate`. It records the learning rates and corresponding losses at each batch, which can be used to analyze the relationship between learning rate and loss.

Args:

model: The Keras model to train.

X: Training features.

y: Training labels.

epochs: Number of epochs to train for the LR search. Defaults to 1.

batch_size: Batch size for training. Defaults to 32.

min_rate: The minimum learning rate to start the search. Defaults to 1e-5.

max_rate: The maximum learning rate to end the search. Defaults to 1.

Returns:

A tuple containing:

- rates (list): A list of learning rates used at each batch.
- losses (list): A list of losses recorded at each batch.

"""

Store the initial weights and learning rate

init_weights = model.get_weights()

init_lr = K.get_value(model.optimizer.learning_rate)

Calculate the total number of iterations (batches) for the given number of epochs

iterations = math.ceil(len(X) / batch_size) * epochs

Calculate the factor to exponentially increase the learning rate from min_rate to max_rate

factor = (max_rate / min_rate) ** (1 / iterations)

model.optimizer.learning_rate = min_rate

exp_lr = ExponentialLearningRate(factor)

Train the model for the specified number of epochs and batch size, using the exponential learning rate

history = model.fit(X, y, epochs=epochs, batch_size=batch_size, callbacks=[exp_lr])

Restore the model's initial weights and learning rate to ensure that this model can be used for other tasks

model.optimizer.learning_rate = init_lr

model.set_weights(init_weights)

Return the recorded learning rates and losses for each batch, which can be used to analyze the relationship between learning rate and loss

return exp_lr.rates, exp_lr.losses

plot_lr_vs_loss(rates, losses) Function

This function visualizes the results of the LR range test:

- It plots the learning rate (on a logarithmic scale) against the corresponding loss.
- The idea is to pick a `max_lr` that is slightly before the point where the loss starts to sharply increase, but after a point where the loss is clearly decreasing.
- The provided code output suggests `0.1` as a good `max_lr`.

```
In [ ]: def plot_lr_vs_loss(rates, losses):  
    """Plots the learning rates against the corresponding losses on a logarithmic scale.  
    plt.plot(rates, losses, "b")  
    plt.gca().set_xscale('log')  
    max_loss = losses[0] + min(losses)  
    plt.hlines(min(losses), min(rates), max(rates), color="k")  
    plt.axis((min(rates), max(rates), 0, max_loss))
```

```
plt.xlabel("Learning rate")
plt.ylabel("Loss")
plt.grid()
```

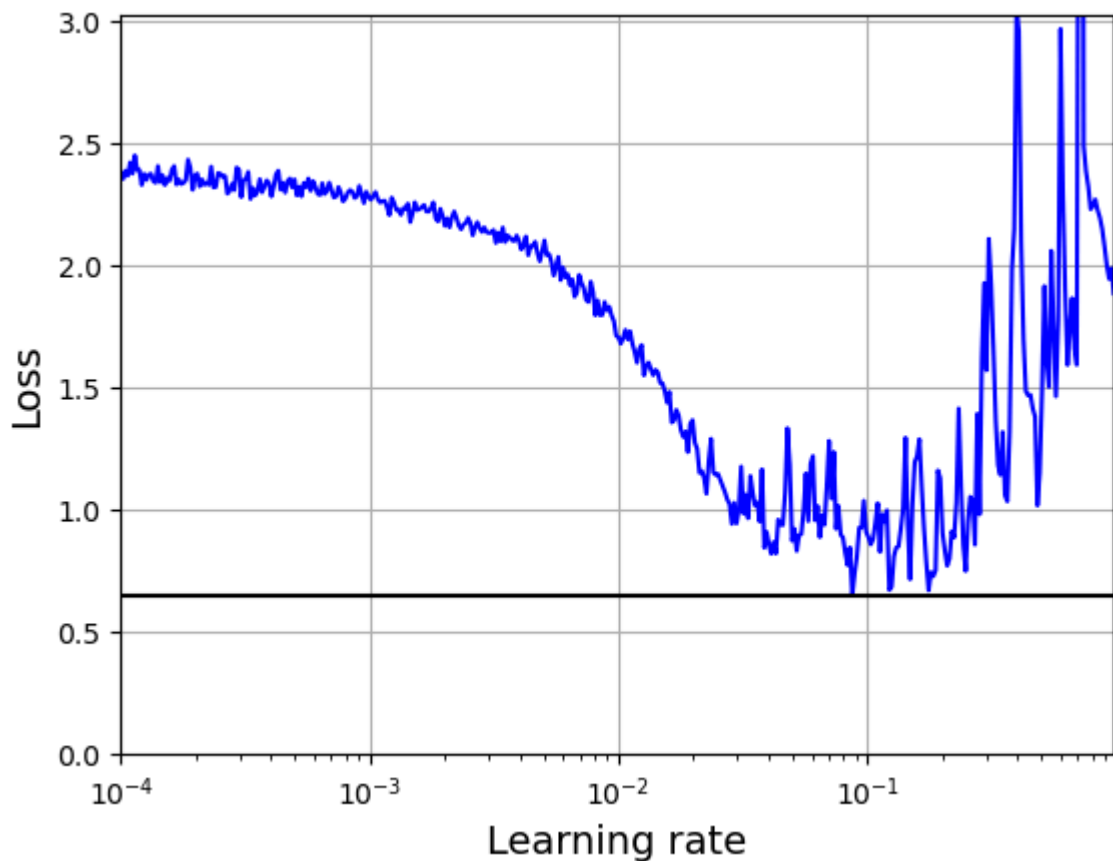
Let's build a simple Fashion MNIST model and compile it:

```
In [139... model = build_model()
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=tf.keras.optimizers.SGD(learning_rate=0.001),
              metrics=["accuracy"])
```

Now let's find the optimal max learning rate for 1 cycle:

```
In [140... batch_size = 128
rates, losses = find_learning_rate(model, X_train, y_train, epochs=1,
                                  batch_size=batch_size)
plot_lr_vs_loss(rates, losses)
```

430/430 ————— 1s 3ms/step - accuracy: 0.3687 - loss: 1.7745



Looks like the max learning rate to use for 1 cycle is around 10^{-1} .

The `OneCycleScheduler` custom callback updates the learning rate at the beginning of each batch. It applies the logic described in the book: increase the learning rate linearly during about half of training, then reduce it linearly back to the initial learning rate, and lastly reduce it down to close to zero linearly for the very last part of training.

2. The `OneCycleScheduler` Custom Callback

This is the core of the 1Cycle implementation. It's a `tf.keras.callbacks.Callback` that dynamically adjusts the learning rate at the end of each batch.

```
__init__(self, iterations, max_lr, ...):
```

- Initializes the scheduler with key hyperparameters: `iterations` (total number of batches for training), `max_lr`, `start_lr` (initial LR, typically `max_lr / 10`), `last_iterations` (number of batches for the final steep drop), and `last_lr` (final LR).
- Calculates `half_iteration`, which determines the split point for the initial increase and subsequent decrease phases.

```
_interpolate(self, iter1, iter2, lr1, lr2):
```

- A helper method that performs linear interpolation. It calculates the learning rate at the current `self.iteration` based on two reference iterations (`iter1`, `iter2`) and their corresponding learning rates (`lr1`, `lr2`).

```
on_batch_begin(self, batch, logs=None):
```

- This method is called at the beginning of every training batch.
- It checks `self.iteration` to determine which phase of the 1Cycle policy the training is currently in (increase, decrease, or final drop).
- It uses `_interpolate` to calculate the appropriate `lr` for the current batch.
- It then directly updates `self.model.optimizer.learning_rate = lr`, ensuring the learning rate is set correctly before the gradient update for the current batch.
- `self.iteration` is incremented to track the global step count.

```
on_epoch_end(self, epoch, logs=None):
```

- This method is called at the end of each epoch.
- It logs the current learning rate (which is the rate at the last batch of the epoch) into the `logs` dictionary, making it available in the `history.history` object for plotting.

In [141...

```
from typing import Any, Dict, Optional
```

```
class OneCycleScheduler(tf.keras.callbacks.Callback):
```

```
    """A Keras callback that implements the 1Cycle learning rate scheduling.
```

```
    It increases the learning rate linearly from a starting value to a maximum value during the first half of training, then decreases it linearly back to the starting value, and finally reduces it linearly to a very low value for the final iterations.
```

```
    Args:
```

```
        iterations (int): Total number of training steps (batches).
```

```
        max_lr (float): The maximum learning rate reached.
```

```
        start_lr (float, optional): The initial learning rate. Defaults to `max_lr / 10`.
```

```

last_iterations (int, optional): The number of steps for the final
    cooldown phase. Defaults to `iterations // 10 + 1`.
last_lr (float, optional): The final learning rate at the end of
    training. Defaults to `start_lr / 1000`.
"""
def __init__(self,
              iterations: int,
              max_lr: float = 1e-3,
              start_lr: float | None = None,
              last_iterations: int | None = None,
              last_lr: float | None = None) -> None:
    """
    Initializes the scheduler.

    Args:
        iterations: Total training steps.
        max_lr: Peak learning rate reached.
        start_lr: Initial learning rate (defaults to max_lr / 10).
        last_iterations: Steps for the final cooldown phase.
        last_lr: Final learning rate at training end.
    """
    super().__init__()
    self.iterations = iterations
    self.max_lr = max_lr
    self.start_lr = start_lr or max_lr / 10
    self.last_iterations = last_iterations or iterations // 10 + 1
    self.half_iteration = (iterations - self.last_iterations) // 2
    self.last_lr = last_lr or self.start_lr / 1000
    self.iteration = 0

def _interpolate(self, iter1: int, iter2: int, lr1: float, lr2: float) -> float:
    """Linear interpolation between two learning rates."""
    return (lr2 - lr1) * (self.iteration - iter1) / (iter2 - iter1) + lr1

def on_batch_begin(self, batch: int, logs: dict | None = None) -> None:
    """Updates the learning rate at the start of every batch."""
    if self.iteration < self.half_iteration:
        lr = self._interpolate(0, self.half_iteration, self.start_lr, self.max_lr)
    elif self.iteration < 2 * self.half_iteration:
        lr = self._interpolate(self.half_iteration, 2 * self.half_iteration,
                               self.max_lr, self.start_lr)
    else:
        lr = self._interpolate(2 * self.half_iteration, self.iterations,
                               self.start_lr, self.last_lr)

    self.iteration += 1
    self.model.optimizer.learning_rate = lr

def on_epoch_end(self, epoch: int, logs: dict | None = None) -> None:
    logs = logs or {}
    logs['lr'] = self.model.optimizer.learning_rate.numpy()

```

Core Concept of 1Cycle Scheduling

The 1Cycle policy, proposed by Leslie N. Smith, involves cyclically varying the learning rate between a minimum and a maximum value. Typically, it consists of three main phases:

- **Linear Increase:** The learning rate is gradually increased from a very low value (`start_lr`) to a relatively high value (`max_lr`) over the first $\approx 30 - 50\%$ of

training iterations. This initial warm-up helps the model quickly traverse flatter regions of the loss landscape.

- **Linear Decrease:** The learning rate is then linearly decreased from `max_lr` back to `start_lr` over the next $\approx 50 - 70\%$ of training. This helps the model settle into a good minimum.
- **Final Steep Drop (Optional but common):** For the very last few iterations, the learning rate is dropped significantly (e.g., to `start_lr / 1000`) to fine-tune the weights in the found minimum.

Let's build and compile a simple Fashion MNIST model, then train it using the `OneCycleScheduler` callback:

In [142...

```
model = build_model()
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=tf.keras.optimizers.SGD(),
              metrics=["accuracy"])
n_epochs = 25
onecycle = OneCycleScheduler(math.ceil(len(X_train) / batch_size) * n_epochs,
                             max_lr=0.1)
history = model.fit(X_train, y_train, epochs=n_epochs, batch_size=batch_size,
                   validation_data=(X_valid, y_valid),
                   callbacks=[onecycle])
```

Epoch 1/25
430/430 ————— 1s 3ms/step - accuracy: 0.6785 - loss: 0.9624 - val_
accuracy: 0.8002 - val_loss: 0.5881 - lr: 0.0180

Epoch 2/25
430/430 ————— 1s 2ms/step - accuracy: 0.8048 - loss: 0.5634 - val_
accuracy: 0.8236 - val_loss: 0.5053 - lr: 0.0260

Epoch 3/25
430/430 ————— 1s 2ms/step - accuracy: 0.8295 - loss: 0.4887 - val_
accuracy: 0.8270 - val_loss: 0.4816 - lr: 0.0340

Epoch 4/25
430/430 ————— 1s 2ms/step - accuracy: 0.8421 - loss: 0.4473 - val_
accuracy: 0.8366 - val_loss: 0.4539 - lr: 0.0420

Epoch 5/25
430/430 ————— 1s 2ms/step - accuracy: 0.8502 - loss: 0.4187 - val_
accuracy: 0.8378 - val_loss: 0.4495 - lr: 0.0500

Epoch 6/25
430/430 ————— 1s 2ms/step - accuracy: 0.8571 - loss: 0.3967 - val_
accuracy: 0.8436 - val_loss: 0.4308 - lr: 0.0580

Epoch 7/25
430/430 ————— 1s 2ms/step - accuracy: 0.8638 - loss: 0.3778 - val_
accuracy: 0.8482 - val_loss: 0.4189 - lr: 0.0660

Epoch 8/25
430/430 ————— 1s 2ms/step - accuracy: 0.8694 - loss: 0.3623 - val_
accuracy: 0.8428 - val_loss: 0.4397 - lr: 0.0740

Epoch 9/25
430/430 ————— 1s 2ms/step - accuracy: 0.8728 - loss: 0.3494 - val_
accuracy: 0.8332 - val_loss: 0.4572 - lr: 0.0820

Epoch 10/25
430/430 ————— 1s 3ms/step - accuracy: 0.8779 - loss: 0.3362 - val_
accuracy: 0.8374 - val_loss: 0.4471 - lr: 0.0900

Epoch 11/25
430/430 ————— 1s 2ms/step - accuracy: 0.8806 - loss: 0.3253 - val_
accuracy: 0.8466 - val_loss: 0.4278 - lr: 0.0980

Epoch 12/25
430/430 ————— 1s 2ms/step - accuracy: 0.8856 - loss: 0.3120 - val_
accuracy: 0.8486 - val_loss: 0.4172 - lr: 0.0940

Epoch 13/25
430/430 ————— 1s 2ms/step - accuracy: 0.8929 - loss: 0.2934 - val_
accuracy: 0.8656 - val_loss: 0.3838 - lr: 0.0860

Epoch 14/25
430/430 ————— 1s 2ms/step - accuracy: 0.8984 - loss: 0.2776 - val_
accuracy: 0.8742 - val_loss: 0.3623 - lr: 0.0780

Epoch 15/25
430/430 ————— 1s 2ms/step - accuracy: 0.9027 - loss: 0.2655 - val_
accuracy: 0.8760 - val_loss: 0.3543 - lr: 0.0700

Epoch 16/25
430/430 ————— 1s 2ms/step - accuracy: 0.9062 - loss: 0.2545 - val_
accuracy: 0.8778 - val_loss: 0.3433 - lr: 0.0620

Epoch 17/25
430/430 ————— 1s 2ms/step - accuracy: 0.9104 - loss: 0.2444 - val_
accuracy: 0.8806 - val_loss: 0.3402 - lr: 0.0540

Epoch 18/25
430/430 ————— 1s 2ms/step - accuracy: 0.9138 - loss: 0.2355 - val_
accuracy: 0.8840 - val_loss: 0.3326 - lr: 0.0460

Epoch 19/25
430/430 ————— 1s 2ms/step - accuracy: 0.9167 - loss: 0.2276 - val_
accuracy: 0.8842 - val_loss: 0.3268 - lr: 0.0380

Epoch 20/25
430/430 ————— 1s 2ms/step - accuracy: 0.9195 - loss: 0.2207 - val_
accuracy: 0.8860 - val_loss: 0.3253 - lr: 0.0300

```

Epoch 21/25
430/430 ————— 1s 2ms/step - accuracy: 0.9218 - loss: 0.2147 - val_
accuracy: 0.8846 - val_loss: 0.3221 - lr: 0.0220
Epoch 22/25
430/430 ————— 1s 2ms/step - accuracy: 0.9239 - loss: 0.2098 - val_
accuracy: 0.8856 - val_loss: 0.3178 - lr: 0.0140
Epoch 23/25
430/430 ————— 1s 2ms/step - accuracy: 0.9254 - loss: 0.2060 - val_
accuracy: 0.8864 - val_loss: 0.3129 - lr: 0.0080
Epoch 24/25
430/430 ————— 1s 2ms/step - accuracy: 0.9266 - loss: 0.2036 - val_
accuracy: 0.8886 - val_loss: 0.3095 - lr: 0.0040
Epoch 25/25
430/430 ————— 1s 2ms/step - accuracy: 0.9268 - loss: 0.2018 - val_
accuracy: 0.8882 - val_loss: 0.3089 - lr: 1.9284e-05

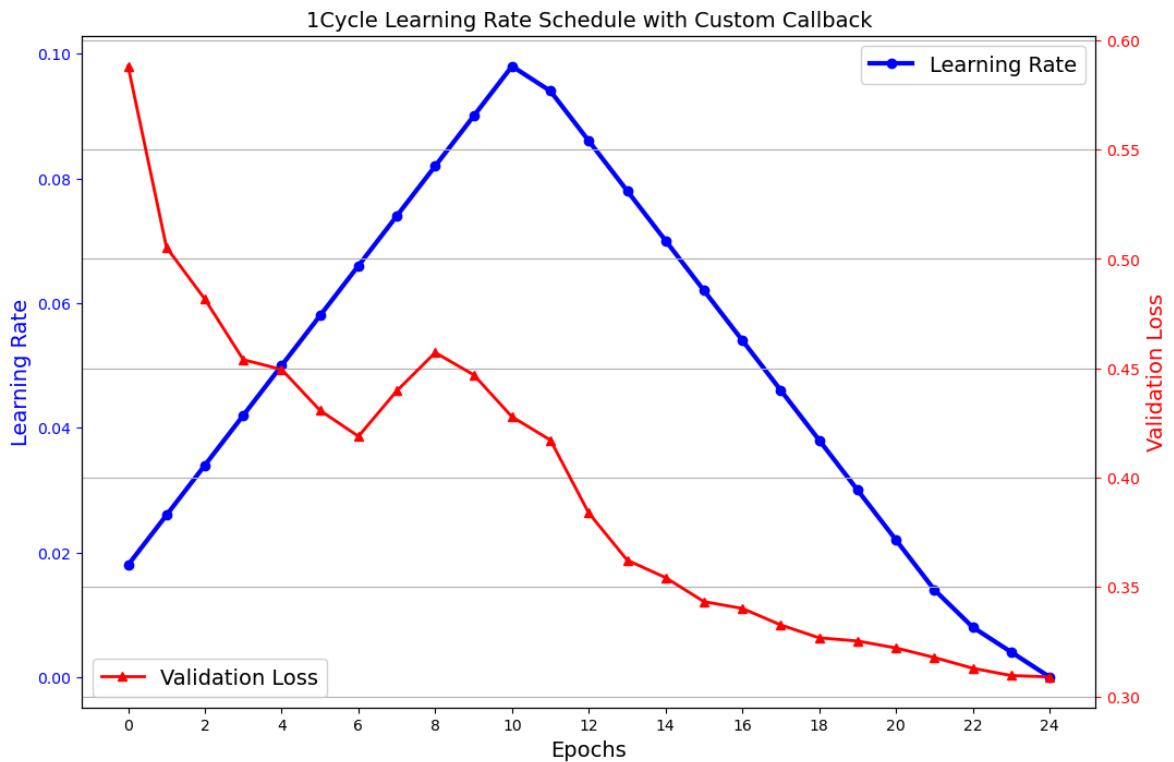
```

3. Training with `OneCycleScheduler`

The final part combines all these elements:

- A model is built and compiled, typically using an `SGD` optimizer. The initial learning rate for the optimizer itself might not be explicitly set, as the `OneCycleScheduler` will manage it dynamically.
- `n_iterations` is calculated, representing the total number of batches across all epochs for the entire training run.
- An instance of `OneCycleScheduler` is created, passing the calculated `n_iterations` and the chosen `max_lr` (e.g., `0.05`, typically determined from the LR range test).
- The `model.fit()` method is called with `callbacks=[onecycle]`, ensuring that the `on_batch_begin` method of `OneCycleScheduler` is invoked before every batch, dynamically updating the learning rate.

```
In [143... plot_lr_and_loss_from_history(history, n_epochs, title="1Cycle Learning Rate Sch
```

The 1Cycle policy is an advanced technique that often leads to "super-convergence."

- Phase 1: Increase LR linearly from a low value to a very high max_lr (about half the training time).
- Phase 2: Decrease LR linearly back to the starting value.
- Phase 3: Drop LR close to zero for the final few iterations.

Why it works: The high learning rate in the middle acts as a regularizer, helping the model find flatter, more stable minima that generalize better.

Why 1Cycle Works and its Benefits

The 1Cycle policy is an advanced training strategy that often leads to "super-convergence," characterized by faster training and improved generalization. Here's why it's effective:

- **Faster Convergence ("Super-convergence"):** The initial phase, where learning rates gradually increase, enables the optimizer to aggressively explore the loss landscape. This helps in quickly escaping saddle points or poor local minima and allows for larger steps during the middle of training, significantly accelerating convergence.
- **Better Generalization:** High learning rates during training act as a form of implicit regularization. They prevent the model from settling into sharp, sub-optimal minima and encourage it to find flatter minima. Flatter minima often correlate with better generalization to unseen data, leading to more robust models.
- **Reduced Hyperparameter Tuning:** The 1Cycle policy simplifies the learning rate tuning process. Instead of meticulously tuning a fixed learning rate or a

complex decay schedule, it primarily requires tuning only the `max_lr` (and `start_lr` if not using the default).

- **Implicit Regularization:** The cyclical nature of the learning rate provides an inherent regularization effect. By constantly varying the learning rate, it implicitly pushes the model out of purely static parameter configurations, similar to other regularization techniques like weight decay.

In the provided example, the 1Cycle policy demonstrates its effectiveness by dramatically speeding up training (e.g., 15 epochs at 2 seconds each compared to 17 epochs at 15 seconds each for a Batch Normalization model) and achieving superior validation accuracy (e.g., 52.0% versus 50.7%).

In summary, we have explored several **Learning Rate Scheduling** techniques, ranging from simple time-based decays to sophisticated adaptive and cyclical strategies:

- **Fixed Schedules:** Power, exponential, and piecewise constant scheduling provide a predictable way to decrease the learning rate as training progresses. In Keras, these are best implemented using `tf.keras.optimizers.schedules`.
- **Adaptive Scheduling:** The `ReduceLROnPlateau` callback allows the optimizer to react to the model's actual performance, reducing the step size only when the validation loss stops improving.
- **1Cycle Scheduling:** This aggressive approach often leads to the fastest convergence by ramping the learning rate up and then down, helping the model find optimal regions of the loss landscape more efficiently.

Choosing the right schedule often depends on the complexity of your dataset and the specific optimizer used. While **1Cycle** is a powerful modern default for speed, simpler **Exponential** or **Performance-based** schedules remain robust choices for many deep learning tasks.

In TensorFlow/Keras, you can control the learning rate (η) either by passing a value/schedule object directly to the **optimizer** or by using a **callback** during the `model.fit()` process.

Comparison Overview

Feature	Optimizer Schedule (<code>tf.keras.optimizers.schedules</code>)	Training Callback (<code>tf.keras.callbacks</code>)
Update Frequency	Usually every step (batch).	Usually every epoch (unless custom).
Logic	Predefined mathematical function of time.	Can be reactive (based on metrics).
Serialization	Saved automatically with the model.	Must be re-defined/attached during training.

Feature	Optimizer Schedule (<code>tf.keras.optimizers.schedules</code>)	Training Callback (<code>tf.keras.callbacks</code>)
Flexibility	Limited to the provided schedule classes.	Highly flexible (Python functions/logic).
Visibility	Hidden inside the optimizer; requires calling the schedule to see current LR.	Explicitly updates <code>optimizer.learning_rate</code> , visible in logs.

1. Learning Rate in the Optimizer

When you pass a schedule object to the optimizer, the learning rate is calculated as a function of the global step t . This is internal to the optimizer's update logic.

Example: Exponential Decay The formula for exponential decay is:

$$\eta_t = \eta_0 \cdot \text{decay_rate}^{t/s}$$

```
exponential_lr_schedule =
tf.keras.optimizers.schedules.ExponentialDecay(
    initial_learning_rate=0.01,
    decay_steps=20_000,
    decay_rate=0.1,
    staircase=False
)
optimizer =
tf.keras.optimizers.SGD(learning_rate=exponential_lr_schedule)
```

2. Learning Rate via Callbacks

Callbacks operate outside the optimizer's internal step-counter. They intercept the training process at specific hooks (e.g., `on_epoch_begin`) to manually overwrite `optimizer.learning_rate`.

Static/Functional Scheduling (`LearningRateScheduler`)

This is used for schedules that depend solely on the epoch index.

```
def exponential_decay_fn(epoch):
    return 0.01 * 0.1 ** (epoch / 20)

lr_scheduler =
tf.keras.callbacks.LearningRateScheduler(exponential_decay_fn)
model.fit(X_train, y_train, epochs=n_epochs, callbacks=[lr_scheduler])
```

Dynamic/Reactive Scheduling (`ReduceLROnPlateau`)

This is a "performance schedule." It does not follow a fixed time-based rule but reacts to the validation loss.

```
lr_scheduler = tf.keras.callbacks.ReduceLROnPlateau(factor=0.5,
patience=5)
model.fit(X_train, y_train, epochs=n_epochs, callbacks=[lr_scheduler])
```

Key "Gotchas"

- **Step vs. Epoch:** Optimizer schedules (like `InverseTimeDecay`) update every single batch, leading to a smooth curve. `LearningRateScheduler` typically updates only once per epoch, creating a "staircase" effect even if the function is continuous.
- **State Visibility:** If you use an optimizer schedule, the learning rate variable in the optimizer is a `LearningRateSchedule` object, not a float. To see the current value, you must call it with the current step:

```
optimizer.learning_rate(optimizer.iterations) .
current_lr = optimizer.learning_rate(optimizer.iterations)
print(f"Current learning rate: {current_lr.numpy()}")
```
- **Priority:** If both are used, the callback will manually overwrite whatever value the optimizer was using at the start of the epoch.

Example Demonstrating Optimizer Schedule with Real-Time Learning Rate Access

This example demonstrates how to implement an **Optimizer Schedule** and programmatically access the "hidden" learning rate value using the `optimizer.learning_rate(optimizer.iterations)` pattern.

I have used a tiny synthetic dataset and a custom callback so you can see the learning rate changing in real-time at every batch.

```
In [144... from math import log

import tensorflow as tf
import numpy as np

# 1. Define an Exponential Decay schedule
# Starts at 0.1, drops by 10% every 100 steps
lr_schedule = tf.keras.optimizers.schedules.ExponentialDecay(
    initial_learning_rate=0.1,
    decay_steps=50,
    decay_rate=0.9
)

# 2. Attach the schedule to an optimizer
optimizer = tf.keras.optimizers.SGD(learning_rate=lr_schedule)

# 3. Create a simple model and dummy data
model = tf.keras.Sequential([
    tf.keras.layers.Input(shape=(10,)),
    tf.keras.layers.Dense(1)
])
model.compile(loss="mse", optimizer=optimizer)

X = np.random.random((1000, 10))
y = np.random.random((1000, 1))

# 4. Custom Callback to "Reveal" and Store the state-dependent Learning rate
class LRVisibilityCallback(tf.keras.callbacks.Callback):
    def __init__(self):
        super().__init__()
        self.rates = [] # Added to store batch-level data
```

```

        self.losses = []          # Loss at every batch

    def on_batch_end(self, batch, logs=None):
        logs = logs or {}
        iteration = self.model.optimizer.iterations
        lr_attr = self.model.optimizer.learning_rate

        if callable(lr_attr):
            current_lr = lr_attr(iteration)
        else:
            current_lr = lr_attr

        current_lr_float = tf.keras.backend.get_value(current_lr)
        self.rates.append(current_lr_float) # Save to our internal list
        self.losses.append(logs["loss"])    # Save the loss at every batch
        if iteration.numpy() % 10 == 0:
            print(f"\n[Step {iteration.numpy()}] Effective Learning Rate: {current_lr_float}")

# 5. Execute training
print("Starting training with schedule visibility...")
lr_config = LRVisibilityCallback()
history = model.fit(X, y, epochs=20, batch_size=32, callbacks=[lr_config], verbose=1)

```

Starting training with schedule visibility...

[Step 10] Effective Learning Rate: 0.097915
[Step 20] Effective Learning Rate: 0.095873
[Step 30] Effective Learning Rate: 0.093874
[Step 40] Effective Learning Rate: 0.091917
[Step 50] Effective Learning Rate: 0.090000
[Step 60] Effective Learning Rate: 0.088123
[Step 70] Effective Learning Rate: 0.086286
[Step 80] Effective Learning Rate: 0.084487
[Step 90] Effective Learning Rate: 0.082725
[Step 100] Effective Learning Rate: 0.081000
[Step 110] Effective Learning Rate: 0.079311
[Step 120] Effective Learning Rate: 0.077657
[Step 130] Effective Learning Rate: 0.076038
[Step 140] Effective Learning Rate: 0.074452
[Step 150] Effective Learning Rate: 0.072900
[Step 160] Effective Learning Rate: 0.071380
[Step 170] Effective Learning Rate: 0.069892
[Step 180] Effective Learning Rate: 0.068434
[Step 190] Effective Learning Rate: 0.067007
[Step 200] Effective Learning Rate: 0.065610
[Step 210] Effective Learning Rate: 0.064242
[Step 220] Effective Learning Rate: 0.062902
[Step 230] Effective Learning Rate: 0.061591
[Step 240] Effective Learning Rate: 0.060306
[Step 250] Effective Learning Rate: 0.059049
[Step 260] Effective Learning Rate: 0.057818
[Step 270] Effective Learning Rate: 0.056612
[Step 280] Effective Learning Rate: 0.055432
[Step 290] Effective Learning Rate: 0.054276

[Step 300] Effective Learning Rate: 0.053144
[Step 310] Effective Learning Rate: 0.052036
[Step 320] Effective Learning Rate: 0.050951
[Step 330] Effective Learning Rate: 0.049889
[Step 340] Effective Learning Rate: 0.048848
[Step 350] Effective Learning Rate: 0.047830
[Step 360] Effective Learning Rate: 0.046832
[Step 370] Effective Learning Rate: 0.045856
[Step 380] Effective Learning Rate: 0.044900
[Step 390] Effective Learning Rate: 0.043963
[Step 400] Effective Learning Rate: 0.043047
[Step 410] Effective Learning Rate: 0.042149
[Step 420] Effective Learning Rate: 0.041270
[Step 430] Effective Learning Rate: 0.040410
[Step 440] Effective Learning Rate: 0.039567
[Step 450] Effective Learning Rate: 0.038742
[Step 460] Effective Learning Rate: 0.037934
[Step 470] Effective Learning Rate: 0.037143
[Step 480] Effective Learning Rate: 0.036369
[Step 490] Effective Learning Rate: 0.035610
[Step 500] Effective Learning Rate: 0.034868
[Step 510] Effective Learning Rate: 0.034141
[Step 520] Effective Learning Rate: 0.033429
[Step 530] Effective Learning Rate: 0.032732
[Step 540] Effective Learning Rate: 0.032049
[Step 550] Effective Learning Rate: 0.031381
[Step 560] Effective Learning Rate: 0.030727
[Step 570] Effective Learning Rate: 0.030086
[Step 580] Effective Learning Rate: 0.029459
[Step 590] Effective Learning Rate: 0.028844

[Step 600] Effective Learning Rate: 0.028243

[Step 610] Effective Learning Rate: 0.027654

[Step 620] Effective Learning Rate: 0.027077

[Step 630] Effective Learning Rate: 0.026513

[Step 640] Effective Learning Rate: 0.025960

Why this works:

- **Global Step Tracking:** The code accesses `optimizer.iterations`, which serves as a global counter that increments automatically every time a batch is processed and weights are updated.
- **Polymorphic Attribute Handling:** The code checks `if callable(lr_attr)`. This is crucial because `optimizer.learning_rate` can either be a static float or a `LearningRateSchedule` object. If it is a schedule (callable), the code invokes it using the current iteration to compute the decaying value.
- **Just-in-Time Calculation:** By passing the iterations into the schedule object, you are asking the mathematical formula: *"Based on the decay rule and the fact that we are at step X, what is the exact step size right now?"*
- **Backend Extraction:** Instead of just using `.numpy()`, the code uses `tf.keras.backend.get_value()`. This is a more robust way to extract the scalar numeric value across different TensorFlow execution modes (Eager vs. Graph) and Keras versions.

In [145...

```
# visualize the learning rate schedule using the list stored in the callback
import matplotlib.pyplot as plt

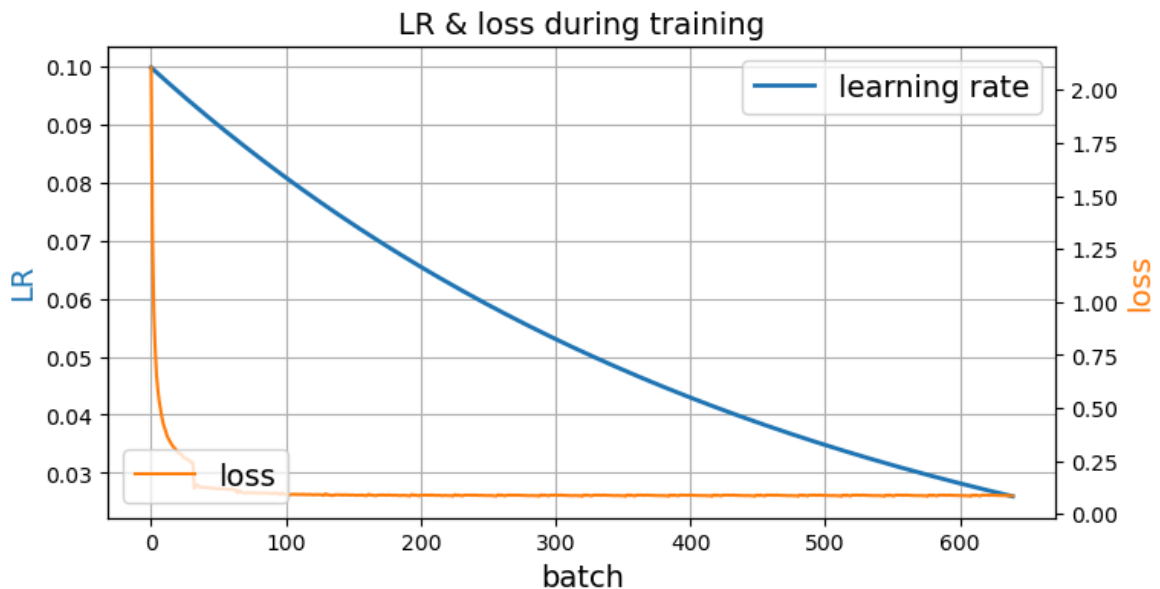
fig, ax1 = plt.subplots(figsize=(8,4))
ax2 = ax1.twinx()

ax1.plot(lr_config.rates, color="C0", label="learning rate", linewidth=2)
ax2.plot(lr_config.losses, color="C1", label="loss")

ax1.set_xlabel("batch")
ax1.set_ylabel("LR", color="C0")
ax2.set_ylabel("loss", color="C1")

ax1.grid(True)
ax1.legend(loc="upper right")
ax2.legend(loc="lower left")

plt.title("LR & loss during training")
plt.show()
```

Summary Recommendation

1. **For Prototyping:** Start with `Adam` and a constant LR.
2. **For Refinement:** Use `SGD` with Momentum and Exponential Decay or `ReduceLROnPlateau`.
3. **For Speed:** Implement the `1Cycle` policy if you have a fixed budget of epochs.

Avoiding Overfitting Through Regularization

Regularization reduces a model's tendency to memorize the training set so it generalizes better to new data. This section summarizes the most effective techniques, when to use them, and a compact troubleshooting recipe you can apply to any model.

Why regularize?

- Overfitting arises when model capacity, training time, or label/noise characteristics let the model fit idiosyncrasies of the training set rather than the underlying signal.
- Regularization trades some training accuracy for lower generalization error.

Common techniques (brief)

- Data: more labeled data, smarter data augmentation (strongest and often cheapest payoff).
- Optimization/early stopping: shorter training / monitor validation loss, `ReduceLROnPlateau`, weight-averaging.
- Weight regularization: L2 (weight decay), L1 (sparsity).
- Architectural: dropout, batch normalization, smaller networks.

- Label/target: label smoothing, mixup, noise injection.
- Ensembling and distillation: expensive but very effective for final models.
- Pruning and quantization: for deployment (reduces capacity / overfitting in some cases).

Practical recipe

1. Verify overfitting with learning curves (train vs val).
2. Fix data issues first (more data, better augmentation, cleaner labels).
3. Add/strengthen weight decay (start $1e-4 \rightarrow 1e-2$) and/or dropout (0.1–0.5) and retune LR.
4. Use early stopping + save best weights; try ReduceLROnPlateau.
5. If needed, simplify the model or use ensembling/distillation for production.
6. Run small ablations (one change at a time) and multiple seeds.

What follows

- Concrete Keras examples: L2 weight decay, dropout, label smoothing, augmentation pipelines, and learning-curve diagnostics — plus recommended hyperparameter ranges and ablation experiments to quantify impact.

ℓ_1 and ℓ_2 regularization

This code creates a dense layer with 100 neurons using the ReLU activation function and L2 weight regularization (weight decay). The

`kernel_regularizer=tf.keras.regularizers.l2(0.01)` argument applies L2 regularization with a factor of 0.01 to the layer's weights, helping to prevent overfitting by penalizing large weights during training.

```
In [99]: layer = tf.keras.layers.Dense(100, activation="relu",
                                         kernel_initializer="he_normal",
                                         kernel_regularizer=tf.keras.regularizers.l2(0.01))
```

Or use `l1(0.1)` for ℓ_1 regularization with a factor of 0.1, or `l1_l2(0.1, 0.01)` for both ℓ_1 and ℓ_2 regularization, with factors 0.1 and 0.01 respectively.

```
In [100... tf.random.set_seed(42) # extra code - for reproducibility
```

```
In [101... from functools import partial

RegularizedDense = partial(tf.keras.layers.Dense,
                           activation="relu",
                           kernel_initializer="he_normal",
                           kernel_regularizer=tf.keras.regularizers.l2(0.01))

model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=[28, 28]),
    RegularizedDense(100),
    RegularizedDense(100),
```

```
RegularizedDense(10, activation="softmax")
])
```

In [102...

```
# extra code - compile and train the model
optimizer = tf.keras.optimizers.SGD(learning_rate=0.02)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
              metrics=["accuracy"])
history = model.fit(X_train, y_train, epochs=2,
                   validation_data=(X_valid, y_valid))
```

Epoch 1/2

1719/1719 ————— 3s 1ms/step - accuracy: 0.7737 - loss: 3.1318 - val_accuracy: 0.8248 - val_loss: 1.8654

Epoch 2/2

1719/1719 ————— 2s 1ms/step - accuracy: 0.8135 - loss: 1.4314 - val_accuracy: 0.8220 - val_loss: 1.1217

Dropout

Dropout randomly zeroes activations during training to prevent co-adaptation of neurons — effectively performing model-averaging and reducing overfitting.

- When to use: large fully-connected layers or when validation gap indicates overfitting; less useful if you already have strong augmentation, large datasets, or heavy weight decay.
- Typical rates: dense layers 0.3–0.5; convolutional layers 0.1–0.3; start small and sweep.
- Placement: usually after the activation of a layer (e.g., Dense→Activation→Dropout). Avoid pairing aggressively with BatchNorm (BatchNorm often reduces the need for dropout).
- Training vs inference: dropout is active only during training; Keras uses inverted dropout so no extra scaling is needed at test time.
- Tuning tips: tune dropout together with learning rate and weight decay; more dropout often requires longer training or a slightly higher LR.
- Debugging: if training accuracy collapses, lower the rate or increase model capacity; if validation is noisy, try larger batch size, stronger augmentation, or ReduceLROnPlateau.
- Advanced: use Monte-Carlo (MC) Dropout for uncertainty estimates by keeping dropout active at inference (run multiple stochastic forward passes and average).

Example (Keras):

```
tf.keras.layers.Dense(100, activation="relu"),
tf.keras.layers.Dropout(0.5)
```

Quick MC-Dropout inference pattern:

```
# run `model(x, training=True)` several times and compute mean/std of outputs
```

```
In [103... tf.random.set_seed(42) # extra code - for reproducibility
```

```
In [104... model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=[28, 28]),
    tf.keras.layers.Dropout(rate=0.2),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dropout(rate=0.2),
    tf.keras.layers.Dense(100, activation="relu",
                           kernel_initializer="he_normal"),
    tf.keras.layers.Dropout(rate=0.2),
    tf.keras.layers.Dense(10, activation="softmax")
])
```

```
In [105... # extra code - compile and train the model
optimizer = tf.keras.optimizers.SGD(learning_rate=0.01, momentum=0.9)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
              metrics=["accuracy"])
history = model.fit(X_train, y_train, epochs=10,
                   validation_data=(X_valid, y_valid))
```

Epoch 1/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.7511 - loss: 0.6787 - val_accuracy: 0.8298 - val_loss: 0.4513

Epoch 2/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8121 - loss: 0.5178 - val_accuracy: 0.8440 - val_loss: 0.4141

Epoch 3/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8254 - loss: 0.4772 - val_accuracy: 0.8512 - val_loss: 0.3939

Epoch 4/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8345 - loss: 0.4502 - val_accuracy: 0.8610 - val_loss: 0.3813

Epoch 5/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8411 - loss: 0.4332 - val_accuracy: 0.8532 - val_loss: 0.3777

Epoch 6/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8436 - loss: 0.4230 - val_accuracy: 0.8576 - val_loss: 0.3636

Epoch 7/10

1719/1719 ————— 4s 2ms/step - accuracy: 0.8487 - loss: 0.4106 - val_accuracy: 0.8640 - val_loss: 0.3685

Epoch 8/10

1719/1719 ————— 4s 2ms/step - accuracy: 0.8531 - loss: 0.4017 - val_accuracy: 0.8624 - val_loss: 0.3565

Epoch 9/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8542 - loss: 0.3977 - val_accuracy: 0.8592 - val_loss: 0.3655

Epoch 10/10

1719/1719 ————— 4s 2ms/step - accuracy: 0.8569 - loss: 0.3896 - val_accuracy: 0.8660 - val_loss: 0.3520

The training accuracy looks like it's lower than the validation accuracy, but that's just because dropout is only active during training. If we evaluate the model on the training set after training (i.e., with dropout turned off), we get the "real" training accuracy, which is very slightly higher than the validation accuracy and the test accuracy:

```
In [106... model.evaluate(X_train, y_train)
```

```
1719/1719 ————— 2s 1ms/step - accuracy: 0.8810 - loss: 0.3127
```

```
Out[106... [0.31272581219673157, 0.8810363411903381]
```

```
In [107... model.evaluate(X_test, y_test)
```

```
313/313 ————— 0s 1ms/step - accuracy: 0.8571 - loss: 0.3739
```

```
Out[107... [0.37386199831962585, 0.8571000099182129]
```

Note: make sure to use `AlphaDropout` instead of `Dropout` if you want to build a self-normalizing neural net using SELU.

MC Dropout

MC Dropout (Monte Carlo Dropout) is a technique used to estimate uncertainty in neural network predictions by applying dropout during both training and inference. This allows the model to generate multiple stochastic forward passes, providing a distribution of outputs that can be analyzed for uncertainty quantification.

To implement MC Dropout in Keras, you can create a custom dropout layer that remains active during inference. Here's an example of how to do this:

```
In [108... tf.random.set_seed(42) # extra code - for reproducibility
```

```
In [109... y_probas = np.stack([model(X_test, training=True)
                      for sample in range(100)])
y_proba = y_probas.mean(axis=0)
```

```
In [110... model.predict(X_test[:1]).round(3)
```

```
1/1 ————— 0s 63ms/step
```

```
Out[110... array([[0.    , 0.    , 0.    , 0.    , 0.    , 0.045, 0.    , 0.13 , 0.    ,
         0.825]], dtype=float32)
```

```
In [111... y_proba[0].round(3)
```

```
Out[111... array([0.001, 0.    , 0.    , 0.    , 0.001, 0.158, 0.001, 0.152, 0.002,
         0.684], dtype=float32)
```

```
In [112... y_std = y_probas.std(axis=0)
y_std[0].round(3)
```

```
Out[112... array([0.003, 0.002, 0.002, 0.001, 0.003, 0.202, 0.005, 0.154, 0.015,
         0.236], dtype=float32)
```

```
In [113... y_pred = y_proba.argmax(axis=1)
accuracy = (y_pred == y_test).sum() / len(y_test)
accuracy
```

```
Out[113... 0.8568
```

```
In [114... class MCDropout(tf.keras.layers.Dropout):
    def call(self, inputs, training=None):
```

```
return super().call(inputs, training=True)
```

```
In [115... # extra code - shows how to convert Dropout to MCDropout in a Sequential model
Dropout = tf.keras.layers.Dropout
mc_model = tf.keras.Sequential([
    MCDropout(layer.rate) if isinstance(layer, Dropout) else layer
    for layer in model.layers
])
mc_model.set_weights(model.get_weights())
```

```
In [116... mc_model.summary()
```

Model: "sequential_25"

Layer (type)	Output Shape	
flatten_22 (Flatten)	(None, 784)	
mc_dropout (MCDropout)	(None, 784)	
dense_89 (Dense)	(None, 100)	
mc_dropout_1 (MCDropout)	(None, 100)	
dense_90 (Dense)	(None, 100)	
mc_dropout_2 (MCDropout)	(None, 100)	
dense_91 (Dense)	(None, 10)	

Total params: 89,610 (350.04 KB)

Trainable params: 89,610 (350.04 KB)

Non-trainable params: 0 (0.00 B)

Now we can use the model with MC Dropout:

```
In [117... # extra code - shows that the model works without retraining
tf.random.set_seed(42)
np.mean([mc_model.predict(X_test[:1])
        for sample in range(100)], axis=0).round(2)
```

1/1	0s	93ms/step
1/1	0s	23ms/step
1/1	0s	22ms/step
1/1	0s	22ms/step
1/1	0s	23ms/step
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```

1/1 ————— 0s 23ms/step
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1/1 ————— 0s 31ms/step
1/1 ————— 0s 24ms/step
1/1 ————— 0s 21ms/step
1/1 ————— 0s 23ms/step

```

```

Out[117...] array([[0. , 0. , 0. , 0. , 0. , 0.13, 0. , 0.16, 0. , 0.71]],
dtype=float32)

```

Max norm

Max norm regularization constrains the maximum norm of the weights in each layer to a specified value, preventing them from growing too large and helping to reduce overfitting. This technique can improve generalization by keeping the model's capacity in check.

```

In [118...] dense = tf.keras.layers.Dense(
    100, activation="relu", kernel_initializer="he_normal",
    kernel_constraint=tf.keras.constraints.max_norm(1.))

```


In [119...

```
# extra code - shows how to apply max norm to every hidden layer in a model

MaxNormDense = partial(tf.keras.layers.Dense,
                        activation="relu", kernel_initializer="he_normal",
                        kernel_constraint=tf.keras.constraints.max_norm(1.))

tf.random.set_seed(42)
model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=[28, 28]),
    MaxNormDense(100),
    MaxNormDense(100),
    tf.keras.layers.Dense(10, activation="softmax")
])
optimizer = tf.keras.optimizers.SGD(learning_rate=0.01, momentum=0.9)
model.compile(loss="sparse_categorical_crossentropy", optimizer=optimizer,
              metrics=["accuracy"])
history = model.fit(X_train, y_train, epochs=10,
                    validation_data=(X_valid, y_valid))
```

Epoch 1/10

c:\Users\yuanh\miniconda3\envs\hom13\lib\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(**kwargs)
```

1719/1719 ————— 3s 1ms/step - accuracy: 0.8015 - loss: 0.5539 - val_accuracy: 0.8294 - val_loss: 0.4500

Epoch 2/10

1719/1719 ————— 2s 1ms/step - accuracy: 0.8496 - loss: 0.4121 - val_accuracy: 0.8438 - val_loss: 0.4080

Epoch 3/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8609 - loss: 0.3785 - val_accuracy: 0.8590 - val_loss: 0.3906

Epoch 4/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8672 - loss: 0.3590 - val_accuracy: 0.8568 - val_loss: 0.3846

Epoch 5/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8712 - loss: 0.3462 - val_accuracy: 0.8576 - val_loss: 0.3960

Epoch 6/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8726 - loss: 0.3380 - val_accuracy: 0.8480 - val_loss: 0.4090

Epoch 7/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8758 - loss: 0.3318 - val_accuracy: 0.8608 - val_loss: 0.3861

Epoch 8/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8783 - loss: 0.3280 - val_accuracy: 0.8560 - val_loss: 0.4003

Epoch 9/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8784 - loss: 0.3249 - val_accuracy: 0.8498 - val_loss: 0.3898

Epoch 10/10

1719/1719 ————— 3s 2ms/step - accuracy: 0.8796 - loss: 0.3198 - val_accuracy: 0.8468 - val_loss: 0.4131

Exercises

1. to 7.

1. Glorot initialization and He initialization were designed to make the output standard deviation as close as possible to the input standard deviation, at least at the beginning of training. This reduces the vanishing/exploding gradients problem.
2. No, all weights should be sampled independently; they should not all have the same initial value. One important goal of sampling weights randomly is to break symmetry: if all the weights have the same initial value, even if that value is not zero, then symmetry is not broken (i.e., all neurons in a given layer are equivalent), and backpropagation will be unable to break it. Concretely, this means that all the neurons in any given layer will always have the same weights. It's like having just one neuron per layer, and much slower. It is virtually impossible for such a configuration to converge to a good solution.
3. It is perfectly fine to initialize the bias terms to zero. Some people like to initialize them just like weights, and that's OK too; it does not make much difference.
4. ReLU is usually a good default for the hidden layers, as it is fast and yields good results. Its ability to output precisely zero can also be useful in some cases (e.g., see Chapter 17). Moreover, it can sometimes benefit from optimized implementations as well as from hardware acceleration. The leaky ReLU variants of ReLU can improve the model's quality without hindering its speed too much compared to ReLU. For large neural nets and more complex problems, GLU, Swish and Mish can give you a slightly higher quality model, but they have a computational cost. The hyperbolic tangent ($\tanh$) can be useful in the output layer if you need to output a number in a fixed range (by default between -1 and 1), but nowadays it is not used much in hidden layers, except in recurrent nets. The sigmoid activation function is also useful in the output layer when you need to estimate a probability (e.g., for binary classification), but it is rarely used in hidden layers (there are exceptions—for example, for the coding layer of variational autoencoders; see Chapter 17). The softplus activation function is useful in the output layer when you need to ensure that the output will always be positive. The softmax activation function is useful in the output layer to estimate probabilities for mutually exclusive classes, but it is rarely (if ever) used in hidden layers.
5. If you set the `momentum` hyperparameter too close to 1 (e.g., 0.99999) when using an `SGD` optimizer, then the algorithm will likely pick up a lot of speed, hopefully moving roughly toward the global minimum, but its momentum will carry it right past the minimum. Then it will slow down and come back, accelerate again, overshoot again, and so on. It may oscillate this way many times before converging, so overall it will take much longer to converge than with a smaller `momentum` value.

6. One way to produce a sparse model (i.e., with most weights equal to zero) is to train the model normally, then zero out tiny weights. For more sparsity, you can apply ℓ_1 regularization during training, which pushes the optimizer toward sparsity. A third option is to use the TensorFlow Model Optimization Toolkit.
 7. Yes, dropout does slow down training, in general roughly by a factor of two. However, it has no impact on inference speed since it is only turned on during training. MC Dropout is exactly like dropout during training, but it is still active during inference, so each inference is slowed down slightly. More importantly, when using MC Dropout you generally want to run inference 10 times or more to get better predictions. This means that making predictions is slowed down by a factor of 10 or more.
-

8. Deep Learning on CIFAR10

In this exercise, you will build and train a deep neural network on the CIFAR10 dataset (60,000 32×32 color images in 10 classes) using TensorFlow/Keras. You will start with a baseline model and progressively apply techniques covered in this chapter to improve generalization and prevent overfitting, such as data augmentation, weight regularization (L2, dropout, max norm), and advanced optimizers and learning rate scheduling. The workflow includes loading and preprocessing the data, defining and compiling the model, training with callbacks (e.g., early stopping, learning rate schedulers), and evaluating performance on the test set. Experiment with hyperparameter tuning and regularization strategies to achieve strong test accuracy, aiming for at least 80%.

a.

Exercise: Build a DNN with 20 hidden layers of 100 neurons each (that's too many, but it's the point of this exercise). Use He initialization and the Swish activation function.

In [117...

```
tf.random.set_seed(42)

model = tf.keras.Sequential()
model.add(tf.keras.layers.Flatten(input_shape=[32, 32, 3]))
for _ in range(20):
    model.add(tf.keras.layers.Dense(100,
                                     activation="swish",
                                     kernel_initializer="he_normal"))
```

b.

Exercise: Using Nadam optimization and early stopping, train the network on the CIFAR10 dataset. You can load it with `tf.keras.datasets.cifar10.load_data()`. The dataset is composed of 60,000 32×32 -pixel color images (50,000 for training, 10,000 for testing) with 10 classes, so you'll need a softmax output layer with 10

neurons. Remember to search for the right learning rate each time you change the model's architecture or hyperparameters.

Let's add the output layer to the model:

```
In [118... model.add(tf.keras.layers.Dense(10, activation="softmax"))
```

Let's use a Nadam optimizer with a learning rate of 5e-5. I tried learning rates 1e-5, 3e-5, 1e-4, 3e-4, 1e-3, 3e-3 and 1e-2, and I compared their learning curves for 10 epochs each (using the TensorBoard callback, below). The learning rates 3e-5 and 1e-4 were pretty good, so I tried 5e-5, which turned out to be slightly better.

```
In [119... optimizer = tf.keras.optimizers.Nadam(learning_rate=5e-5)
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=optimizer,
              metrics=["accuracy"])
```

Let's load the CIFAR10 dataset. We also want to use early stopping, so we need a validation set. Let's use the first 5,000 images of the original training set as the validation set:

```
In [120... cifar10 = tf.keras.datasets.cifar10.load_data()
(X_train_full, y_train_full), (X_test, y_test) = cifar10

X_train = X_train_full[5000:]
y_train = y_train_full[5000:]
X_valid = X_train_full[:5000]
y_valid = y_train_full[:5000]
```

Now we can create the callbacks we need and train the model:

```
In [121... early_stopping_cb = tf.keras.callbacks.EarlyStopping(patience=20,
                                                       restore_best_weights=True)
model_checkpoint_cb = tf.keras.callbacks.ModelCheckpoint("my_cifar10_model.keras",
                                                         save_best_only=True)

run_index = 1 # increment every time you train the model
run_logdir = Path() / "my_cifar10_logs" / f"run_{run_index:03d}"
tensorboard_cb = tf.keras.callbacks.TensorBoard(run_logdir)
callbacks = [early_stopping_cb, model_checkpoint_cb, tensorboard_cb]
```

```
In [122... %load_ext tensorboard
%tensorboard --logdir=./my_cifar10_logs
```

C:\ 索引

名稱	大小	修改日期
\$AV_ASW/		2022/5/12 晚上9:44
\$Recycle.Bin/		2023/4/22 下午5:36
avast! sandbox/		2025/12/19 下午2:46
Config.Msi/		2026/3/10 上午8:54
Documents and Settings/		2022/4/4 清晨6:20
ETAX/		2025/7/7 下午6:06
hp/		2022/3/4 下午4:43
inetpub/		2025/4/9 晚上9:57
Intel/		2026/3/5 上午8:57
KNSHCrossPlatform/		2024/10/31 晚上7:54
Log/		2025/6/6 晚上7:27
OneDriveTemp/		2024/5/5 下午1:18
path/		2025/7/14 下午3:28
PerfLogs/		2024/4/1 下午3:26
Program Files/		2026/3/10 上午8:54
Program Files (x86)/		2025/10/27 上午11:40
ProgramData/		2026/2/28 中午12:44
Python313/		2025/4/9 晚上8:29
Recovery/		2025/1/15 晚上7:34
swigwin-4.3.1/		2025/8/11 凌晨12:29
SWSetup/		2025/11/9 下午6:17
System Volume Information/		2026/3/9 下午2:57
System.sav/		2022/3/5 上午9:14
Users/		2024/12/18 上午10:44
Voiceover/		2024/11/20 下午4:44
Windows/		2026/3/9 上午11:06
workspace/		2025/5/17 下午6:13
.rnd	1.0 kB	2023/5/10 晚上11:19
DumpStack.log	12.0 kB	2025/8/10 中午12:19

In [123...

```
model.fit(X_train, y_train, epochs=100,  
          validation_data=(X_valid, y_valid),  
          callbacks=callbacks)
```

Epoch 1/100
1404/1407 [=====>.] - ETA: 0s - loss: 4.0493 - accuracy: 0.1598INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 17s 10ms/step - loss: 4.0462 - accuracy: 0.1597 - val_loss: 2.1441 - val_accuracy: 0.2036
Epoch 2/100
1407/1407 [=====] - ETA: 0s - loss: 2.0667 - accuracy: 0.2320INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 12s 9ms/step - loss: 2.0667 - accuracy: 0.2320 - val_loss: 2.0134 - val_accuracy: 0.2472
Epoch 3/100
1407/1407 [=====] - ETA: 0s - loss: 1.9472 - accuracy: 0.2819INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.9472 - accuracy: 0.2819 - val_loss: 1.9427 - val_accuracy: 0.2796
Epoch 4/100
1405/1407 [=====>.] - ETA: 0s - loss: 1.8636 - accuracy: 0.3182INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.8637 - accuracy: 0.3182 - val_loss: 1.8934 - val_accuracy: 0.3222
Epoch 5/100
1402/1407 [=====>.] - ETA: 0s - loss: 1.7975 - accuracy: 0.3464INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.7974 - accuracy: 0.3465 - val_loss: 1.8389 - val_accuracy: 0.3284
Epoch 6/100
1407/1407 [=====] - 9s 7ms/step - loss: 1.7446 - accuracy: 0.3664 - val_loss: 2.0006 - val_accuracy: 0.3030
Epoch 7/100
1407/1407 [=====] - ETA: 0s - loss: 1.6974 - accuracy: 0.3852INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 12s 8ms/step - loss: 1.6974 - accuracy: 0.3852 - val_loss: 1.7075 - val_accuracy: 0.3738
Epoch 8/100
1405/1407 [=====>.] - ETA: 0s - loss: 1.6605 - accuracy: 0.3984INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 12s 8ms/step - loss: 1.6604 - accuracy: 0.3984 - val_loss: 1.6788 - val_accuracy: 0.3836
Epoch 9/100
1405/1407 [=====>.] - ETA: 0s - loss: 1.6322 - accuracy: 0.4114INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.6321 - accuracy: 0.4114 - val_loss: 1.6477 - val_accuracy: 0.4014
Epoch 10/100
1407/1407 [=====] - 12s 8ms/step - loss: 1.6065 - accuracy: 0.4205 - val_loss: 1.6623 - val_accuracy: 0.3980
Epoch 11/100
1401/1407 [=====>.] - ETA: 0s - loss: 1.5843 - accuracy: 0.4287INFO:tensorflow:Assets written to: my_cifar10_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.5845 - accuracy: 0.4285 - val_loss: 1.6032 - val_accuracy: 0.4198
Epoch 12/100
1407/1407 [=====] - 9s 6ms/step - loss: 1.5634 - accuracy: 0.4367 - val_loss: 1.6063 - val_accuracy: 0.4258
Epoch 13/100
1401/1407 [=====>.] - ETA: 0s - loss: 1.5443 - accuracy: 0.4420INFO:tensorflow:Assets written to: my_cifar10_model/assets
<<47 more lines>>
1407/1407 [=====] - 12s 8ms/step - loss: 1.3247 - accuracy: 0.5256 - val_loss: 1.5130 - val_accuracy: 0.4616

```

Epoch 33/100
1407/1407 [=====] - 13s 9ms/step - loss: 1.3164 - accuracy: 0.5286 - val_loss: 1.5284 - val_accuracy: 0.4686
Epoch 34/100
1407/1407 [=====] - 12s 9ms/step - loss: 1.3091 - accuracy: 0.5303 - val_loss: 1.5208 - val_accuracy: 0.4682
Epoch 35/100
1407/1407 [=====] - 12s 8ms/step - loss: 1.3026 - accuracy: 0.5319 - val_loss: 1.5479 - val_accuracy: 0.4604
Epoch 36/100
1407/1407 [=====] - 11s 8ms/step - loss: 1.2930 - accuracy: 0.5378 - val_loss: 1.5443 - val_accuracy: 0.4580
Epoch 37/100
1407/1407 [=====] - 12s 8ms/step - loss: 1.2833 - accuracy: 0.5406 - val_loss: 1.5165 - val_accuracy: 0.4710
Epoch 38/100
1407/1407 [=====] - 11s 8ms/step - loss: 1.2763 - accuracy: 0.5433 - val_loss: 1.5345 - val_accuracy: 0.4672
Epoch 39/100
1407/1407 [=====] - 12s 9ms/step - loss: 1.2687 - accuracy: 0.5437 - val_loss: 1.5162 - val_accuracy: 0.4712
Epoch 40/100
1407/1407 [=====] - 11s 7ms/step - loss: 1.2623 - accuracy: 0.5490 - val_loss: 1.5717 - val_accuracy: 0.4566
Epoch 41/100
1407/1407 [=====] - 14s 10ms/step - loss: 1.2580 - accuracy: 0.5467 - val_loss: 1.5296 - val_accuracy: 0.4738
Epoch 42/100
1407/1407 [=====] - 13s 9ms/step - loss: 1.2469 - accuracy: 0.5532 - val_loss: 1.5179 - val_accuracy: 0.4690
Epoch 43/100
1407/1407 [=====] - 11s 8ms/step - loss: 1.2404 - accuracy: 0.5542 - val_loss: 1.5542 - val_accuracy: 0.4566
Epoch 44/100
1407/1407 [=====] - 12s 8ms/step - loss: 1.2292 - accuracy: 0.5605 - val_loss: 1.5536 - val_accuracy: 0.4608
Epoch 45/100
1407/1407 [=====] - 12s 9ms/step - loss: 1.2276 - accuracy: 0.5606 - val_loss: 1.5522 - val_accuracy: 0.4624
Epoch 46/100
1407/1407 [=====] - 13s 9ms/step - loss: 1.2200 - accuracy: 0.5637 - val_loss: 1.5339 - val_accuracy: 0.4794
Epoch 47/100
1407/1407 [=====] - 13s 9ms/step - loss: 1.2080 - accuracy: 0.5677 - val_loss: 1.5451 - val_accuracy: 0.4688
Epoch 48/100
1407/1407 [=====] - 15s 10ms/step - loss: 1.2050 - accuracy: 0.5675 - val_loss: 1.5209 - val_accuracy: 0.4770
Epoch 49/100
1407/1407 [=====] - 10s 7ms/step - loss: 1.1947 - accuracy: 0.5718 - val_loss: 1.5435 - val_accuracy: 0.4736

```

Out[123... <keras.callbacks.History at 0x7fb9f02fc070>

In [124... `model.evaluate(X_valid, y_valid)`

```

157/157 [=====] - 0s 2ms/step - loss: 1.5062 - accuracy: 0.4676

```

Out[124... [1.5061508417129517, 0.4675999879837036]

The model with the lowest validation loss gets about 46.8% accuracy on the validation set. It took 29 epochs to reach the lowest validation loss, with roughly 10 seconds per epoch on my laptop (without a GPU). Let's see if we can improve the model using Batch Normalization.

C.

Exercise: Now try adding Batch Normalization and compare the learning curves: Is it converging faster than before? Does it produce a better model? How does it affect training speed?

The code below is very similar to the code above, with a few changes:

- I added a BN layer after every Dense layer (before the activation function), except for the output layer.
- I changed the learning rate to 5e-4. I experimented with 1e-5, 3e-5, 5e-5, 1e-4, 3e-4, 5e-4, 1e-3 and 3e-3, and I chose the one with the best validation performance after 20 epochs.
- I renamed the run directories to run_bn_* and the model file name to `my_cifar10_bn_model`.

In [125...

```
tf.random.set_seed(42)

model = tf.keras.Sequential()
model.add(tf.keras.layers.Flatten(input_shape=[32, 32, 3]))
for _ in range(20):
    model.add(tf.keras.layers.Dense(100, kernel_initializer="he_normal"))
    model.add(tf.keras.layers.BatchNormalization())
    model.add(tf.keras.layers.Activation("swish"))

model.add(tf.keras.layers.Dense(10, activation="softmax"))

optimizer = tf.keras.optimizers.Nadam(learning_rate=5e-4)
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=optimizer,
              metrics=["accuracy"])

early_stopping_cb = tf.keras.callbacks.EarlyStopping(patience=20,
                                                    restore_best_weights=True)
model_checkpoint_cb = tf.keras.callbacks.ModelCheckpoint("my_cifar10_bn_model.keras",
                                                         save_best_only=True)

run_index = 1 # increment every time you train the model
run_logdir = Path() / "my_cifar10_logs" / f"run_bn_{run_index:03d}"
tensorboard_cb = tf.keras.callbacks.TensorBoard(run_logdir)
callbacks = [early_stopping_cb, model_checkpoint_cb, tensorboard_cb]

model.fit(X_train, y_train, epochs=100,
        validation_data=(X_valid, y_valid),
        callbacks=callbacks)

model.evaluate(X_valid, y_valid)
```


Epoch 1/100
1403/1407 [=====>.] - ETA: 0s - loss: 2.0377 - accuracy: 0.2523INFO:tensorflow:Assets written to: my_cifar10_bn_model/assets
1407/1407 [=====] - 32s 18ms/step - loss: 2.0374 - accuracy: 0.2525 - val_loss: 1.8766 - val_accuracy: 0.3154
Epoch 2/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.7874 - accuracy: 0.3542 - val_loss: 1.8784 - val_accuracy: 0.3268
Epoch 3/100
1407/1407 [=====] - 20s 15ms/step - loss: 1.6806 - accuracy: 0.3969 - val_loss: 1.9764 - val_accuracy: 0.3252
Epoch 4/100
1403/1407 [=====>.] - ETA: 0s - loss: 1.6111 - accuracy: 0.4229INFO:tensorflow:Assets written to: my_cifar10_bn_model/assets
1407/1407 [=====] - 24s 17ms/step - loss: 1.6112 - accuracy: 0.4228 - val_loss: 1.7087 - val_accuracy: 0.3750
Epoch 5/100
1402/1407 [=====>.] - ETA: 0s - loss: 1.5520 - accuracy: 0.4478INFO:tensorflow:Assets written to: my_cifar10_bn_model/assets
1407/1407 [=====] - 21s 15ms/step - loss: 1.5521 - accuracy: 0.4476 - val_loss: 1.6272 - val_accuracy: 0.4176
Epoch 6/100
1406/1407 [=====>.] - ETA: 0s - loss: 1.5030 - accuracy: 0.4659INFO:tensorflow:Assets written to: my_cifar10_bn_model/assets
1407/1407 [=====] - 23s 16ms/step - loss: 1.5030 - accuracy: 0.4660 - val_loss: 1.5401 - val_accuracy: 0.4452
Epoch 7/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.4559 - accuracy: 0.4812 - val_loss: 1.6990 - val_accuracy: 0.3952
Epoch 8/100
1403/1407 [=====>.] - ETA: 0s - loss: 1.4169 - accuracy: 0.4987INFO:tensorflow:Assets written to: my_cifar10_bn_model/assets
1407/1407 [=====] - 21s 15ms/step - loss: 1.4168 - accuracy: 0.4987 - val_loss: 1.5078 - val_accuracy: 0.4652
Epoch 9/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.3863 - accuracy: 0.5123 - val_loss: 1.5513 - val_accuracy: 0.4470
Epoch 10/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.3514 - accuracy: 0.5216 - val_loss: 1.5208 - val_accuracy: 0.4562
Epoch 11/100
1407/1407 [=====] - 16s 12ms/step - loss: 1.3220 - accuracy: 0.5314 - val_loss: 1.7301 - val_accuracy: 0.4206
Epoch 12/100
1404/1407 [=====>.] - ETA: 0s - loss: 1.2933 - accuracy: 0.5410INFO:tensorflow:Assets written to: my_cifar10_bn_model/assets
1407/1407 [=====] - 25s 18ms/step - loss: 1.2931 - accuracy: 0.5410 - val_loss: 1.4909 - val_accuracy: 0.4734
Epoch 13/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.2702 - accuracy: 0.5490 - val_loss: 1.5256 - val_accuracy: 0.4636
Epoch 14/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.2424 - accuracy: 0.5591 - val_loss: 1.5569 - val_accuracy: 0.4624
Epoch 15/100
<<12 more lines>>
Epoch 21/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.1174 - accuracy: 0.6066 - val_loss: 1.5241 - val_accuracy: 0.4828
Epoch 22/100

```

1407/1407 [=====] - 18s 13ms/step - loss: 1.0978 - accur
acy: 0.6128 - val_loss: 1.5313 - val_accuracy: 0.4772
Epoch 23/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.0844 - accur
acy: 0.6198 - val_loss: 1.4993 - val_accuracy: 0.4924
Epoch 24/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.0677 - accur
acy: 0.6244 - val_loss: 1.4622 - val_accuracy: 0.5078
Epoch 25/100
1407/1407 [=====] - 18s 13ms/step - loss: 1.0571 - accur
acy: 0.6297 - val_loss: 1.4917 - val_accuracy: 0.4990
Epoch 26/100
1407/1407 [=====] - 19s 14ms/step - loss: 1.0395 - accur
acy: 0.6327 - val_loss: 1.4888 - val_accuracy: 0.4896
Epoch 27/100
1407/1407 [=====] - 18s 13ms/step - loss: 1.0298 - accur
acy: 0.6370 - val_loss: 1.5358 - val_accuracy: 0.5024
Epoch 28/100
1407/1407 [=====] - 18s 13ms/step - loss: 1.0150 - accur
acy: 0.6444 - val_loss: 1.5219 - val_accuracy: 0.5030
Epoch 29/100
1407/1407 [=====] - 16s 12ms/step - loss: 1.0100 - accur
acy: 0.6456 - val_loss: 1.4933 - val_accuracy: 0.5098
Epoch 30/100
1407/1407 [=====] - 20s 14ms/step - loss: 0.9956 - accur
acy: 0.6492 - val_loss: 1.4756 - val_accuracy: 0.5012
Epoch 31/100
1407/1407 [=====] - 16s 11ms/step - loss: 0.9787 - accur
acy: 0.6576 - val_loss: 1.5181 - val_accuracy: 0.4936
Epoch 32/100
1407/1407 [=====] - 17s 12ms/step - loss: 0.9710 - accur
acy: 0.6565 - val_loss: 1.7510 - val_accuracy: 0.4568
Epoch 33/100
1407/1407 [=====] - 20s 14ms/step - loss: 0.9613 - accur
acy: 0.6628 - val_loss: 1.5576 - val_accuracy: 0.4910
Epoch 34/100
1407/1407 [=====] - 19s 14ms/step - loss: 0.9530 - accur
acy: 0.6651 - val_loss: 1.5087 - val_accuracy: 0.5046
Epoch 35/100
1407/1407 [=====] - 19s 13ms/step - loss: 0.9388 - accur
acy: 0.6701 - val_loss: 1.5534 - val_accuracy: 0.4950
Epoch 36/100
1407/1407 [=====] - 17s 12ms/step - loss: 0.9331 - accur
acy: 0.6743 - val_loss: 1.5033 - val_accuracy: 0.5046
Epoch 37/100
1407/1407 [=====] - 19s 14ms/step - loss: 0.9144 - accur
acy: 0.6808 - val_loss: 1.5679 - val_accuracy: 0.5028
157/157 [=====] - 0s 2ms/step - loss: 1.4236 - accuracy:
0.5074

```

Out[125... [1.4236289262771606, 0.5073999762535095]

- *Is the model converging faster than before?* Much faster! The previous model took 29 epochs to reach the lowest validation loss, while the new model achieved that same loss in just 12 epochs and continued to make progress until the 17th epoch. The BN layers stabilized training and allowed us to use a much larger learning rate, so convergence was faster.

- *Does BN produce a better model?* Yes! The final model is also much better, with 50.7% validation accuracy instead of 46.7%. It's still not a very good model, but at least it's much better than before (a Convolutional Neural Network would do much better, but that's a different topic, see chapter 14).
- *How does BN affect training speed?* Although the model converged much faster, each epoch took about 15s instead of 10s, because of the extra computations required by the BN layers. But overall the training time (wall time) to reach the best model was shortened by about 10%.

d.

Exercise: Try replacing Batch Normalization with SELU, and make the necessary adjustments to ensure the network self-normalizes (i.e., standardize the input features, use LeCun normal initialization, make sure the DNN contains only a sequence of dense layers, etc.).

```
In [126... tf.random.set_seed(42)

model = tf.keras.Sequential()
model.add(tf.keras.layers.Flatten(input_shape=[32, 32, 3]))
for _ in range(20):
    model.add(tf.keras.layers.Dense(100,
                                     kernel_initializer="lecun_normal",
                                     activation="selu"))

model.add(tf.keras.layers.Dense(10, activation="softmax"))

optimizer = tf.keras.optimizers.Nadam(learning_rate=7e-4)
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=optimizer,
              metrics=["accuracy"])

early_stopping_cb = tf.keras.callbacks.EarlyStopping(
    patience=20, restore_best_weights=True)
model_checkpoint_cb = tf.keras.callbacks.ModelCheckpoint(
    "my_cifar10_selu_model.keras", save_best_only=True)
run_index = 1 # increment every time you train the model
run_logdir = Path() / "my_cifar10_logs" / f"run_selu_{run_index:03d}"
tensorboard_cb = tf.keras.callbacks.TensorBoard(run_logdir)
callbacks = [early_stopping_cb, model_checkpoint_cb, tensorboard_cb]

X_means = X_train.mean(axis=0)
X_stds = X_train.std(axis=0)
X_train_scaled = (X_train - X_means) / X_stds
X_valid_scaled = (X_valid - X_means) / X_stds
X_test_scaled = (X_test - X_means) / X_stds

model.fit(X_train_scaled, y_train, epochs=100,
          validation_data=(X_valid_scaled, y_valid),
          callbacks=callbacks)

model.evaluate(X_valid_scaled, y_valid)
```

Epoch 1/100
1403/1407 [=====>.] - ETA: 0s - loss: 1.9386 - accuracy: 0.3045
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 20s 13ms/step - loss: 1.9385 - accuracy: 0.3046 - val_loss: 1.8175 - val_accuracy: 0.3510
Epoch 2/100
1405/1407 [=====>.] - ETA: 0s - loss: 1.7241 - accuracy: 0.3869
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 16s 11ms/step - loss: 1.7241 - accuracy: 0.3869 - val_loss: 1.7677 - val_accuracy: 0.3614
Epoch 3/100
1407/1407 [=====] - ETA: 0s - loss: 1.6272 - accuracy: 0.4263
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 18s 13ms/step - loss: 1.6272 - accuracy: 0.4263 - val_loss: 1.6878 - val_accuracy: 0.4054
Epoch 4/100
1406/1407 [=====>.] - ETA: 0s - loss: 1.5644 - accuracy: 0.4492
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 18s 13ms/step - loss: 1.5643 - accuracy: 0.4492 - val_loss: 1.6589 - val_accuracy: 0.4304
Epoch 5/100
1404/1407 [=====>.] - ETA: 0s - loss: 1.5080 - accuracy: 0.4712
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 16s 11ms/step - loss: 1.5080 - accuracy: 0.4712 - val_loss: 1.5651 - val_accuracy: 0.4538
Epoch 6/100
1404/1407 [=====>.] - ETA: 0s - loss: 1.4611 - accuracy: 0.4873
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 17s 12ms/step - loss: 1.4613 - accuracy: 0.4872 - val_loss: 1.5305 - val_accuracy: 0.4678
Epoch 7/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.4174 - accuracy: 0.5077 - val_loss: 1.5346 - val_accuracy: 0.4558
Epoch 8/100
1406/1407 [=====>.] - ETA: 0s - loss: 1.3781 - accuracy: 0.5175
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 17s 12ms/step - loss: 1.3781 - accuracy: 0.5175 - val_loss: 1.4773 - val_accuracy: 0.4882
Epoch 9/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.3413 - accuracy: 0.5345 - val_loss: 1.5021 - val_accuracy: 0.4764
Epoch 10/100
1407/1407 [=====] - 15s 10ms/step - loss: 1.3182 - accuracy: 0.5422 - val_loss: 1.5709 - val_accuracy: 0.4762
Epoch 11/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.2832 - accuracy: 0.5571 - val_loss: 1.5345 - val_accuracy: 0.4868
Epoch 12/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.2557 - accuracy: 0.5667 - val_loss: 1.5024 - val_accuracy: 0.4900
Epoch 13/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.2373 - accuracy: 0.5710 - val_loss: 1.5114 - val_accuracy: 0.5028
Epoch 14/100
1404/1407 [=====>.] - ETA: 0s - loss: 1.2071 - accuracy: 0.5846
INFO:tensorflow:Assets written to: my_cifar10_selu_model/assets
1407/1407 [=====] - 17s 12ms/step - loss: 1.2073 - accuracy: 0.5847 - val_loss: 1.4608 - val_accuracy: 0.5026
Epoch 15/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.1843 - accuracy:

acy: 0.5940 - val_loss: 1.4962 - val_accuracy: 0.5038
Epoch 16/100
1407/1407 [=====] - 16s 12ms/step - loss: 1.1617 - accur
acy: 0.6026 - val_loss: 1.5255 - val_accuracy: 0.5062
Epoch 17/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.1452 - accur
acy: 0.6084 - val_loss: 1.5057 - val_accuracy: 0.5036
Epoch 18/100
1407/1407 [=====] - 17s 12ms/step - loss: 1.1297 - accur
acy: 0.6145 - val_loss: 1.5097 - val_accuracy: 0.5010
Epoch 19/100
1407/1407 [=====] - 16s 12ms/step - loss: 1.1004 - accur
acy: 0.6245 - val_loss: 1.5218 - val_accuracy: 0.5014
Epoch 20/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0971 - accur
acy: 0.6304 - val_loss: 1.5253 - val_accuracy: 0.5090
Epoch 21/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.0670 - accur
acy: 0.6345 - val_loss: 1.5006 - val_accuracy: 0.5034
Epoch 22/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.0544 - accur
acy: 0.6407 - val_loss: 1.5244 - val_accuracy: 0.5010
Epoch 23/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0338 - accur
acy: 0.6502 - val_loss: 1.5355 - val_accuracy: 0.5096
Epoch 24/100
1407/1407 [=====] - 14s 10ms/step - loss: 1.0281 - accur
acy: 0.6514 - val_loss: 1.5257 - val_accuracy: 0.5164
Epoch 25/100
1407/1407 [=====] - 14s 10ms/step - loss: 1.4097 - accur
acy: 0.6478 - val_loss: 1.8203 - val_accuracy: 0.3514
Epoch 26/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.3733 - accur
acy: 0.5157 - val_loss: 1.5600 - val_accuracy: 0.4664
Epoch 27/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.2032 - accur
acy: 0.5814 - val_loss: 1.5367 - val_accuracy: 0.4944
Epoch 28/100
1407/1407 [=====] - 16s 11ms/step - loss: 1.1291 - accur
acy: 0.6121 - val_loss: 1.5333 - val_accuracy: 0.4852
Epoch 29/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0734 - accur
acy: 0.6317 - val_loss: 1.5475 - val_accuracy: 0.5032
Epoch 30/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0294 - accur
acy: 0.6469 - val_loss: 1.5400 - val_accuracy: 0.5052
Epoch 31/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0081 - accur
acy: 0.6605 - val_loss: 1.5617 - val_accuracy: 0.4856
Epoch 32/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0109 - accur
acy: 0.6603 - val_loss: 1.5727 - val_accuracy: 0.5124
Epoch 33/100
1407/1407 [=====] - 17s 12ms/step - loss: 0.9646 - accur
acy: 0.6762 - val_loss: 1.5333 - val_accuracy: 0.5174
Epoch 34/100
1407/1407 [=====] - 16s 11ms/step - loss: 0.9597 - accur
acy: 0.6789 - val_loss: 1.5601 - val_accuracy: 0.5016
157/157 [=====] - 0s 1ms/step - loss: 1.4608 - accuracy:
0.5026

Out[126... [1.4607702493667603, 0.5026000142097473]

This model reached the first model's validation loss in just 8 epochs. After 14 epochs, it reached its lowest validation loss, with about 50.3% accuracy, which is better than the original model (46.7%), but not quite as good as the model using batch normalization (50.7%). Each epoch took only 9 seconds. So it's the fastest model to train so far.

e.

Exercise: Try regularizing the model with alpha dropout. Then, without retraining your model, see if you can achieve better accuracy using MC Dropout.

Warning: there are now two versions of `AlphaDropout`. One is deprecated and also broken in some recent versions of TF, and unfortunately that's the version in the `tensorflow` library. Luckily, there's a perfectly fine version in the `keras` library (i.e., `keras`, not `tf.keras`). It's neither deprecated nor broken, so let's import and use that one:

In [127... `import keras.layers`

```
In [128... tf.random.set_seed(42)

model = tf.keras.Sequential()
model.add(tf.keras.layers.Flatten(input_shape=[32, 32, 3]))
for _ in range(20):
    model.add(tf.keras.layers.Dense(100,
                                     kernel_initializer="lecun_normal",
                                     activation="selu"))

model.add(keras.layers.AlphaDropout(rate=0.1))
model.add(tf.keras.layers.Dense(10, activation="softmax"))

optimizer = tf.keras.optimizers.Nadam(learning_rate=5e-4)
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=optimizer,
              metrics=["accuracy"])

early_stopping_cb = tf.keras.callbacks.EarlyStopping(
    patience=20, restore_best_weights=True)
model_checkpoint_cb = tf.keras.callbacks.ModelCheckpoint(
    "my_cifar10_alpha_dropout_model.keras", save_best_only=True)
run_index = 1 # increment every time you train the model
run_logdir = Path() / "my_cifar10_logs" / f"run_alpha_dropout_{run_index:03d}"
tensorboard_cb = tf.keras.callbacks.TensorBoard(run_logdir)
callbacks = [early_stopping_cb, model_checkpoint_cb, tensorboard_cb]

X_means = X_train.mean(axis=0)
X_stds = X_train.std(axis=0)
X_train_scaled = (X_train - X_means) / X_stds
X_valid_scaled = (X_valid - X_means) / X_stds
X_test_scaled = (X_test - X_means) / X_stds

model.fit(X_train_scaled, y_train, epochs=100,
```

```
validation_data=(X_valid_scaled, y_valid),  
callbacks=callbacks)  
  
model.evaluate(X_valid_scaled, y_valid)
```

Epoch 1/100
1405/1407 [=====>.] - ETA: 0s - loss: 1.8953 - accuracy: 0.3240
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 18s 11ms/step - loss: 1.8950 - accuracy: 0.3239 - val_loss: 1.7556 - val_accuracy: 0.3812

Epoch 2/100
1403/1407 [=====>.] - ETA: 0s - loss: 1.6618 - accuracy: 0.4129
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 16s 11ms/step - loss: 1.6618 - accuracy: 0.4130 - val_loss: 1.6563 - val_accuracy: 0.4114

Epoch 3/100
1402/1407 [=====>.] - ETA: 0s - loss: 1.5772 - accuracy: 0.4431
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 16s 11ms/step - loss: 1.5770 - accuracy: 0.4432 - val_loss: 1.6507 - val_accuracy: 0.4232

Epoch 4/100
1406/1407 [=====>.] - ETA: 0s - loss: 1.5081 - accuracy: 0.4673
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 15s 10ms/step - loss: 1.5081 - accuracy: 0.4672 - val_loss: 1.5892 - val_accuracy: 0.4566

Epoch 5/100
1403/1407 [=====>.] - ETA: 0s - loss: 1.4560 - accuracy: 0.4902
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.4561 - accuracy: 0.4902 - val_loss: 1.5382 - val_accuracy: 0.4696

Epoch 6/100
1401/1407 [=====>.] - ETA: 0s - loss: 1.4095 - accuracy: 0.5050
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 16s 11ms/step - loss: 1.4094 - accuracy: 0.5050 - val_loss: 1.5236 - val_accuracy: 0.4818

Epoch 7/100
1401/1407 [=====>.] - ETA: 0s - loss: 1.3634 - accuracy: 0.5234
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 14s 10ms/step - loss: 1.3636 - accuracy: 0.5232 - val_loss: 1.5139 - val_accuracy: 0.4840

Epoch 8/100
1405/1407 [=====>.] - ETA: 0s - loss: 1.3297 - accuracy: 0.5377
INFO:tensorflow:Assets written to: my_cifar10_alpha_dropout_model/assets
1407/1407 [=====] - 15s 11ms/step - loss: 1.3296 - accuracy: 0.5378 - val_loss: 1.4780 - val_accuracy: 0.4982

Epoch 9/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.2907 - accuracy: 0.5485 - val_loss: 1.5151 - val_accuracy: 0.4854

Epoch 10/100
1407/1407 [=====] - 13s 10ms/step - loss: 1.2559 - accuracy: 0.5646 - val_loss: 1.4980 - val_accuracy: 0.4976

Epoch 11/100
1407/1407 [=====] - 14s 10ms/step - loss: 1.2221 - accuracy: 0.5767 - val_loss: 1.5199 - val_accuracy: 0.4990

Epoch 12/100
1407/1407 [=====] - 13s 9ms/step - loss: 1.1960 - accuracy: 0.5870 - val_loss: 1.5167 - val_accuracy: 0.5030

Epoch 13/100
1407/1407 [=====] - 14s 10ms/step - loss: 1.1684 - accuracy: 0.5955 - val_loss: 1.5815 - val_accuracy: 0.5014

Epoch 14/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.1463 - accuracy: 0.6025 - val_loss: 1.5427 - val_accuracy: 0.5112

Epoch 15/100
1407/1407 [=====] - 13s 9ms/step - loss: 1.1125 - accuracy:


```

cy: 0.6169 - val_loss: 1.5868 - val_accuracy: 0.5212
Epoch 16/100
1407/1407 [=====] - 12s 8ms/step - loss: 1.0854 - accuracy: 0.6243 - val_loss: 1.6234 - val_accuracy: 0.5090
Epoch 17/100
1407/1407 [=====] - 15s 11ms/step - loss: 1.0668 - accuracy: 0.6328 - val_loss: 1.6162 - val_accuracy: 0.5072
Epoch 18/100
1407/1407 [=====] - 15s 10ms/step - loss: 1.0440 - accuracy: 0.6442 - val_loss: 1.5748 - val_accuracy: 0.5162
Epoch 19/100
1407/1407 [=====] - 12s 9ms/step - loss: 1.0272 - accuracy: 0.6477 - val_loss: 1.6518 - val_accuracy: 0.5200
Epoch 20/100
1407/1407 [=====] - 13s 10ms/step - loss: 1.0007 - accuracy: 0.6594 - val_loss: 1.6224 - val_accuracy: 0.5186
Epoch 21/100
1407/1407 [=====] - 15s 10ms/step - loss: 0.9824 - accuracy: 0.6639 - val_loss: 1.6972 - val_accuracy: 0.5136
Epoch 22/100
1407/1407 [=====] - 12s 9ms/step - loss: 0.9660 - accuracy: 0.6714 - val_loss: 1.7210 - val_accuracy: 0.5278
Epoch 23/100
1407/1407 [=====] - 13s 10ms/step - loss: 0.9472 - accuracy: 0.6780 - val_loss: 1.6436 - val_accuracy: 0.5006
Epoch 24/100
1407/1407 [=====] - 14s 10ms/step - loss: 0.9314 - accuracy: 0.6819 - val_loss: 1.7059 - val_accuracy: 0.5160
Epoch 25/100
1407/1407 [=====] - 13s 9ms/step - loss: 0.9172 - accuracy: 0.6888 - val_loss: 1.6926 - val_accuracy: 0.5200
Epoch 26/100
1407/1407 [=====] - 14s 10ms/step - loss: 0.8990 - accuracy: 0.6947 - val_loss: 1.7705 - val_accuracy: 0.5148
Epoch 27/100
1407/1407 [=====] - 13s 9ms/step - loss: 0.8758 - accuracy: 0.7028 - val_loss: 1.7023 - val_accuracy: 0.5198
Epoch 28/100
1407/1407 [=====] - 12s 8ms/step - loss: 0.8622 - accuracy: 0.7090 - val_loss: 1.7567 - val_accuracy: 0.5184
157/157 [=====] - 0s 1ms/step - loss: 1.4780 - accuracy: 0.4982

```

```
Out[128...] [1.4779616594314575, 0.498199999332428]
```

The model reaches 48.1% accuracy on the validation set. That's worse than without dropout (50.3%). With an extensive hyperparameter search, it might be possible to do better (I tried dropout rates of 5%, 10%, 20% and 40%, and learning rates 1e-4, 3e-4, 5e-4, and 1e-3), but probably not much better in this case.

Let's use MC Dropout now. We will need the `MCA1phaDropout` class we used earlier, so let's just copy it here for convenience:

```
In [129...] class MCA1phaDropout(keras.layers.AlphaDropout):
    def call(self, inputs):
        return super().call(inputs, training=True)
```



```

model.add(keras.layers.AlphaDropout(rate=0.1))
model.add(tf.keras.layers.Dense(10, activation="softmax"))

optimizer = tf.keras.optimizers.SGD()
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=optimizer,
              metrics=["accuracy"])

```

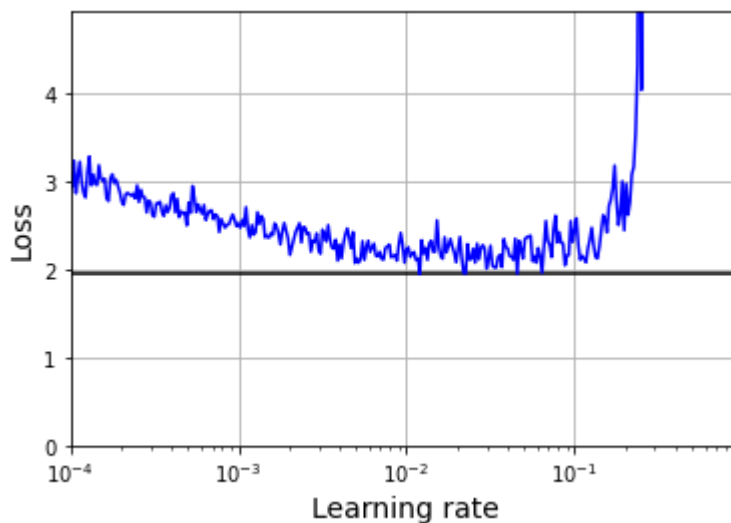
In [134...

```

batch_size = 128
rates, losses = find_learning_rate(model, X_train_scaled, y_train, epochs=1,
                                  batch_size=batch_size)
plot_lr_vs_loss(rates, losses)

```

352/352 [=====] - 3s 8ms/step - loss: nan - accuracy: 0.1706



In [135...

```

tf.random.set_seed(42)

model = tf.keras.Sequential()
model.add(tf.keras.layers.Flatten(input_shape=[32, 32, 3]))
for _ in range(20):
    model.add(tf.keras.layers.Dense(100,
                                    kernel_initializer="lecun_normal",
                                    activation="selu"))

model.add(keras.layers.AlphaDropout(rate=0.1))
model.add(tf.keras.layers.Dense(10, activation="softmax"))

optimizer = tf.keras.optimizers.SGD(learning_rate=2e-2)
model.compile(loss="sparse_categorical_crossentropy",
              optimizer=optimizer,
              metrics=["accuracy"])

```

In [136...

```

n_epochs = 15
n_iterations = math.ceil(len(X_train_scaled) / batch_size) * n_epochs
onecycle = OneCycleScheduler(n_iterations, max_lr=0.05)
history = model.fit(X_train_scaled, y_train, epochs=n_epochs, batch_size=batch_size,
                   validation_data=(X_valid_scaled, y_valid),
                   callbacks=[onecycle])

```

```

Epoch 1/15
352/352 [=====] - 3s 9ms/step - loss: 2.0559 - accuracy:
0.2839 - val_loss: 1.7917 - val_accuracy: 0.3768
Epoch 2/15
352/352 [=====] - 3s 8ms/step - loss: 1.7596 - accuracy:
0.3797 - val_loss: 1.6566 - val_accuracy: 0.4258
Epoch 3/15
352/352 [=====] - 3s 8ms/step - loss: 1.6199 - accuracy:
0.4247 - val_loss: 1.6395 - val_accuracy: 0.4260
Epoch 4/15
352/352 [=====] - 3s 9ms/step - loss: 1.5451 - accuracy:
0.4524 - val_loss: 1.6202 - val_accuracy: 0.4408
Epoch 5/15
352/352 [=====] - 3s 8ms/step - loss: 1.4952 - accuracy:
0.4691 - val_loss: 1.5981 - val_accuracy: 0.4488
Epoch 6/15
352/352 [=====] - 3s 9ms/step - loss: 1.4541 - accuracy:
0.4842 - val_loss: 1.5720 - val_accuracy: 0.4490
Epoch 7/15
352/352 [=====] - 3s 9ms/step - loss: 1.4171 - accuracy:
0.4967 - val_loss: 1.6035 - val_accuracy: 0.4470
Epoch 8/15
352/352 [=====] - 3s 9ms/step - loss: 1.3497 - accuracy:
0.5194 - val_loss: 1.4918 - val_accuracy: 0.4864
Epoch 9/15
352/352 [=====] - 3s 9ms/step - loss: 1.2788 - accuracy:
0.5459 - val_loss: 1.5597 - val_accuracy: 0.4672
Epoch 10/15
352/352 [=====] - 3s 9ms/step - loss: 1.2070 - accuracy:
0.5707 - val_loss: 1.5845 - val_accuracy: 0.4864
Epoch 11/15
352/352 [=====] - 3s 10ms/step - loss: 1.1433 - accurac
y: 0.5926 - val_loss: 1.5293 - val_accuracy: 0.4998
Epoch 12/15
352/352 [=====] - 3s 9ms/step - loss: 1.0745 - accuracy:
0.6182 - val_loss: 1.5118 - val_accuracy: 0.5072
Epoch 13/15
352/352 [=====] - 3s 10ms/step - loss: 1.0030 - accurac
y: 0.6413 - val_loss: 1.5388 - val_accuracy: 0.5204
Epoch 14/15
352/352 [=====] - 3s 10ms/step - loss: 0.9388 - accurac
y: 0.6654 - val_loss: 1.5547 - val_accuracy: 0.5210
Epoch 15/15
352/352 [=====] - 3s 9ms/step - loss: 0.8989 - accuracy:
0.6805 - val_loss: 1.5835 - val_accuracy: 0.5242

```

One cycle allowed us to train the model in just 15 epochs, each taking only 2 seconds (thanks to the larger batch size). This is several times faster than the fastest model we trained so far. Moreover, we improved the model's performance (from 50.7% to 52.0%).